CLIMATE ACTION PLAN FOR MAKINDYE-SSABAGABO MUNICIPALITY

2025-2030



Approved by the Council under Minute Number MSMC/COU/03/26/06/25

Executive Summary

Introduction

Uganda's greenhouse gas (GHG) emissions are dominated by the Agriculture, Forestry, and Other Land Use (AFOLU) sector, accounting for about 86% of the total (NDC, 2022). Uganda's greenhouse gas (GHG) emissions are projected to increase from 90.1 MtCO2e in 2015 to 148.8 MtCO2e in 2030 and 235.7 MtCO2e by 2050 under the Business as Usual (BAU) Scenario

Objectives

The objectives of this climate action plan are:

1. To establish a climate change and vulnerability baseline for Makindye-Ssabagabo Municipality
2. To provide climate actions that address key vulnerabilities, and build adaptive capacities and resilience at all levels
3. To facilitate climate risk decision-making processes with guidance related to budgeting, climate action implementation arrangements, resource mobilization, and dissemination of climate information
4. To promote a low-carbon and climate-resilient society by increasing public awareness and integrating climate actions into broader policies

Vision Statement:

"A resilient, sustainable, and climate-smart Makindye-Ssabagabo Municipality where communities thrive in harmony with the environment, leveraging innovative solutions and local knowledge to mitigate climate impacts, adapt to changing conditions, and foster inclusive socio-economic development."

The purpose of this plan is to:

1. Strengthen the resilience of communities and infrastructure to withstand and recover from climate-related disasters such as floods, heat waves, and drought.
2. Increase community awareness of human-induced climate change impacts and promote responsible environmental practices.
3. Implement sustainable waste management, water conservation, and energy efficiency measures to reduce carbon emissions and protect natural ecosystems.
4. Promote climate-smart agriculture and adaptive livelihood strategies to improve food security and reduce vulnerability to climate variability.
5. Invest in green, eco-friendly urban planning and infrastructure to minimize risks like flooding, heat stress, and environmental degradation.
6. Foster inclusive, community-driven climate action by engaging residents, leaders, and organizations in adaptation and mitigation initiatives.
7. Create sustainable green jobs and livelihoods that support economic development while safeguarding the environment.
8. Leverage technology and indigenous knowledge for climate monitoring, adaptation, and emissions reduction.
9. Develop robust municipal policies and frameworks that integrate climate resilience, sustainability, and urban development.

Greenhouse Gas Emission and climate change

The energy sector has the largest source of emissions, followed by waste sector. The alarming source of emissions for the various sectors or sub-sectors is the solid waste generated from the city, followed by on-road transportation and non-specified sources.

The communities were asked whether they have experienced climate change or variability in their areas. About 89.5% of the respondents reported experiencing climate change or variability in their respective wards. The indicators of climate change or variability reported by community members include increased temperatures (15.3%), increased extreme events such as floods and droughts (12.4%), prolonged dry spells (11.2%), unpredictable and erratic rainfall patterns (10.2%), increased amount of rainfall (8.6%), change of planting seasons (7.1%), and increased occurrence of strong winds (6.8%)

The observed average annual maximum temperatures for Makindye Ssabagabo municipality are about 24.6 °C, with a minimum of 24.1 °C in August and a maximum of 25.9 °C in February. For the climatological period 1994–2024, the maximum temperature in Makindye Ssabagabo municipality has increased at a rate of 0.23°C per decade. Between 1994 and 2024, the highest maximum temperature of 25.3°C was recorded in 2024. The projections of maximum temperature (2025–2030) indicate an increase of 0.41°C and 0.64°C under moderate (RCP 4.5) and high (RCP 8.5) greenhouse gas (GHG) concentration scenarios, respectively, relative to the 1994–2024 average.

Climate change actions and coordination

The climate action options proposed in this plan are meant to reduce greenhouse gas emissions (mitigation) and facilitate climate adaptation in Makindye-Ssabagabo Municipality. These actions are categorised into 12 result areas: Waste management, Energy, Transportation, Trade, Manufacturing, Agriculture, Environment, ICT, Land and housing, social development (Communities), Education and Sport and Health.

Makindye Ssabagabo Municipal Council will serve as the lead entity responsible for implementing the Climate Action Plan. A Municipal Climate and Disaster Taskforce, chaired by the Department of Natural Resources, will be established. This taskforce will comprise technical representatives from relevant departments including Works, Health, Physical Planning, Community Development, and Education. The taskforce will be responsible for overseeing plan implementation, mobilizing resources, coordinating activities, conducting monitoring and evaluation, and reporting progress. It will report directly to the Town Clerk.

Foreword

Climate change presents an urgent and complex challenge that affects every sector of our municipality, from agriculture and health to infrastructure and livelihoods. As the frontline government institution serving the people of Makindye Ssabagabo, we recognize our responsibility to act boldly and collaboratively in building a climate-resilient and sustainable future for our communities.

This Climate Action Plan for the period 2025-2030 is the result of a highly participatory and inclusive process involving local leaders, technical experts, community members, and development partners. It provides a comprehensive roadmap to mitigate greenhouse gas emissions, adapt to the changing climate, and reduce the vulnerability of our population to climate-related shocks such as floods, droughts, and heatwaves.

The plan aligns with Uganda's national commitments under the Paris Agreement, the updated Nationally Determined Contributions (NDCs), and the Sustainable Development Goals (SDGs), while remaining tailored to the unique realities of our municipality. It outlines practical climate actions across 12 priority sectors, including waste management, energy, transportation, agriculture, land use, and social development.

Through this document, we commit to fostering a green economy, integrating climate considerations into our planning frameworks, strengthening early warning systems, and empowering our communities with the knowledge.

On behalf of Makindye Ssabagabo Municipal Council, I extend my gratitude to all stakeholders who contributed to the development of this plan. It is our collective vision, and it will require our collective effort to implement. Let us embrace this opportunity to ensure that Makindye Ssabagabo remains not only habitable, but prosperous, inclusive, and climate-smart for generations to come.



Ssemwanga Godfrey

His Worship, Mayor

Makindye Ssabagabo Municipality

Acknowledgement

The development of the Makindye Ssabagabo Climate Action Plan (2025–2030) would not have been possible without the invaluable support, collaboration, and commitment of numerous individuals and institutions.

First and foremost, we extend our sincere appreciation to the political leadership of Makindye Ssabagabo Municipal Council for their unwavering commitment and for creating an enabling environment for this process to take place. We particularly thank the Office of the Mayor, the Speaker, and all Municipal Councilors for their strategic guidance and support.

Special thanks go to the Department of Natural Resources for leading the technical coordination of this process and for ensuring that all stakeholder voices were meaningfully included.

We also recognize the valuable contributions from community leaders, private sector actors, and residents who participated in the consultations across the municipality. Your shared experiences and insights shaped the priorities and strategies outlined in this plan.

I wish to extend our heartfelt appreciation to the World Bank for their generous support through the Greater Kampala Metropolitan Area Urban Development Program (GKMA-UDP). The funding provided under this program was instrumental in facilitating the development and finalization of this Climate Action Plan.

Finally, we appreciate the contributions of the environment and earth consultants, who supported the documentation and compilation of this Climate Action Plan.



Richard K Monday

Town Clerk

Makindye Ssabagabo Municipality

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Acronyms

AFOLU	Agriculture, forestry and other land use
BAU	Business as Usual
CHIRPS	Climate Hazards Group InfraRed Precipitation with Station data
CIRIS	City Inventory Reporting and Information System
CoM SSA	Covenant of Mayors in Sub-Saharan Africa
DRM	Disaster Risk Management
DRR	Disaster Risk Reduction
EF	Emission Factor
GDP	Gross domestic product
GHG	Greenhouse gas
GHG	GreenHouse Gas
GHG	Greenhouse Gas Emission
GPC	Global Protocol for Community-Scale Greenhouse Gas Emission Inventories
IPPU	Industrial processes and product use
NDC	Nationally Determined Contribution
RCP	Representative Concentration Pathway
SEACAP	Sustainable Energy Access and Climate Action Plan
tCO_2e	Metric tonnes of carbondioxide equivalent
UNFCCC	United Nations Framework Convention on Climate Change
WASH	Water Sanitation and Hygiene

1. INTRODUCTION

1.1 Background

Uganda's greenhouse gas (GHG) emissions are dominated by the Agriculture, Forestry, and Other Land Use (AFOLU) sector, accounting for about 86% of the total (NDC, 2022). Uganda's greenhouse gas (GHG) emissions are projected to increase from 90.1 MtCO2e in 2015 to 148.8 MtCO2e in 2030 and 235.7 MtCO2e by 2050 under the Business as Usual (BAU) Scenario (Figure 1). This projection is in consideration of Makindye-Ssabagabo Municipality, an urban settling with vibrant sectors (energy, transport, agriculture, waste and industries among others) that contribute to this cause with significant amounts of Greenhouse Gases.

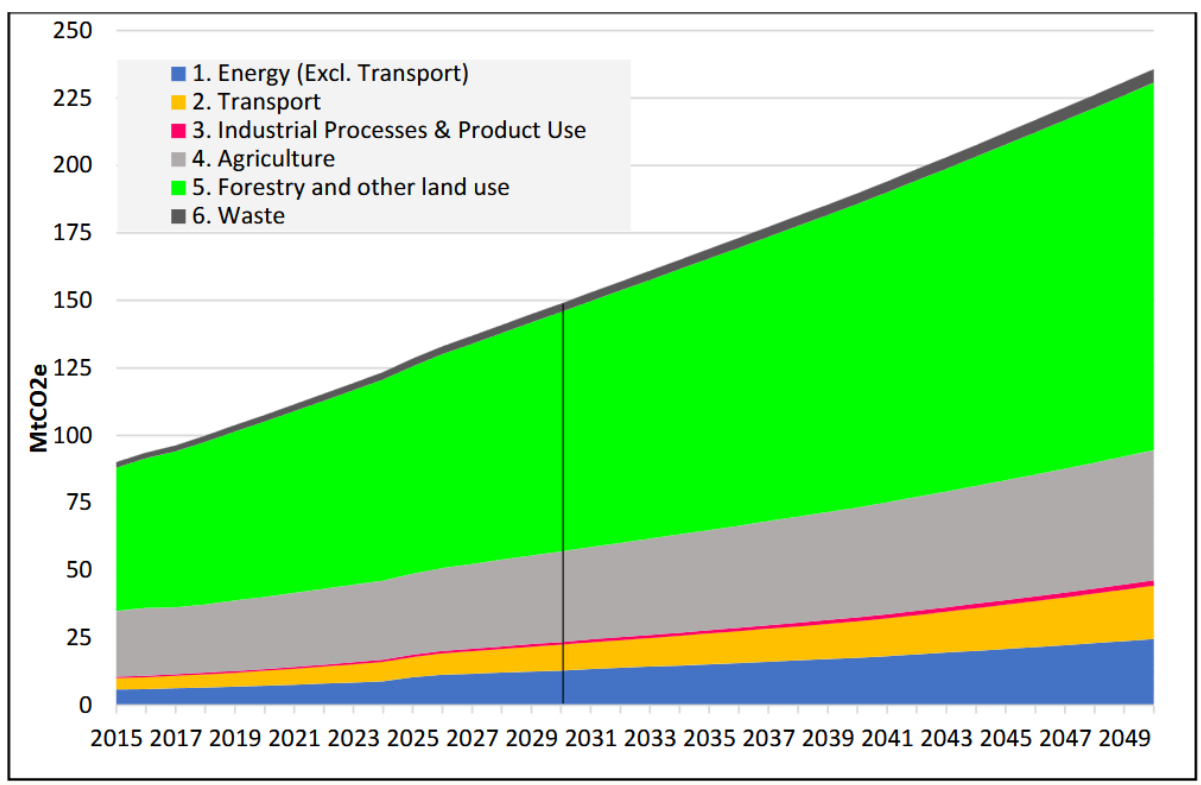


Figure 1: Uganda's BAU GHG emissions trajectory (Source: NDC, 2022)

It is against this backdrop, that Makindye-Ssabagabo Municipal Climate Action Plan has been developed to inform decision making processes geared towards the planning and implementation of enhanced climate mitigation and adaptation actions in the short, medium and long term up to 2030 across the different sectors, aimed at improving their resilience.

1.2 Objectives

The objectives of this climate action plan are:

5. To establish a climate change and vulnerability baseline for Makindye-Ssabagabo Municipality
6. To provide climate actions that address key vulnerabilities, and build adaptive capacities and resilience at all levels
7. To facilitate climate risk decision making processes with guidance related to budgeting, climate action implementation arrangements, resource mobilization and dissemination of climate information
8. To promote a low carbon and climate resilient society by increasing public awareness and integrating climate actions into broader policies

1.3 Vision of the plan

Vision Statement:

"A resilient, sustainable, and climate-smart Makindye-Ssabagabo Municipality where communities thrive in harmony with the environment, leveraging innovative solutions and local knowledge to mitigate climate impacts, adapt to changing conditions, and foster inclusive socio-economic development."

1.4 Purpose of the plan

The purpose of this plan is to:

10. Strengthen the resilience of communities and infrastructure to withstand and recover from climate-related stocks such as floods, heat waves, and drought.
11. Increase community awareness of human-induced climate change impacts and promote responsible environmental practices.
12. Implement sustainable waste management, water conservation, and energy efficiency measures to reduce carbon emissions and protect natural ecosystems.
13. Promote climate-smart agriculture and adaptive livelihood strategies to improve food security and reduce vulnerability to climate variability.
14. Invest in green, eco-friendly urban planning and infrastructure to minimize risks like flooding, heat stress, and environmental degradation.
15. Foster inclusive, community-driven climate action by engaging local residents, leaders, and organizations in adaptation and mitigation initiatives.
16. Create sustainable green jobs and livelihoods that support economic development while safeguarding the environment.
17. Leverage technology and indigenous knowledge for climate monitoring, adaptation, and emissions reduction.
18. Develop robust municipal policies and frameworks that integrate climate resilience, sustainability, and urban development.

1.5 Planning Process

This plan was developed through a consultative process, indicating that it was stakeholder driven. The steps undertaken to develop this plan included stakeholder consultations (at the municipal, sub division and parish levels), analysis of climate change and vulnerabilities and validation of climate mitigation and adaptation actions. The process was entirely participatory, suggesting that the development of the plan vision, objectives and climate actions were stakeholder led. The participants were selected based on the principles of fairness, equity, gender and social inclusion.

The time frame for this plan is between 2025 and 2030, implying it is also aligned with Uganda's Updated Nationally Determined Contribution (NDC) which was developed in fulfilment of Article 4 of the Paris Agreement. In the updated NDC, Uganda presents an ambitious economy-wide mitigation target in 2030 of 24.7% reduction below the Business As Usual (BAU), a progression from the 22% reduction target communicated in the first NDC in 2016.

While developing this plan, the datasets collected and analysed to demonstrate climate change variability and vulnerabilities included meteorological (rainfall and temperature), social economic, digital elevation model, rivers, health, schools, roads and water sources among others. The results are presented in form of maps, tables and graph displaying climate change mitigation and adaptation options and adaptive capacities within Makindye-Ssabagabo Municipal. The climate actions are presented to reflect 12 result areas and these include Waste management, Energy, Transportation, Trade, Manufacturing, Agriculture (crop and livestock), Environment, ICT, Land and housing, social development (Communities), Education and Sport and Health.

1.6 Policy, Regulatory and Planning Frameworks

This Plan has been developed to integrate multiple objectives, sectoral, catchment, and local planning for climate-resilient adaptation, in alignment with international, national, and strategic commitments. The frameworks aim to reduce emissions, build resilience to climate impacts, enhance climate-resilient infrastructure and services, and strengthen the knowledge base. Table 1 outlines the policy, regulatory, and planning frameworks that guide the planning and implementation of climate-resilient mitigation and adaptation strategies.

Table 1: Strategies, International and national policy and regulatory frameworks

Frameworks, Policies, Plans and Acts	Relevance to Climate actions
International Frameworks	
United Nations Framework Convention on Climate Change (UNFCCC) 1992	Uganda signed and ratified the UNFCCC in 1992. The UNFCCC primarily focuses on developing NDCs, promotes integration of climate change into national planning, and enables access to international climate finance, capacity building and technology to align actions with global climate goals and strengthens institutional frameworks for coordinated climate action.
The Sendai Framework for Disaster Risk Reduction (2015-2030)	The Sendai Framework emphasizes the need to reduce disaster risks that are exacerbated by climate change, including droughts, floods, and extreme weather events, and encourages the integration of DRR into climate change policies and plans. With an aim of building climate resilience, improving EWS,

Frameworks, Policies, Plans and Acts	Relevance to Climate actions
	and ensuring that climate adaptation efforts contribute to reducing vulnerability to climate-related disasters.
The United Nations Convention to Combat Desertification (1994)	The UNCCD focuses on promoting sustainable land management practices that prevent desertification, land degradation, and drought; implementing climate-resilient agricultural practices, improving soil health, and restoring degraded lands; mitigating the impacts of climate change, enhancing carbon sequestration, and increasing agricultural productivity in vulnerable areas, with the goal of contributing to climate adaptation and greenhouse gas reduction.
The Paris Agreement 2016	The Paris Agreement is core in limiting global temperature rise to well below 2°C and pursuing efforts to stay within 1.5°C; it guides the formulation and implementation of NDCs, which outline sector-specific strategies for climate mitigation and adaptation; enhances access to international climate finance, capacity building, and technology transfer; and supports the transition to a low-carbon and climate-resilient development pathway.
Kyoto Protocol (2005)	The Kyoto Protocol's primary focus is on reducing greenhouse gas emissions; implementing emission-reduction projects, accessing climate finance, and promoting low-carbon technologies, supporting climate action goals is easy through the Clean Development Mechanism (CDM).
Sustainable Development Goals (2015).	The SDGs, specifically Goal 13 (Climate Action), focus on strengthening resilience and adaptive capacity to climate-related hazards; promoting the integration of climate change measures into national policies, strategies, and planning; encouraging access to climate finance; investing in renewable energy; and promoting sustainable land and resource use to achieve a low-carbon, climate-resilient future.
African Union Agenda (2063)	Focuses on promoting a prosperous and sustainable Africa by supporting inclusive green growth, environmental sustainability, and climate-resilient economies; advocating for renewable energy adoption, sustainable land use, and climate-smart agriculture; and encouraging regional cooperation and resource mobilization to tackle climate change effectively.
c	The strategy focuses on promoting climate-resilient agriculture, sustainable land management, and ecosystem restoration; addressing land degradation and reducing vulnerability to climate shocks; promoting cross-border collaboration, capacity building, and the use of climate-smart practices; and contributing to greenhouse gas reduction, enhanced adaptive capacity, and improved food security in drought-prone areas.
UN 2030 Agenda	The agenda adopted in 2015 outlines 17 SDGs, with Goal 13 (Climate Action) directly calling for urgent steps to combat climate change and its impacts; guides the integration of climate change into national planning, encourages resilient development, and supports access to international climate finance and technology.
The Ramsar Convention on Wetlands (1971)	The Ramsar Convention promotes the conservation and wise use of wetlands, which advocates for carbon storage, climate regulation, and biodiversity conservation by buffering climate impacts and supporting climate-resilient livelihoods, including fishing, ecotourism, and sustainable agriculture.
National Planning frameworks	

Frameworks, Policies, Plans and Acts	Relevance to Climate actions
Uganda Vision (2040)	The Vision acknowledges climate change as a hindrance to development and thus promotes a shift to a green economy focusing on low-carbon development, renewable energy, and sustainable natural resource use; it also supports integration of climate change adaptation and mitigation into all sectors of the economy to ensure long-term sustainability.
Forth National Development Plan	NDPIV focuses on integrating climate change into national and local development plans. It promotes sustainable use of natural resources, environmental protection, and low-emission development pathways through Program 9. It has supported investment in climate-smart technologies and encourages coordination among sectors to strengthen climate resilience.
Updated Nationally Determined Contribution (2022)	The updated NDC outlines the country's enhanced commitments to reduce greenhouse gas emissions by 24.7% by 2030, prioritizing sectors including agriculture, energy, forestry, and waste management. It promotes transition to clean energy, climate-smart agriculture, and afforestation while advocating for inclusive participation and gender-responsive approaches.
Uganda Disaster Preparedness Plan (2005–2009)	Although it was formulated before Uganda's formal climate change policy frameworks, the plan indirectly supports climate goals by enhancing national capacity to prepare for climate-induced hazards such as floods and droughts.
National Disaster Risk Management (DRM) Plan (2011–2028)	The DRM Plan supports the country's climate resilience efforts by promoting risk-informed development planning. It acknowledges the growing impact of climate-induced hazards, including droughts, floods, and landslides, and advocates for adaptation measures that reduce long-term vulnerability to climate change.
NRM manifesto (2021–2026)	The manifesto fronts the importance of sustainable natural resource management as a foundation for Uganda's future development. It profiles a commitment to climate-smart agriculture, which aims to enhance productivity while minimizing environmental degradation. It promotes the adoption of innovative technologies that can reduce carbon emissions and enhance environmental sustainability. It also calls for the integration of climate change adaptation and mitigation strategies across all sectors, including agriculture, energy, infrastructure, and transport to ensure climate resilience in both urban and rural communities.
Uganda National Adaptation Programmes of Action (2007)	NAPA establishes a strategic approach to climate change adaptation by identifying the most vulnerable communities and sectors. It focuses on actions that enhance resilience in sectors including health, agriculture, and water resources. It proposes adaptation measures including promoting drought-resistant crops, improving water storage, and enhancing EWS for extreme weather events.
Health Sector Development Plan (2015-2020)	The plan recognizes the growing impact of climate change on public health, basically through climate-sensitive diseases including cholera, malaria, and waterborne diseases. It aims to enhance the need for climate adaptation strategies to protect health systems and improve healthcare access for vulnerable populations. The plan also supports initiatives such as vector control in malaria-prone areas and strengthening health sector capacity to handle climate-related health crises.

Frameworks, Policies, Plans and Acts	Relevance to Climate actions
The Water Action Plan (1995)	The plan centers on improving water resource management by ensuring sustainable water use in the face of growing climate change impacts. It emphasizes the development of water infrastructure, water conservation, and promotion of efficient irrigation practices that are important for adapting to climate-induced water scarcity and droughts. It also advocates for better management of wetlands to improve water retention and resilience to climate-related shifts in water availability.
National Policies	
The 1995 Constitution of Uganda (1995)	The Constitution lays the foundation for environmental protection, recognizing the duty of the state and citizens to protect natural resources, such as water, land, wetlands, and biodiversity. It mandates that development must be ecologically sustainable, which supports climate action by providing a legal basis for integrating environmental and climate considerations into national and local policies.
National Environment Management Policy (1995)	NEMP establishes a framework that focuses on integrating climate considerations and environmental sustainability into Uganda's development agenda by promoting sustainable use of natural resources, supporting climate change mitigation and adaptation measures, and advocating for integration of climate issues into sectoral plans, policies, and budgets. NEMP encourages the mobilization of financial and technical resources for environmental protection and resilience building.
Uganda National Land Policy (2013)	The policy advocates for climate-resilient land use planning by promoting sustainable land management practices that prevent degradation and enhance productivity. It encourages protection of fragile ecosystems, including forests, wetlands, and mountainous areas, which are crucial for carbon sequestration and climate regulation. The policy also emphasizes the integration of climate change adaptation strategies into land tenure and utilization systems.
Uganda National Climate Change Policy (2015)	The policy is a central framework for addressing climate change, guiding both mitigation and adaptation efforts, promoting low-carbon development, encouraging sustainable energy use, and advocating for the integration of climate change into all national and sectoral plans; it supports green economy initiatives and provides direction for mobilizing climate finance, research, and technology transfer to build national climate resilience.
Disaster Preparedness and Management Policy (2010)	Although the policy primarily focuses on disaster management, it also recognizes the increasing climate-related hazards, including floods, droughts, and landslides, and their impacts; promotes the mainstreaming of climate change adaptation into disaster risk reduction strategies; and supports actions that build community resilience to the long-term impacts of climate change.
The Uganda Forestry Policy	The policy seeks to promote sustainable forest management for climate change mitigation through biodiversity conservation and carbon sequestration and encourage afforestation, reforestation, and restoration of degraded forest ecosystems, all of which contribute to greenhouse gas emissions reduction and enhancing climate resilience across landscapes.
The Water Statute (1995)	The Water Statute advocates for sustainable use, conservation, efficient water use, protection of water resources, and resilience building in water supply systems, which is key for climate adaptation. It establishes the legal basis for

Frameworks, Policies, Plans and Acts	Relevance to Climate actions
	the integrated water resources management (IWRM) approach to ensure water availability amidst changing climate patterns.
The Gender Policy (1997)	The policy primarily emphasizes equity and inclusiveness in climate change planning and response. It recognizes that women and men experience climate impacts differently and promotes the integration of gender perspectives into environmental and climate policies; it advocates for women's participation in climate actions, mainly in sectors including agriculture, water, and energy, which are highly climate-sensitive
National Acts	
National Forestry and Tree Planting Act (2003)	The Act establishes a legal framework that prioritizes sustainable expansion of forests and management for climate change mitigation through carbon sequestration. It also promotes afforestation, tree planting, and reforestation in degraded areas and supports the conservation of natural forest reserves to reduce greenhouse gas emissions and protect biodiversity.
National Climate Change Act (2021)	The Act primarily seeks to address climate change. It mandates the integration of climate change into all national and sectoral plans, policies, and budgets. It has established institutional structures for climate governance, enforces commitments under international agreements (e.g., the Paris Agreement), and promotes emissions tracking, low-carbon development, and climate finance mechanisms to support climate-resilient growth.
National Disaster Preparedness and Management Act (2021)	The Act supports climate change adaptation by integrating disaster risk reduction (DRR) into national development plans, which helps address the vulnerabilities of sectors including water resources, agriculture, and health to climate change impacts such as droughts and floods. The Act enhances Uganda's capacity to prepare for and manage climate-induced hazards, promoting the sustainability of communities through EWS, resource mobilization, and adaptive measures.
The National Environment Act (NEA) 2019	The NEA provides a framework for environmental management that emphasizes sustainable development and the integration of environmental concerns into national planning. It established the NEMA, a coordinating, monitoring, regulatory, and supervisory body for environmental activities. The Act seeks to address emerging environmental issues, including climate change, by promoting strategic environmental assessments and the management of hazardous chemicals.
The National Water Policy (1999)	The policy provides a framework that prioritizes sustainable management and development of water resources. It has promoted IWRM for adapting to climate variability and ensuring the equitable and efficient use of water across all sectors. It also supports water conservation and sustainable water supply for rain-fed agricultural production affected by climate change; it addresses the need for efficient use of water in industries and urban areas to reduce the carbon and water footprints of these sectors.
The Water Act (1997)	The Act establishes a framework for the use, protection, and management of water resources. It supports climate adaptation efforts, ensuring that water use for various purposes, including agricultural, domestic, industrial, and energy production, is conducted sustainably. This helps mitigate the impacts of climate change by maintaining the availability and integrity of water resources.

Frameworks, Policies, Plans and Acts	Relevance to Climate actions
Land Act, 1998	The Act establishes a framework for land tenure, including customary, freehold, mailo, and leasehold; ownership; and management. Although the Act does not explicitly address climate change issues, it encourages sustainable land-use practices and long-term investments in land conservation, which are important in climate change mitigation and adaptation strategies.
The Local Governments Act (1997)	The Act entrusts local governments with the management of natural resources within their jurisdictions. It empowers local authorities to implement environmental policies and programs customized to their specific contexts, facilitating more effective responses to climate change challenges; LGs can develop and enforce ordinances related to environmental conservation, promote sustainable land use practices, and engage communities in climate adaptation initiatives, enabling grassroots-level climate action, fostering resilience, and sustainable development.
Public Finance Management Act, 2015	PFMA carries out the systematic integration of climate change in planning and budgeting by emphasizing prudent financial management, which ensures that resources are efficiently allocated toward climate-related initiatives; it facilitates the government's ability to fund climate adaptation and mitigation programs, which help build resilience against climate change impacts.
The National Water & Sewerage Corporation Act, 1995	The policy gave rise to the establishment of the National Water and Sewerage Corporation (NWSC). The NWSC has been responsible for piped water provision and focus on expanding water and sewerage services in large urban centers and small urban centers. In this context, the Act supports the implementation of IWRM, encouraging the sustainable management and conservation of water resources, contributing to enhancing climate resilience in urban areas, and helping communities adapt to the impacts of climate change, such as altered precipitation patterns and water scarcity.

2. CLIMATE CHANGE AND VULNERABILITY

2.1 Greenhouse Gas Emission Analysis

The energy sector has the largest source of emissions, followed by waste sector as shown in (Table 2 & Figure 2). Notably, most of the emissions are scope 1 emissions. Whereas, scope 2 emission only form a small portion of the overall total. On the other hand, scope 3 emissions were not estimated in this assessment due to lack of this data.

Table 2: GHG Emissions Summary

GHG Emissions Source (By Sector)		Total GHGs (metric tonnes CO_2e)		
		Scope 1	Scope 2	BASIC
STATIONARY ENERGY	Energy use	308031	476	308507
	Energy generation supplied to the grid			
TRANSPORTATION		121625		121625
WASTE	Waste generated in the city	185227		185227
	Waste generated outside city			
IPPU		NE		
AFOLU		NE		
OTHER SCOPE 3		NE		
TOTAL		614883	476	615359

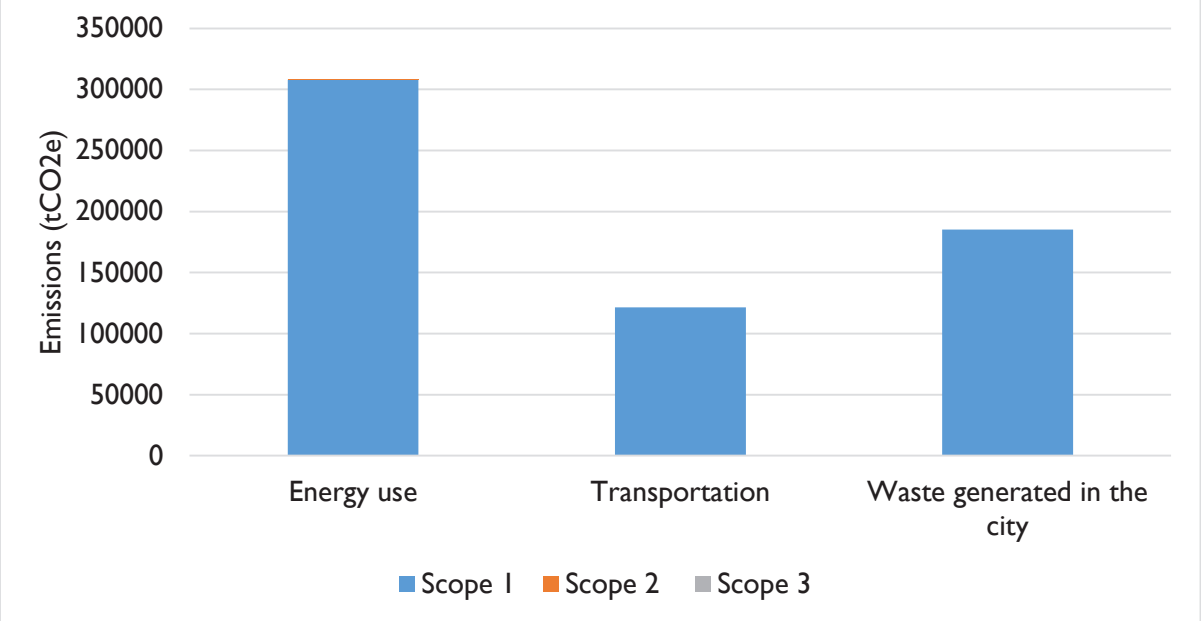


Figure 2: Makindye Ssabagabo GHG emissions summary

Furthermore, the emissions by sector and sub-sector are presented below (Table 3): The alarming source of emissions for the various sectors or sub-sectors is the solid waste generated from the city, followed by on-road transportation and non-specified sources. The contribution of each sector/ sub-sector is illustrated in (Table 3 & Figure 3) below:

Table 3: Emissions by sector and sub-sector

GHG Emissions Source (By Sector and Sub-sector)	Total GHGs (metric tonnes CO2e)		
	Scope 1	Scope 2	Total
I. STATIONARY ENERGY			
Residential buildings	21563	143	21706
Commercial and institutional buildings and facilities	37862	58	37920
Manufacturing industries and construction	98501	274	98775
Energy industries	19026	0	19026
Agriculture, forestry and fishing activities	12422	0	12422
Non-specified sources	118658	0	118658
SUB-TOTAL	**308031**	**475**	**308506**
II. TRANSPORTATION			0
On-road transportation	121625	NO	121625
SUB-TOTAL	**121625**		**121625**
III. WASTE			0
Solid waste generated in the city	124801		124801
Wastewater generated in the city	60426		60426
SUB-TOTAL	**185227**		**185227**
TOTAL	**1229766**	**950**	**1230716**

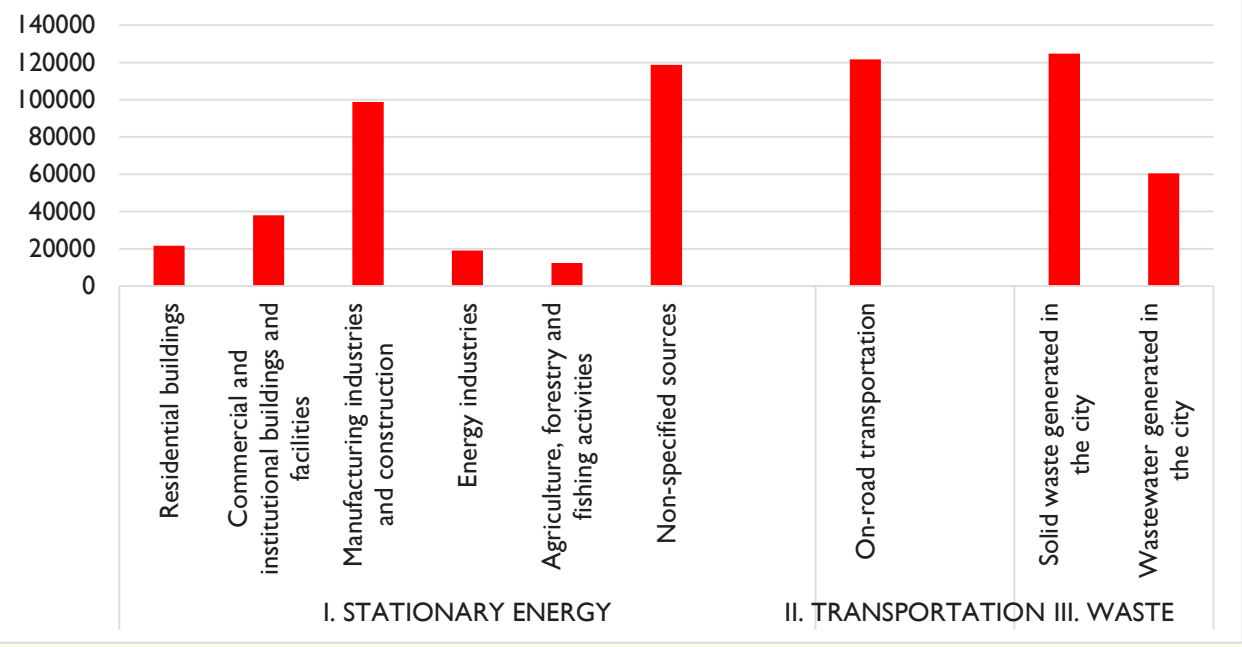


Figure 3: Emissions summary by each sector and sub-sector

There are numerous fuels used to generate energy for transportation or stationary uses. The summary is presented in (Table 4). Results indicate the primary sources of inventory are wood or waste wood (38.5%), diesel (33.94%) and charcoal (15.83%) as shown (FFigure 4).

Table 4: Emissions by fuel for the various sectors or subsectors

Sector	Subsector	Fuel	Emissions (tCO2e)
Stationary Energy	Residential	Wood or wood waste	10271.73
		Charcoal	11291.4
		Electricity	142.68
	Commercial and Institutional	Wood or wood waste	35597.43
		Charcoal	2264.43
		Electricity	57.81
	Manufacturing and Construction	Petrol	11546.01
		Diesel	44720.34
		Paraffin	6731.79
		Charcoal	35502.72
		Electricity	274.29
	Energy Industries	Charcoal	19025.64
	Agriculture, Forestry and Fisheries	Petrol	12421.77
	Non-Specified	Wood or wood waste	118658.1
		Electricity	0
Transportation	On-road Transportation	Petrol	20360.19
		Diesel	101264.67

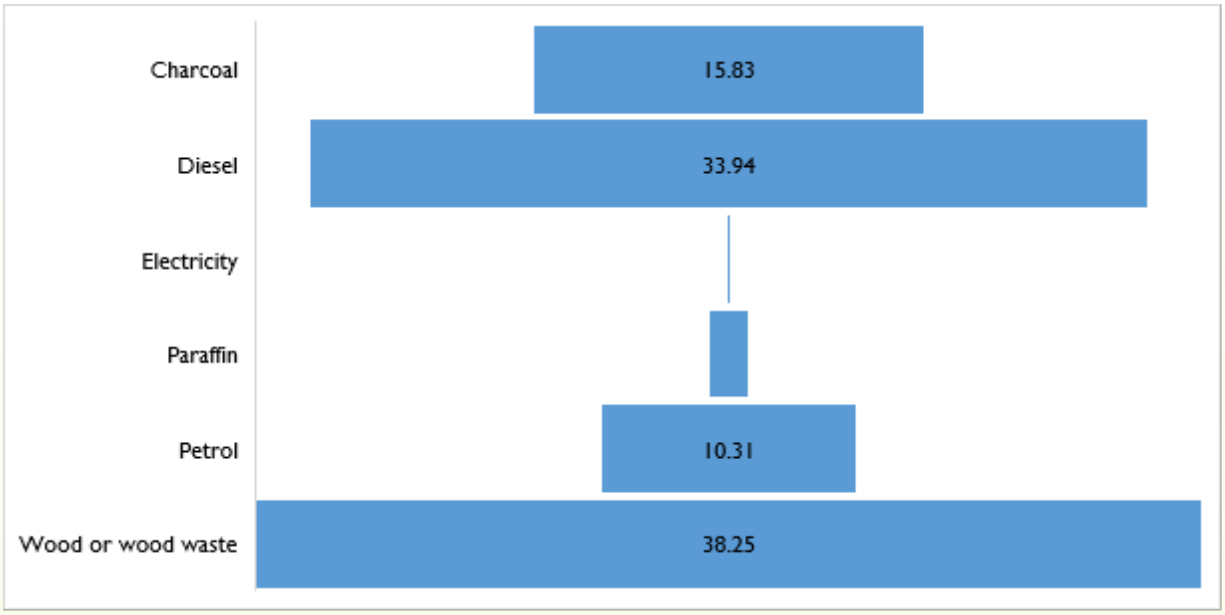


Figure 4: Fuel share percentages

2.2 Indicators of Climate Change

The communities were asked whether they have experienced climate change or variability in their areas. About 89.5% of the respondents reported experiencing climate change or variability in their respective wards. The indicators of climate change or variability reported by community members include increased temperatures (15.3%), increased extreme events such as floods and droughts (12.4%), prolonged dry spells (11.2%), unpredictable and erratic rainfall patterns (10.2%), increased amount of rainfall (8.6%), change of planting seasons (7.1%), and increased occurrence of strong winds (6.8%) (Figure 5). The key informants also revealed that climate change in Makindye Ssabagabo has manifested through increased prolonged dry spells, increased temperatures characterized by too much heat, and changes in planting seasons.

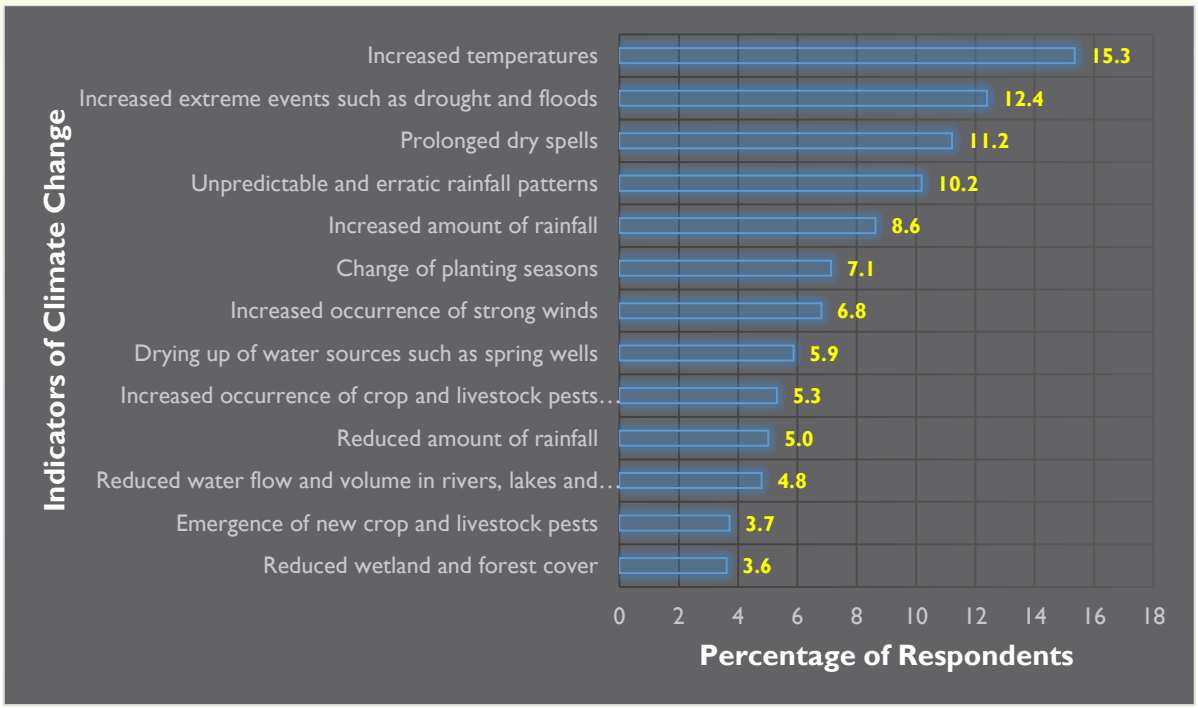


Figure 5: Indicators of climate change in Makindye Ssabagabo Municipality

2.3 Climate Trends and Projections

2.2.1 Observed and Projected Maximum Temperature Trends in Makindye Ssabagabo Municipality

The observed average annual maximum temperatures for Makindye Ssabagabo municipality are about 24.6 °C, with a minimum of 24.1 °C in August and a maximum of 25.9 °C in February. For the climatological period 1994–2024, the maximum temperature in Makindye Ssabagabo municipality has increased at a rate of 0.23°C per decade. Between 1994 and 2024, the highest maximum temperature of 25.3°C was recorded in 2024 (**Figure 6**). The projections of maximum temperature (2025–2030) indicate an increase of 0.41°C and 0.64°C under moderate (RCP 4.5) and high (RCP 8.5) greenhouse gas (GHG) concentration scenarios, respectively, relative to the 1994–2024 average.

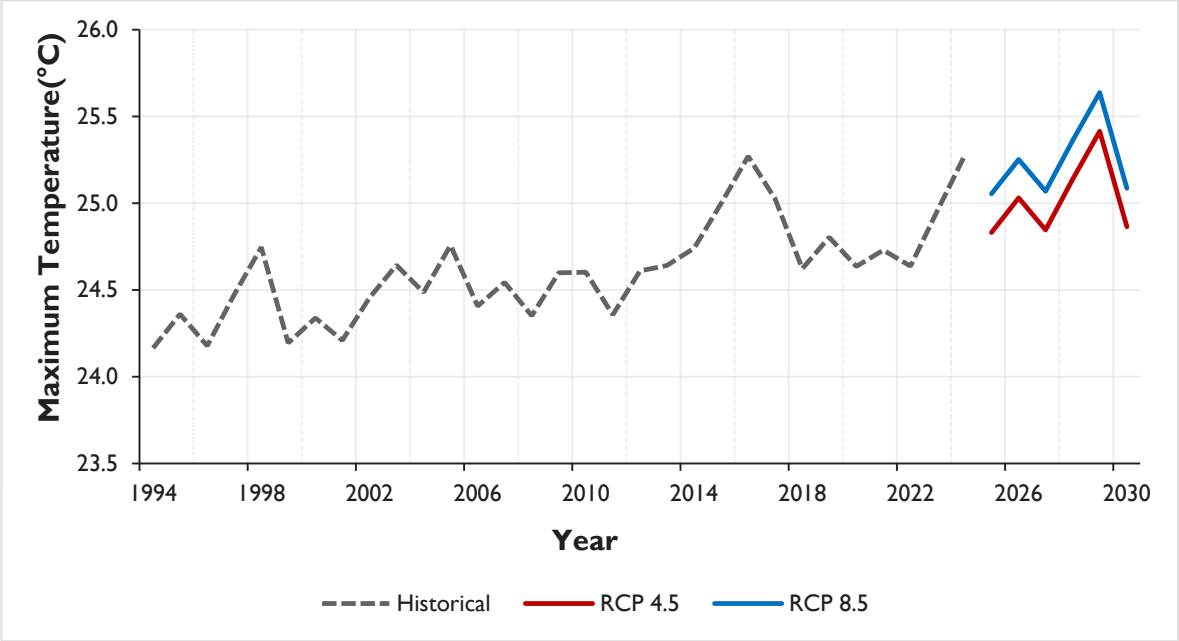


Figure 6: Observed and projected maximum temperatures for Makindye Ssabagabo municipality from 1994 to 2030

The highest projected maximum temperatures between 2025 and 2030 are expected to be experienced in Mutundwe, Namasuba, Masajja, Bunamwaya, Busabala, and Ndejje wards, especially under the high (RCP 8.5) greenhouse gas (GHG) concentration scenario (**Figure 7**).

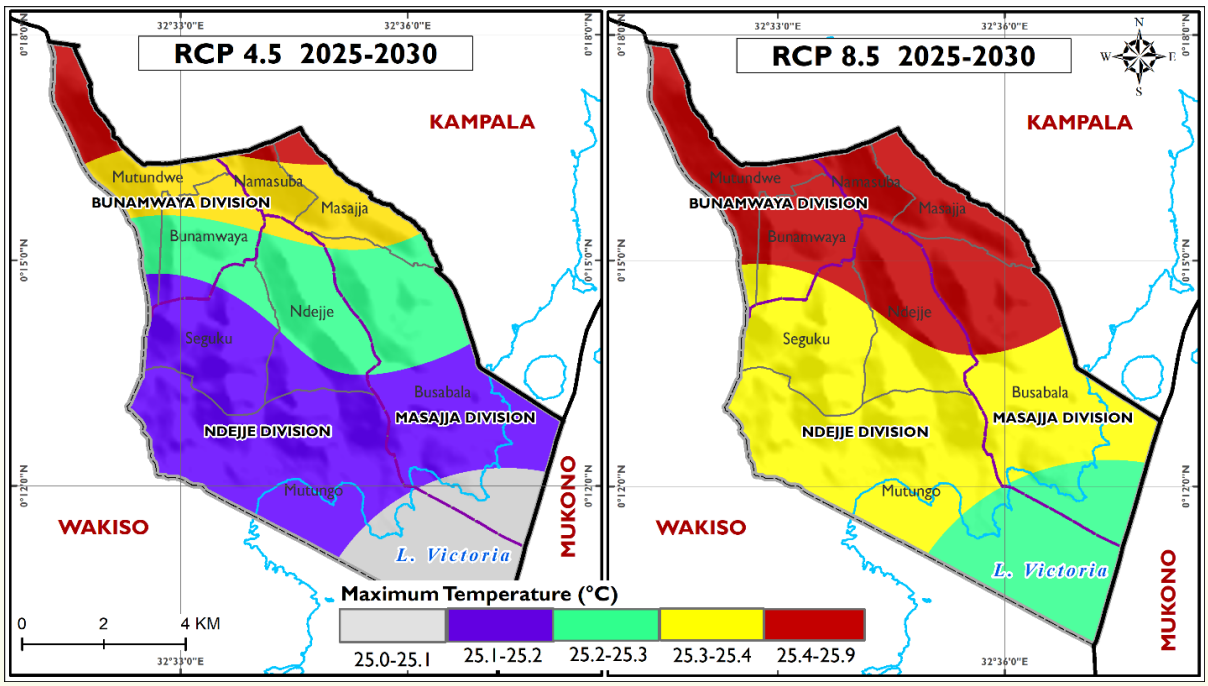


Figure 7: Projected maximum temperature for the 2025-2030 near-term projection period in Makindye Ssabagabo municipality

2.2.2 Observed and Projected Minimum Temperature Trends in Makindye Ssabagabo Municipality

The observed average annual minimum temperatures for Makindye Ssabagabo municipality are about 20.8°C, with a minimum of 19.6°C in August and a maximum of 21.2 °C in February. Over the climatological period 1994-2024, the minimum temperature has increased at a rate of 0.22 °C per decade. The highest minimum temperature of 20.9 °C was recorded in 2024 (**Figure 8**). Projections for the period 2025-2030 show an expected increase in minimum temperature of about 0.7 °C under RCP 4.5 and 0.6 °C under RCP 8.5, relative to the 1994-2024 average.

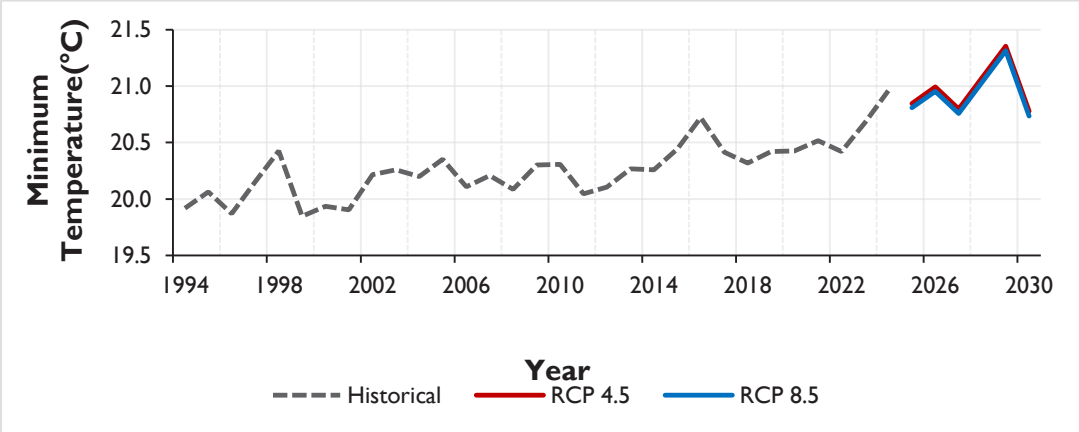


Figure 8: Observed and projected minimum temperatures for Makindye Ssabagabo municipality from 1994 to 2030

The areas near the shores of Lake Victoria (Busabala and Mutungo wards) are projected to have the highest minimum temperatures, especially under the high (RCP 8.5) greenhouse gas (GHG) concentration scenario (**Figure 9**).

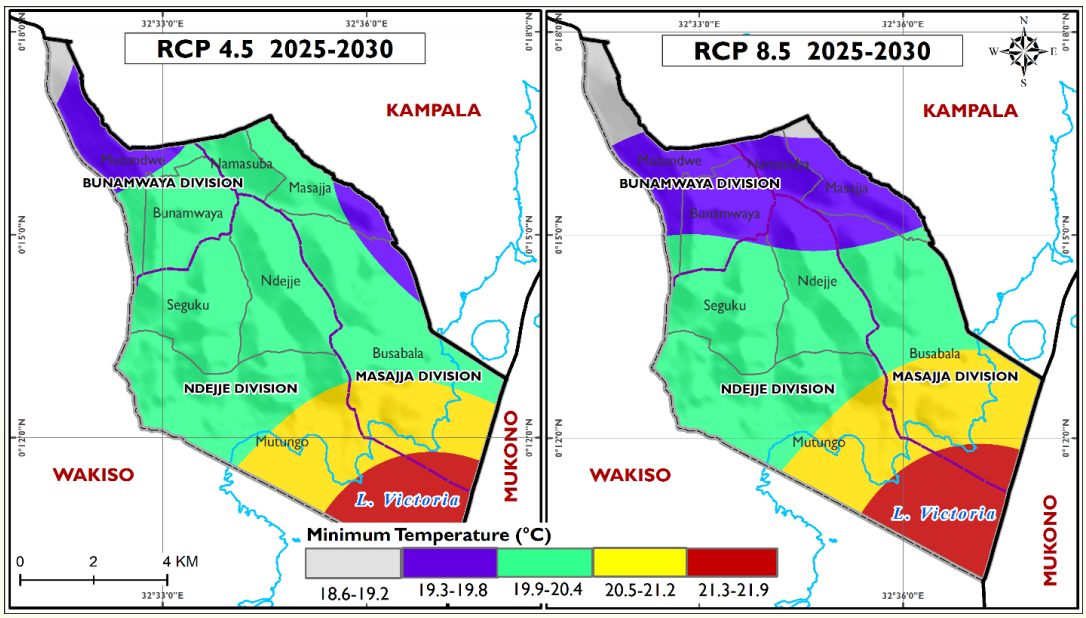


Figure 9: Projected minimum temperature for the 2025-2030 near-term projection period in Makindye Ssabagabo municipality

2.2.3 Observed and Projected Rainfall Trends in Makindye Ssabagabo Municipality

Between 1994 and 2024, the total annual average rainfall for Makindye Ssabagabo Municipality was 1,411.4 mm, and mean monthly rainfall varied from 59.8 mm in February to 218.9 mm in April. For the climatological period 1994-2024, rainfall in Makindye Ssabagabo municipality has decreased at a rate of 24.2 mm per decade. Between 1994 and 2024, the highest annual rainfall of 1,756.4 mm was recorded in 1997 (**Figure 10**). The projections (2025-2030) indicate an increase in mean rainfall for Makindye Ssabagabo Municipality. Under RCP 4.5, mean rainfall is projected to increase by 271.1 mm (19.2%) relative to the 1994-2024 average. Similarly, under RCP 8.5, mean rainfall is projected to increase by 245.6 mm (17.4%) relative to the 1994-2024 average. These projections suggest a potential shift toward wetter conditions in the near future under both climate change scenarios.

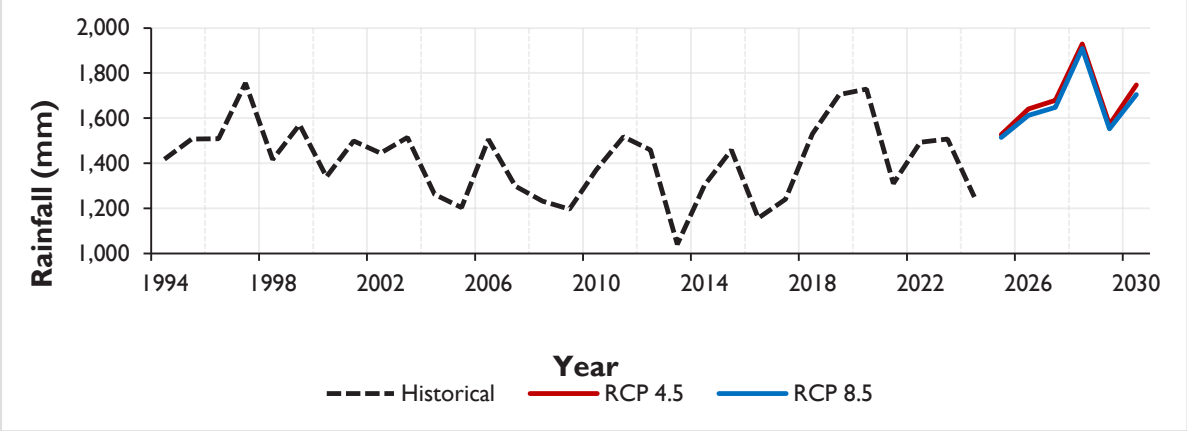


Figure 10: Observed and projected rainfall for Makindye Ssabagabo municipality from 1994 to 2030

The highest projected mean rainfall in Makindye Ssabagabo Municipality is expected around the shores of Lake Victoria in Mutungo and Busabala wards (**Figure 11**).

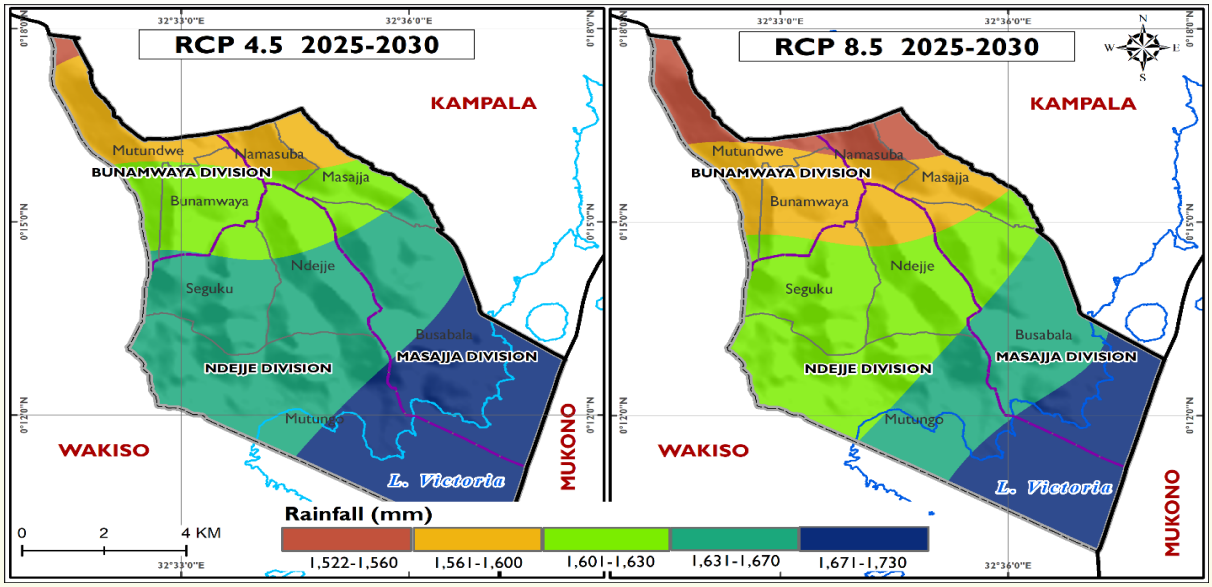


Figure 11: Projected rainfall for the 2025-2030 near-term projection period in Makindye Ssabagabo municipality

2.4 Climate Hazards

Climate hazards refer to events or conditions resulting from climate- or weather-related phenomena that have the potential to cause harm to people, property, and the environment (Zscheischler et al., 2020). The climate hazards that affect Makindye Ssabagabo municipality include drought, floods, hailstorms, lightning, and windstorms. These climate hazards are briefly described below.

2.3.1 Drought Hazard

Drought is defined as a recurrent feature of climate that occurs when there is an extended period of abnormal deficiency in precipitation (relative to what is considered normal) (Sivakumar & Wilhite, 2002). Drought in Makindye Ssabagabo Municipality has been attributed to environmental degradation, majorly cutting down trees, and encroachment of wetland ecosystems. The impacts of drought include water crises, reduced crop and livestock productivity, and increased food prices. The warmer temperatures also favour the survival of some pests and parasites, thus an increase in pests, parasites, and diseases.

Drought in Makindye Ssabagabo Municipality was assessed using the Standardized Precipitation Index (SPI). Rainfall data (1994-2024) from CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data) was used to compute the SPI. The SPI was computed using the Drought Indices Calculator (DrinC). Drought was then categorized based on SPI values as described by Guttman (1999) and McKee et al. (1993).

The biggest part of the municipality (61.3%) is exposed to low drought hazard. However, wards such as Namasuba, Masajja, and parts of Mutundwe, Bunamwaya, Ndejje, and Busabala are exposed to moderate drought hazard (**Figure 12**).

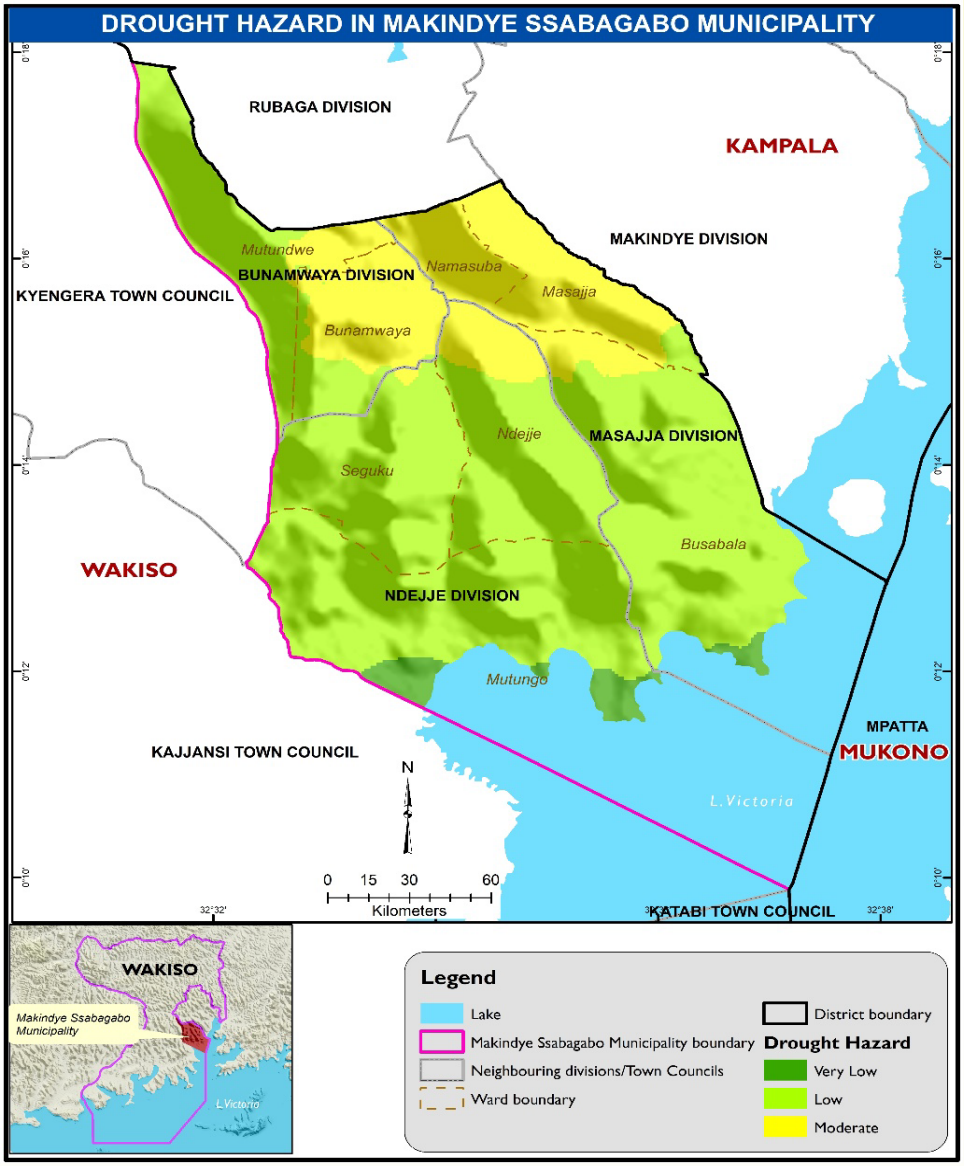


Figure 12: Drought hazard zonation in Makindye Ssabagabo Municipality

2.3.2 Flood Hazard

Flooding is an overflow of water that submerges land that is usually dry (Farida & Maswanku, 2022). The floods in Makindye Ssabagabo Municipality are majorly attributed to excessive rainfall due to climate change, clearing of trees and vegetation, encroachment of wetlands, municipal waste disposal in drainage channels, and increased farming of the riverbanks. The impacts of floods in Makindye Ssabagabo Municipality include loss of lives; destruction of houses and household property; destruction of croplands; contamination of water sources; and destruction of WASH facilities and infrastructure such as health centers, schools, roads, bridges and culverts. The biggest part of the municipality (29.5%) is exposed to floods of moderate intensity.

Flooding in Makindye Ssabagabo Municipality was delineated using the HEC-GeoRAS tool. The input datasets included rainfall, elevation, land use/cover, and stream geometry data. The flood extents were then categorized into flood hazard intensities based on water depth: very high (>2m), high (1.5m-2m), moderate (1m-1.5m), and low (0.5m-1m). The areas that are prone to

floods in Makindye Ssabagabo Municipality include Lufuka cell, Mirembe and Kisigula cells in Bunamwaya division, and Lweza A and Seguku cells in Ndejje division. The flood modeling showed that due to wetland encroachment, most of the low-lying areas in Makindye Ssabagabo Municipality are prone to floods (**Figure 13)**.

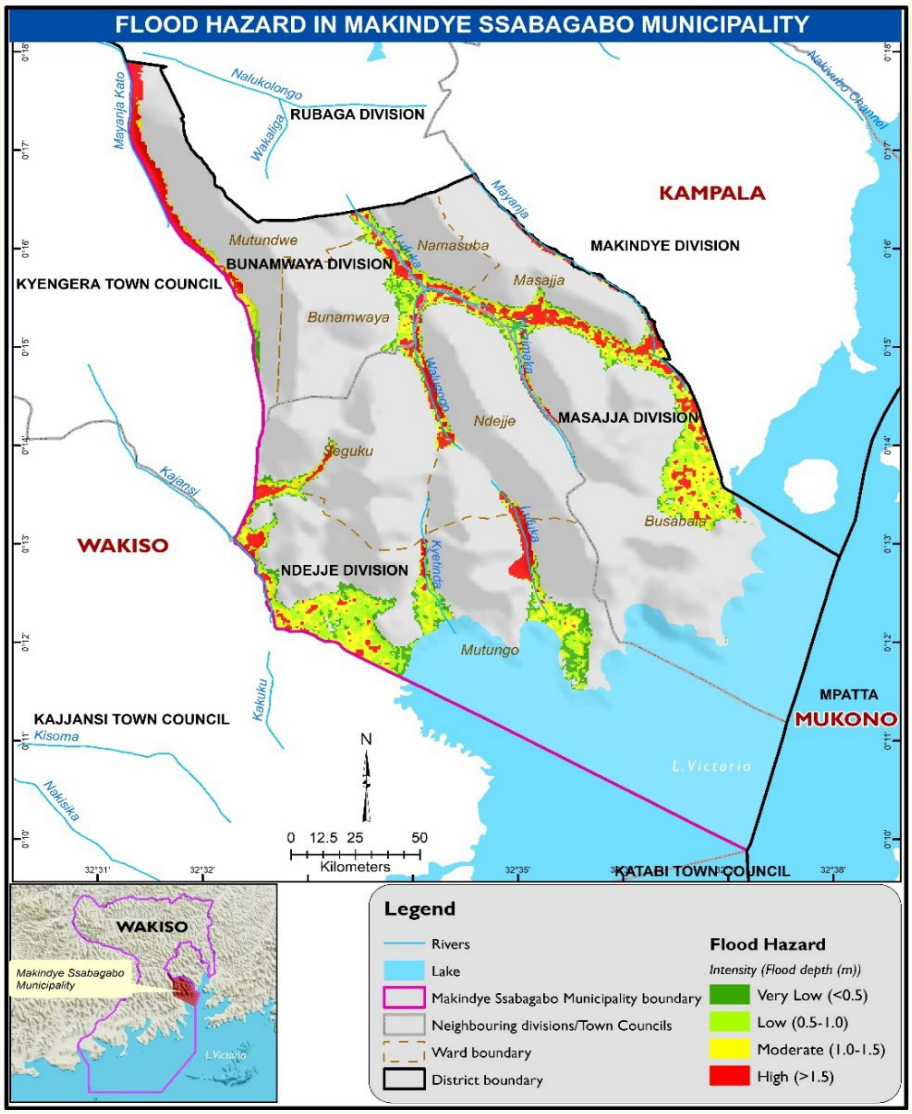


Figure 13: Flood hazard zonation in Makindye Ssabagabo Municipality

2.3.3 Hailstorm Hazard

Hail is a form of solid precipitation. It consists of balls or irregular lumps of ice, each of which is called a hailstone. Hailstones consist mostly of water ice and measure between 0.2 inches and 6 inches in diameter. Hail develops when rising air in a thunderstorm, known as the updraft, lifts water droplets high into the atmosphere where temperatures are below freezing. This causes the water droplets to turn into hailstones before falling down to earth. The stronger the updraft, the larger the hailstones can become before succumbing to the pull of gravity and falling to the earth's surface. Depending on their intensity, hailstorms can destroy crops also cause temporary damage to WASH infrastructure (such as water points, and latrines). The entire municipality is moderately vulnerable to hailstorm hazard (**Figure 14**).

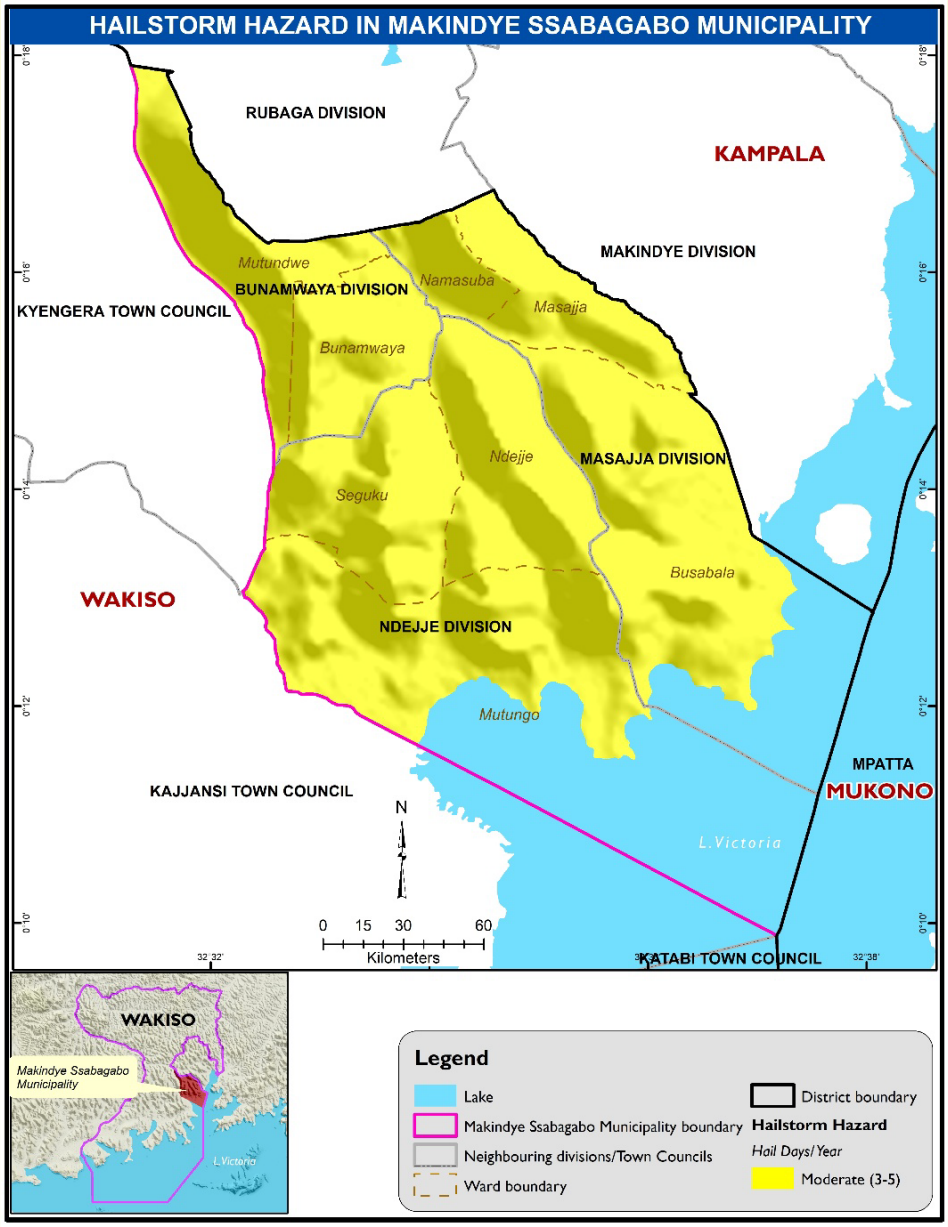


Figure 14: Hailstorm hazard zonation in Makindye Ssabagabo Municipality
Data Source: *OPM, 2019*

2.3.4 Lightning Hazard

Lightning is the occurrence of a natural electrical discharge of very short duration and high voltage between a cloud and the ground or within a cloud (Agrawal & Nigam, 2014). The frequencies and severities of lightning incidences have increased, resulting in significant loss of life and property. Lightning happens under favourable conditions, where electrical discharges occur from a charge center in a cloud either (i) to (i) the induced charge on the earth, (ii) to the charge centers of another cloud, or (iii) to a charge center of the same cloud.

Thunder day data was used to profile lightning, and this was supplemented with Over-shooting Tops (OT) data acquired from NASA Meteosat Second Generation (MSG) 3km resolution Spin-Enhanced Visible Infrared Imager (SEVIRI). The lightning hazard zonation was generated with reference to flashes per square kilometer per year and classified as low, moderate, and high (***Table 5***).

Table 5: Lightning hazard intensity scale

Intensity Class	OT Counts/month	Flash density (strikes/yr/km^2)
High	10-15	50-75
Moderate	5-10	10-50
Low	2-5	< 10

The biggest part of Makindye Ssabagabo Municipality (46.7%) is exposed to moderate lightning hazard. The areas exposed to moderate lightning hazard include Mutungo, Ndejje, Busabala, and Masajja wards (**Figure 15**).

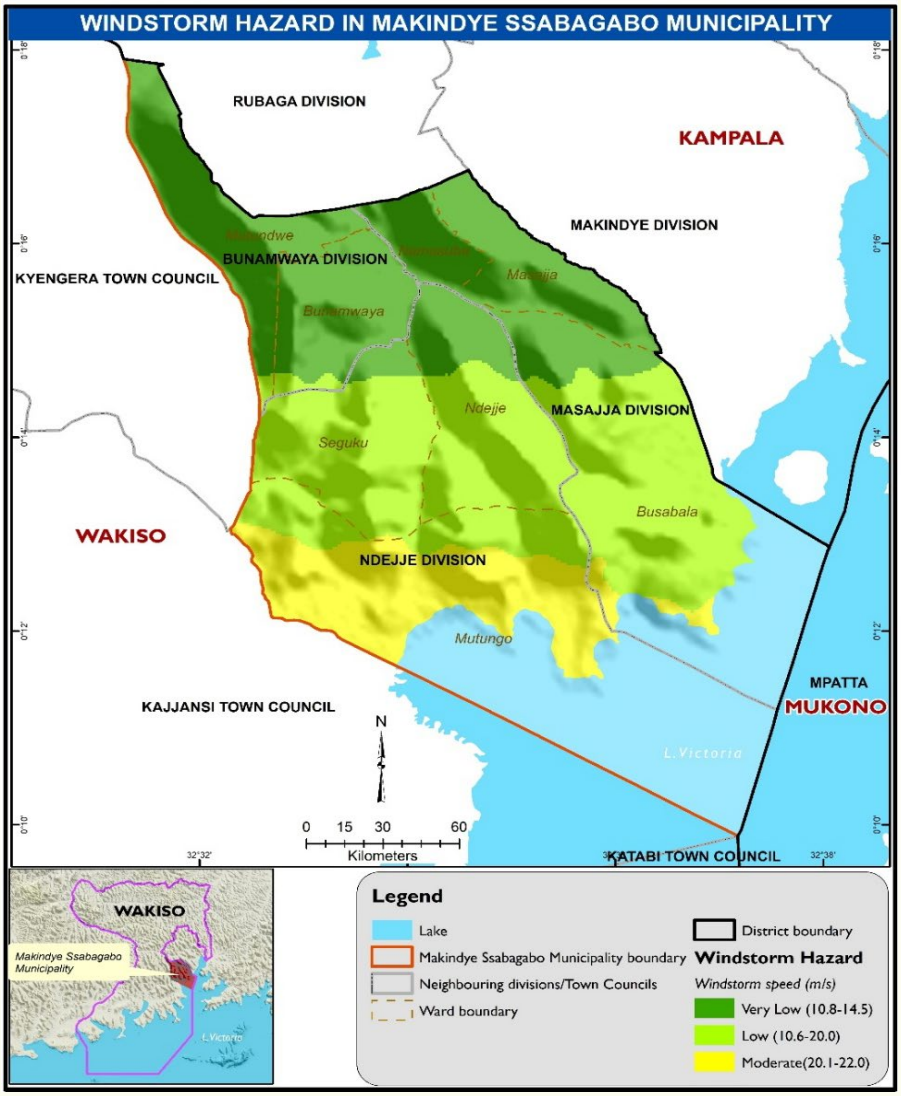


Figure 15: Lightning hazard zonation in Makindye Ssabagabo Municipality

2.3.5 Windstorm Hazard

Wind is the perceptible natural movement of air. Wind is caused by differences in the atmospheric pressure. When a difference in atmospheric pressure exists, air moves from the higher to the lower pressure area, resulting in winds of various speeds. The impacts of windstorms in Makindye Ssabagabo Municipality include air pollution, the spread of human diseases, uprooting of trees, and considerable structural damage to buildings, power and telephone lines, communication masts, and other urban infrastructure.

The assessment adopted the wind speed modelling approach by Morjani (2011) to develop the windstorm hazard of Makindye Ssabagabo Municipality. Wind speed data for the municipality was obtained from the climate engine for the period 1994-2024. The spatial distribution of wind hazard intensity was done by conducting a kriging interpolation analysis and applying the Beaufort wind scale (**Table 6**).

Table 6: Beaufort wind hazard intensity scale

Beaufort scale	Description	Wind speed (m/s)	Conditions	Windstorm Hazard Class
6	Strong breeze	10.8-14.5	Large tree branches/crops in motion; whistling heard in electric/telegraph wires; difficulty using umbrellas	Very low
7	High wind, moderate gale	14.6-20.0	Whole trees in motion; difficulty when walking against the wind	Low
8	Gale	20.1-22.0	Twigs break off trees; generally, impedes progress	Moderate
9	Strong/severe gale	22.1-28.0	Can cause slight structural damages	High
10	Storm, whole gale	> 28.1	Trees/crops uprooted; considerable structural damage, may be accompanied by widespread damage	Very high

The biggest part of the municipality (37.2%) is exposed to low windstorm hazard. However, wards such as sections of Mutungo and Busabala are exposed to moderate windstorm hazard (**Figure 16**).

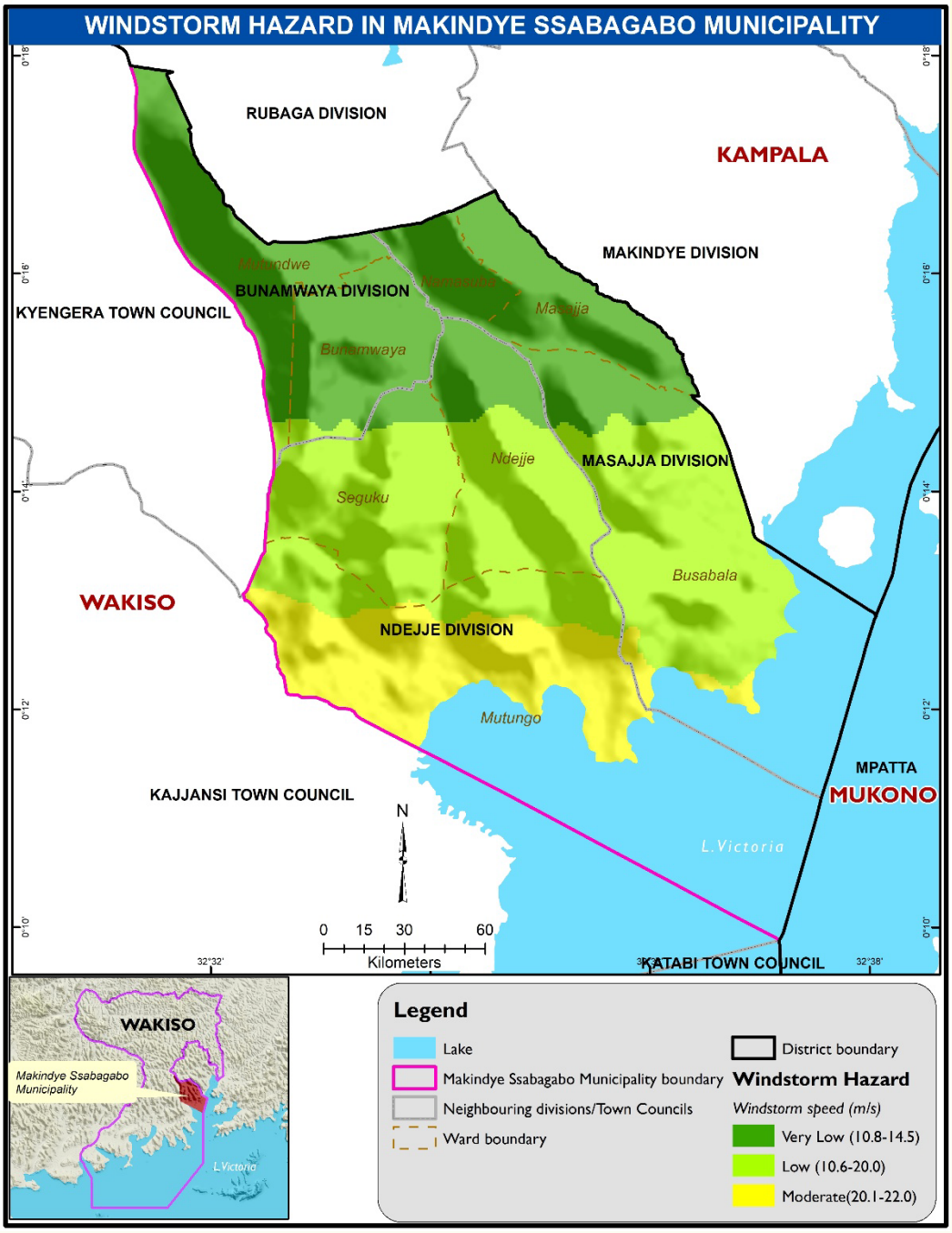


Figure 16: Windstorm hazard zonation in Makindye Ssabagabo Municipality

2.5 Exposure Assessment

Exposure is defined as the situation of people, infrastructure, housing, production capacities and other tangible human assets located in hazard-prone areas (UNISDR, 2017). The criterion considered for exposure assessment of elements at risk was location in moderate, high, and very high hazard intensities (OPM, 2019). Exposure of the elements at risk was done based on the availability of spatial datasets. **Table 7** indicates the elements at risk and their sensitivity to the different climate hazards.

Table 7: Sensitivity of elements at risk to the different climate hazards

Sector and Element at Risk	Data Source	Drought	Floods	Hailstorms	Lightning	Windstorms
1. Agriculture						
1.1 Farmlands	Satellite Imagery, 2025	High	High	Moderate	Low	Moderate
2. Education						
2.1 Schools	MoES, 2018	Low	High	Low	Moderate	High
3. Energy						
3.1 Substations	UMEME, 2024	Low	High	Low	High	Low
3.2 Transformers	UMEME, 2024	Low	Moderate	Low	High	High
3.3 Distribution lines	UMEME, 2024	Low	Low	Low	High	High
4. Health						
4.1 Health facilities	MoH, 2024	Low	High	Low	Moderate	High
5. Land and Housing						
5.1 Buildings	OSM & Satellite Imagery, 2025	Low	High	Low	High	Moderate
6. Social development						
6.1 Human population	UBOS Census, 2014	High	High	High	High	High
7. Transportation						
7.1 Roads	UNRA, 2022	Low	High	Low	Low	Low
8. Water and Environment						
8.1 Wetlands	WMD, 2021	Moderate	High	Low	Low	Low
8.3 Water sources	MWE, 2024	High	High	Low	Low	Low

2.4.1 Exposure of the agriculture sector to climate hazards

The agriculture sector element exposed to climate hazards in Makindye Ssabagabo Municipality is farmlands.

The assessment showed that 0.01 Km^2 (0.4%) of farmlands are exposed to drought hazard in Makindye Ssabagabo Municipality. The farmlands exposed to drought hazard are located in Bunamwaya division (1.8%) (

Table *8*).

A total of 1.1 Km^2 (48.7%) of farmlands are exposed to flood hazard in Makindye Ssabagabo Municipality. Masajja (69.1%) and Ndejje (44.1%) divisions have the biggest proportion of farmlands exposed to flood hazard (

Table *8*).

All the farmlands (100%) in Makindye Ssabagabo Municipality are exposed to hailstorm hazard. Masajja (1.1 Km^2) and Ndejje (0.6 Km^2) divisions have the biggest acreage of farmlands exposed to hailstorms (***Table 8***). In Makindye Ssabagabo Municipality, 0.4 Km^2 (18.6%) of farmlands are exposed to windstorms whereas 1.8 Km^2 (81.4%) are not exposed to this hazard. The exposed farmlands to windstorm hazard are located in Masajja and Ndejje divisions.

Table 8: Exposure of farmlands to drought, flood, hailstorm, and windstorm hazards in Makindye Ssabagabo Municipality

Climate Hazards	**Drought**				**Floods**				**Hailstorms**		**Windstorms**			
Divisions	Exposed		Not Exposed		Exposed		Not Exposed		Exposed		Exposed		Not Exposed	
	Area (Km^2)	%	Area (Km^2)	%	Area (Km^2)	%	Area (Km^2)	%	Area (Km^2)	%	Area (Km^2)	%	Area (Km^2)	%
Bunamwaya	0.01	1.8	0.5	98.2	0.1	14.1	0.5	85.9	0.5	24.9	0.0	0.0	0.5	100
Masajja	0.0	0.0	1.1	100	0.7	69.1	0.3	30.9	1.1	48.1	0.01	1.0	1.1	99.0
Ndejje	0.0	0.0	0.6	100	0.3	44.1	0.3	55.9	0.6	27.1	0.4	67.2	0.2	32.8
Total	**0.01**	**0.4**	**2.2**	**99.6**	**1.1**	**48.7**	**1.1**	**51.3**	**2.2**	**100**	**0.4**	**18.6**	**1.8**	**81.4**

Figure 17 shows the exposure of farmlands to drought, flood, hailstorm, and windstorm hazards in Makindye Ssabagabo Municipality.

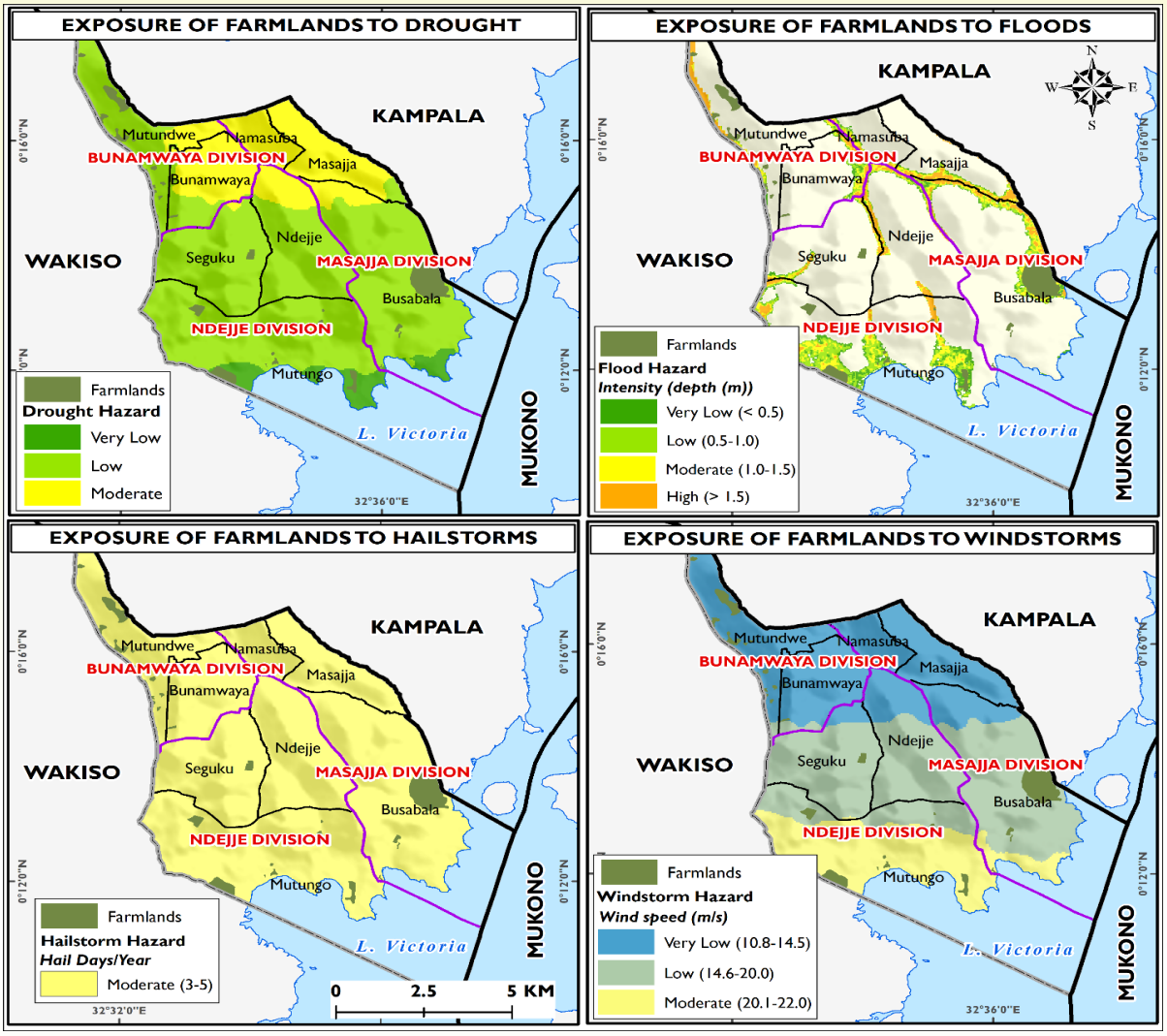


Figure 17: Exposure of farmlands to drought, flood, windstorm, and hailstorm hazards in Makindye Ssabagabo Municipality

2.4.2 Exposure of the education sector to climate hazards

The education sector element exposed to climate hazards in Makindye Ssabagabo Municipality is schools.

The analysis showed that 16 schools (7.3%) in Makindye Ssabagabo Municipality are exposed to floods whereas 203 schools (92.7%) are not exposed to floods (**Table 9**). Bunamwaya division (8.6%) has the biggest percentage of schools exposed to floods. Schools such as Ndejje View Kindergarten Infant School in Lufuka cell; MB Academic war P.S in Masajja B; as well as Africa S.S, St. Kizito Lubbe, and New Betheny P/S in Wabiyinja cell; are highly exposed to floods.

A total of 12 schools (5.5%) are exposed to lightning hazard in Makindye Ssabagabo Municipality (**Table 9**). The schools exposed to lightning in Makindye Ssabagabo Municipality are located in Masajja (9.2%) and Ndejje (6.5%) divisions.

In Makindye Ssabagabo Municipality, 12 schools (5.5%) are exposed to windstorms whereas 207 schools (94.5%) are not exposed to this climate hazard (**Table 9**). The schools exposed to windstorms in Makindye Ssabagabo Municipality are located in Masajja (1.1%) and Ndejje (17.7%) divisions.

Table 9: Exposure of schools to floods, lightning, and windstorms in Makindye Ssabagabo Municipality

Climate Hazards	Lightning				Floods				Windstorms			
Divisions	Exposed		Not Exposed		Exposed		Not Exposed		Exposed		Not Exposed	
	Number	%	Number	%	Number	%	Number	%	Number	%	Number	%
Bunamwaya	0	0.0	70	100	6	8.6	64	91.4	0	0.0	70	100
Masajja	8	9.2	79	90.8	7	8.0	80	92.0	1	1.1	86	98.9
Ndejje	4	6.5	58	93.5	3	4.8	59	95.2	11	17.7	51	82.3
Total	**12**	**5.5**	**207**	**94.5**	**16**	**7.3**	**203**	**92.7**	**12**	**5.5**	**207**	**94.5**

Figure 18 presents the schools exposed to floods, lightning and windstorms in Makindye Ssabagabo Municipality.

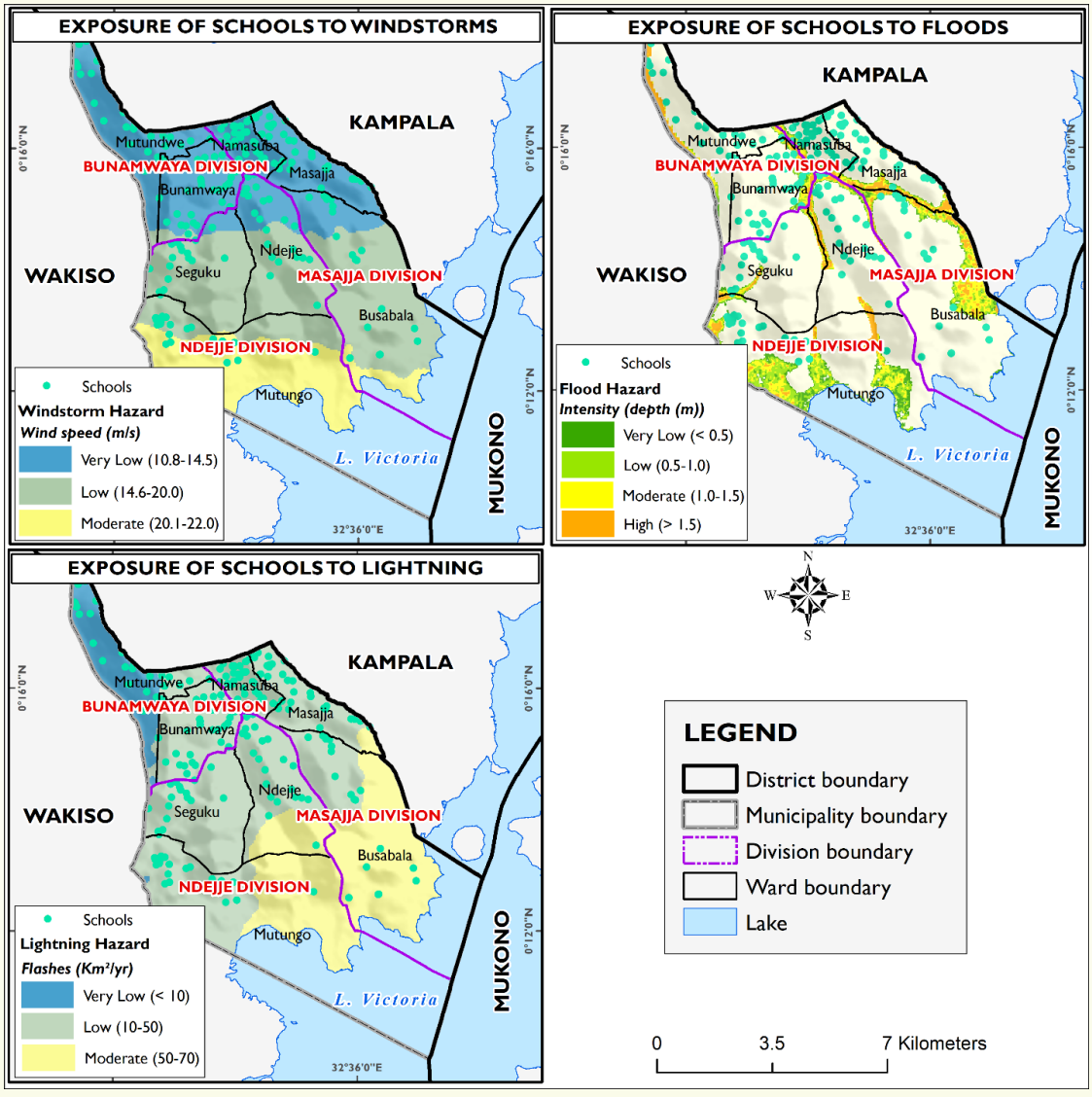


Figure 18: Exposure of schools to lightning, flood, and windstorm hazards in Makindye Ssabagabo Municipality

2.4.3 Exposure of the energy sector to climate hazards

The energy sector elements exposed to climate hazards in Makindye Ssabagabo Municipality include substations, transformers, and distribution lines.

Exposure of distribution substations to flood and lightning hazards

The assessment showed that 1 distribution substation of Kajjansi (50%) is highly exposed to floods whereas another distribution substation of Lubowa (50%) is not exposed to floods. Both substations are not exposed to lightning hazard. The exposure of distribution substations to floods and lightning is presented in **Figure 19**.

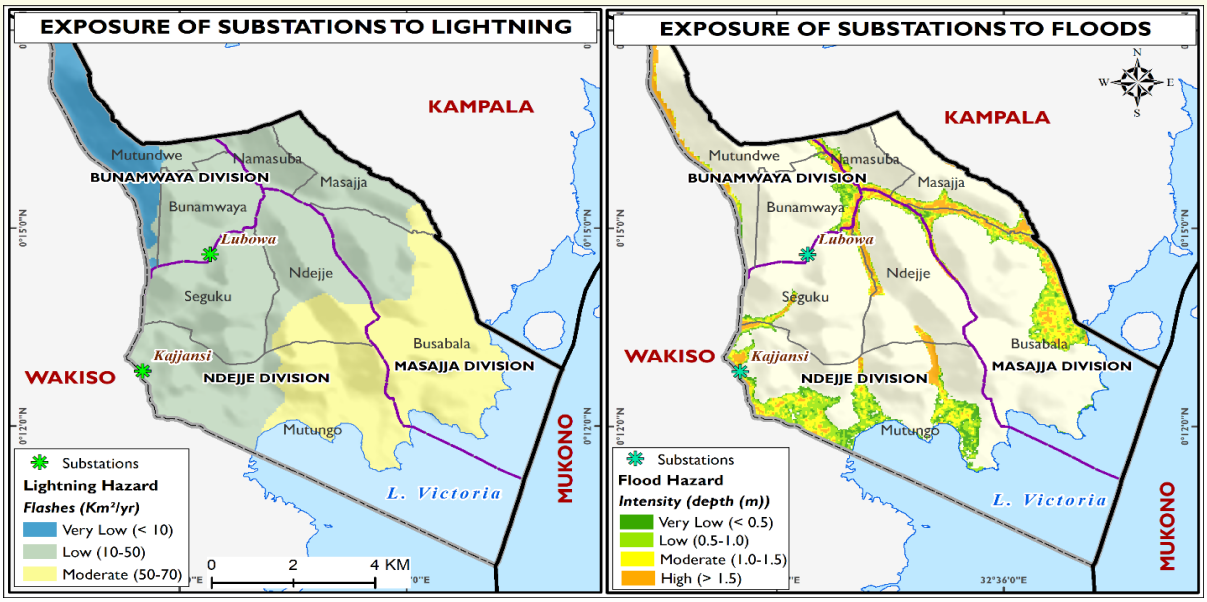


Figure 19: Exposure of distribution substations to floods and lightning in Makindye Ssabagabo Municipality

Exposure of distribution transformers to lightning and windstorm hazards

The analysis showed that 40 distribution transformers (16%) are exposed to lightning hazard whereas 210 distribution transformers (84%) are not exposed to lightning hazard (**Table 10**). The exposed distribution transformers to lightning are located in Masajja (24.3%) and Ndejje (20.4%) divisions.

A total of 26 distribution transformers (10.4%) are exposed to windstorm hazard while 224 distribution transformers (89.6%) are not exposed to this hazard (**Table 10**). Ndejje (23.1%) and Masajja (1.4%) divisions have the distribution transformers exposed to windstorm hazard.

Table 10: Exposure of distribution transformers to lightning and windstorm hazards

Climate Hazards	Lightning				Windstorms			
Divisions	Exposed		Not Exposed		Exposed		Not Exposed	
	Number	%	Number	%	Number	%	Number	%
Bunamwaya	0	0.0	68	100	0	0.0	68	100
Masajja	18	24.3	56	75.7	1	1.4	73	98.6
Ndejje	22	20.4	86	79.6	25	23.1	83	76.9
Total	**40**	**16.0**	**210**	**84.0**	**26**	**10.4**	**224**	**89.6**

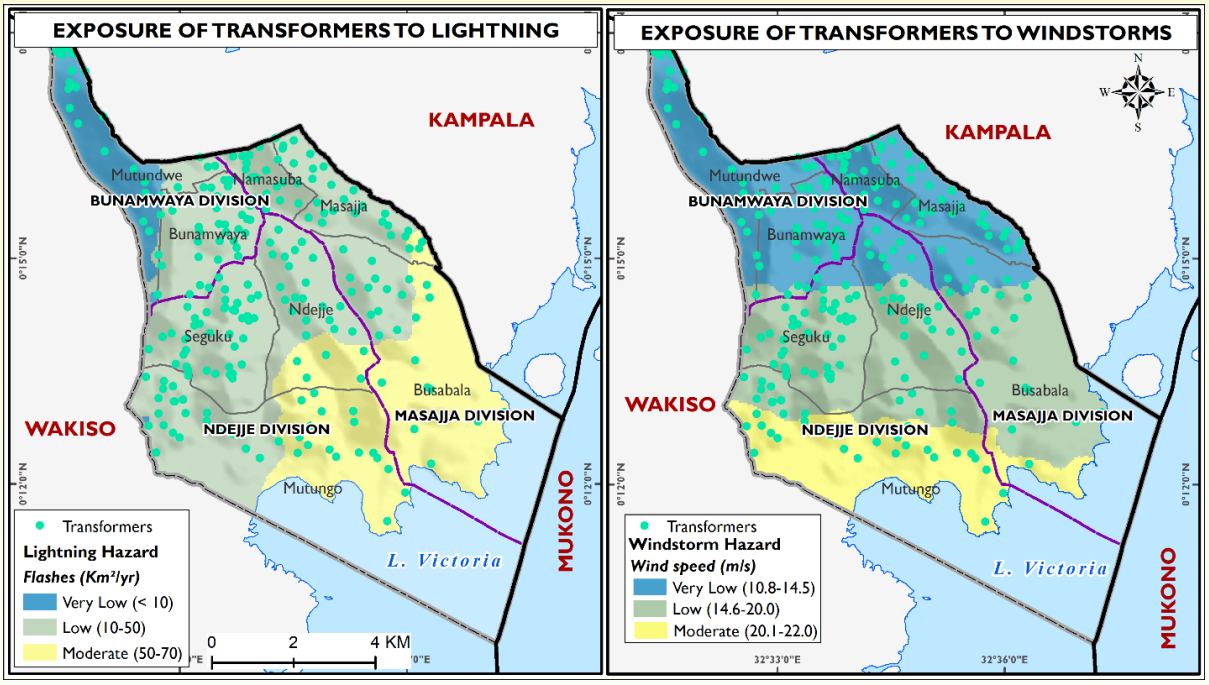


Figure 20: Exposure of distribution transformers to windstorm and lightning hazards in Makindye Ssabagabo Municipality

Exposure of distribution lines to lightning and windstorm hazards

The assessment showed that 21.8 kilometers (17.4%) of distribution lines in Makindye Ssabagabo Municipality are exposed to lightning hazard (**Table 11**). The exposed distribution lines to lightning hazard in Makindye Ssabagabo Municipality are found in Ndejje (22.5%) and Masajja (24.5%) divisions.

A total of 16.6 kilometers (13.2%) of distribution lines are exposed to windstorms whereas 108.9 kilometers (86.8%) of distribution lines are not exposed to distribution lines (**Table 11**). Ndejje (26.5%) and Masajja (2.0%) divisions have the distribution lines exposed to windstorm hazard.

Table 11: Exposure of distribution lines to lightning and windstorm hazards

Climate Hazards	Lightning				Windstorms			
Divisions	Exposed		Not Exposed		Exposed		Not Exposed	
	Length (Km)	%	Length (Km)	%	Length (Km)	%	Length (Km)	%
Bunamwaya	0.0	0.0	31.6	100	0.0	0.0	31.6	100
Masajja	8.3	24.5	25.6	75.5	0.7	2.0	33.3	98.0
Ndejje	13.5	22.5	46.5	77.5	15.9	26.5	44.1	73.5
Total	**21.8**	**17.4**	**103.7**	**82.6**	**16.6**	**13.2**	**108.9**	**86.8**

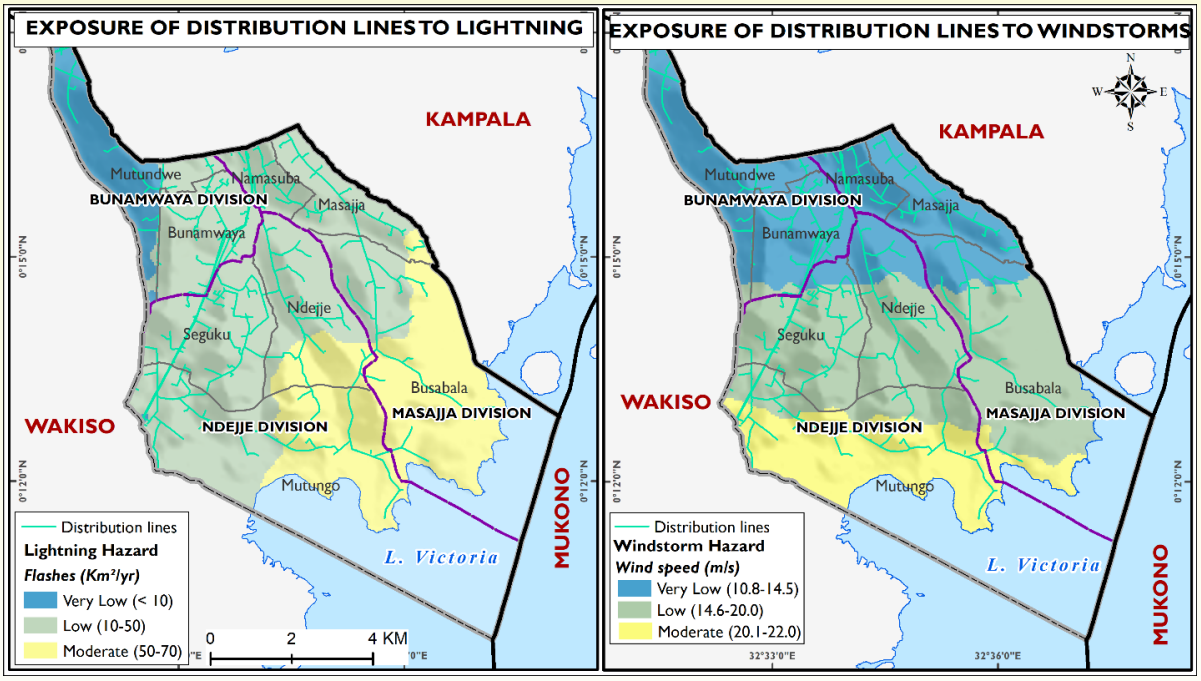


Figure 21: Exposure of distribution lines to windstorm and lightning hazards in Makindye Ssabagabo Municipality

2.4.4 Exposure of the health sector to climate hazards

The health sector element exposed to climate hazards in Makindye Ssabagabo Municipality is health facilities.

In Makindye Ssabagabo Municipality, 2 health facilities (2.7%) in Bunamwaya division are exposed to floods whereas 73 health facilities (97.3%) are not exposed to flood hazard. The health facilities exposed to floods include General Health Care HC II in Kirimanya cell, and Pea Clinic HC II in Nakinga cell.

The assessment showed that 4 health facilities (5.5%) in Makindye Ssabagabo Municipality are exposed to lightning whereas 69 health facilities (94.5%) are not exposed to this hazard

. The exposed health facilities to lightning are found in Ndejje (15%) and Masajja (3%) divisions. The exposed health facilities include Kigo Main Prisons HC III and Kigo Women's Prisons HC II in Kigo Prison cell, Mutungo HC II in Mutungo cell, and Dianah Joy Medical Clinic HC II in Kibira B cell.

A total of 5 health facilities (6.8%) are exposed to windstorms in Makindye Ssabagabo Municipality while 68 health facilities (93.2%) are not exposed to this hazard.

The health facilities exposed to windstorms are located in Ndejje (25%) division; and these include Kigo Main Prisons HC III and Kigo Women's Prisons HC II in Kigo Prison cell, Le Memorial Medical Services Hospital in Birongo cell, Mutungo HC II in Mutungo cell, and N.A.P Domicilliary Clinic HC III in Lweza A cell.

Table 12: Exposure of health facilities to floods, lightning, and windstorms in Makindye Ssabagabo Municipality

Climate Hazards	Floods				Lightning				Windstorms			
Divisions	Exposed		Not Exposed		Exposed		Not Exposed		Exposed		Not Exposed	
	Number	%	Number	%	Number	%	Number	%	Number	%	Number	%
Bunamwaya	2	9.1	20	90.9	0	0.0	20	100	0	0.0	20	100
Masajja	0	0.0	33	100	1	3.0	32	97.0	0	0.0	33	100
Ndejje	0	0.0	20	100	3	15.0	17	85.0	5	25.0	15	75
Total	**2**	**2.7**	**73**	**97.3**	**4**	**5.5**	**69**	**94.5**	**5**	**6.8**	**68**	**93.2**

The exposure of health facilities to the different climate hazards is shown in **Figure 22**.

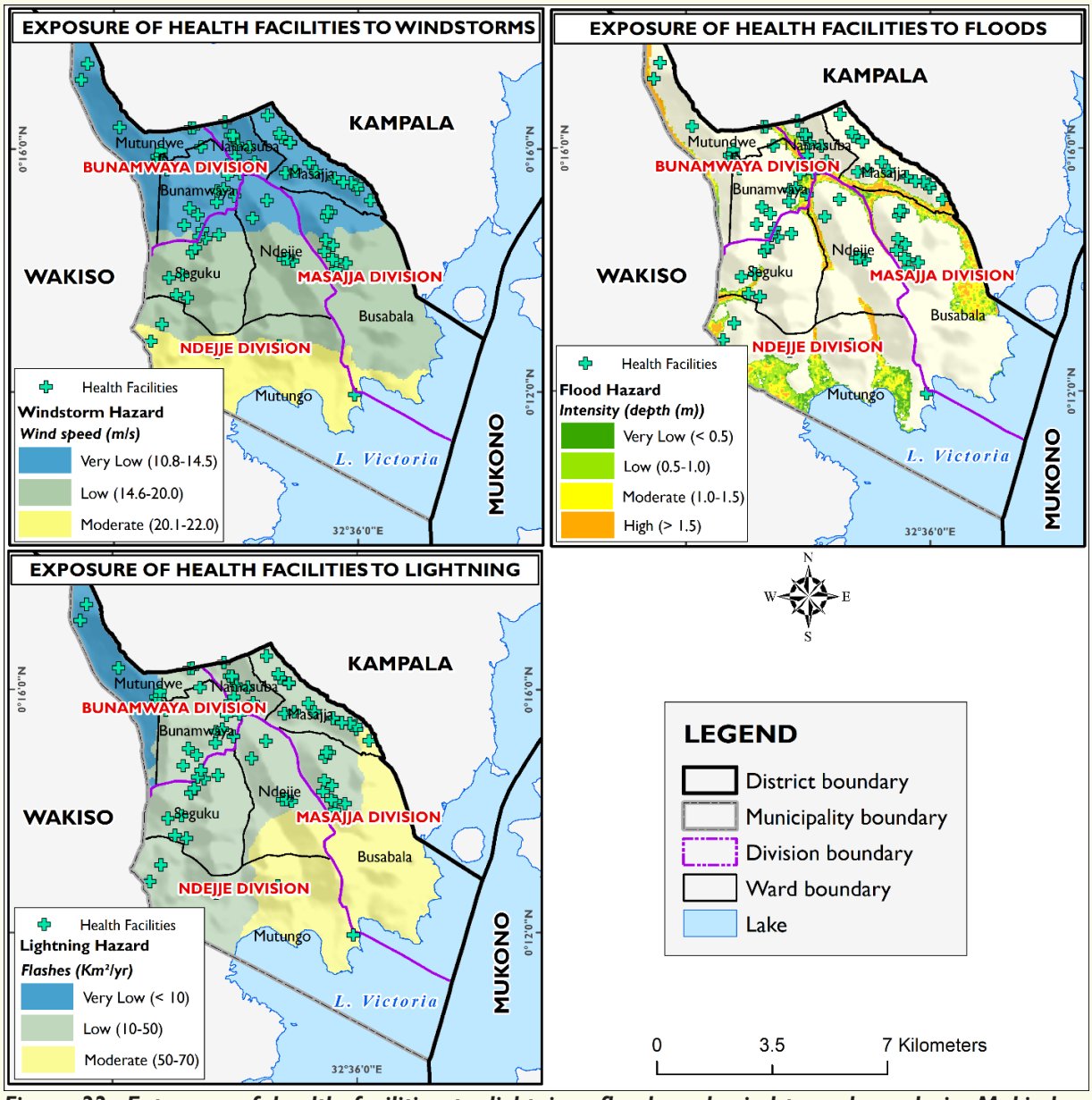


Figure 22: Exposure of health facilities to lightning, flood, and windstorm hazards in Makindye Ssabagabo Municipality

2.4.5 Exposure of the land and housing sector to climate hazards

The land and housing sector element exposed to climate hazards in Makindye Ssabagabo Municipality is buildings (both commercial and residential).

The analysis showed that 6,894 buildings (12%) are exposed to floods whereas 50,416 buildings (88%) are not exposed to flood hazard (**Table 13**). Masajja (13.3%) and Bunamwaya (13.1%) divisions have the biggest proportion of buildings exposed to floods.

A total of 16,213 buildings (30.9%) are exposed to lightning hazard while 36,286 buildings (69.1%) are not exposed to this hazard (**Table 13**). The buildings exposed to lightning are found in Masajja (47.3%) and Ndejje (31.3%) divisions.

In Makindye Ssabagabo Municipality, 7.677 buildings (14.6%) are exposed to windstorm hazard whereas 44,897 buildings (85.4%) are not exposed to this hazard (**Table 13**). The buildings exposed to windstorms in Makindye Ssabagabo Municipality are found in Masajja (4.1%) and Ndejje (29.2%) divisions.

Table 13: Exposure of buildings to floods, lightning, and windstorms in Makindye Ssabagabo Municipality

Climate Hazards	Floods				Lightning				Windstorms			
Divisions	Exposed		Not Exposed		Exposed		Not Exposed		Exposed		Not Exposed	
	Number	%	Number	%	Number	%	Number	%	Number	%	Number	%
Bunamwaya	1,445	13.1	9,577	86.9	0	0.0	10,251	100	0	0.0	10,212	100
Masajja	2,737	13.3	17,812	86.7	8,850	47.3	9,867	52.7	776	4.1	17,958	95.9
Ndejje	2,601	10.2	23,015	89.8	7,363	31.3	16,168	68.7	6,901	29.2	16,727	70.8
Total	**6,894**	**12.0**	**50,416**	**88.0**	**16,213**	**30.9**	**36,286**	**69.1**	**7,677**	**14.6**	**44,897**	**85.4**

The exposure of buildings to floods, lightning, and windstorms in Makindye Ssabagabo Municipality is presented in **Figure 23**.

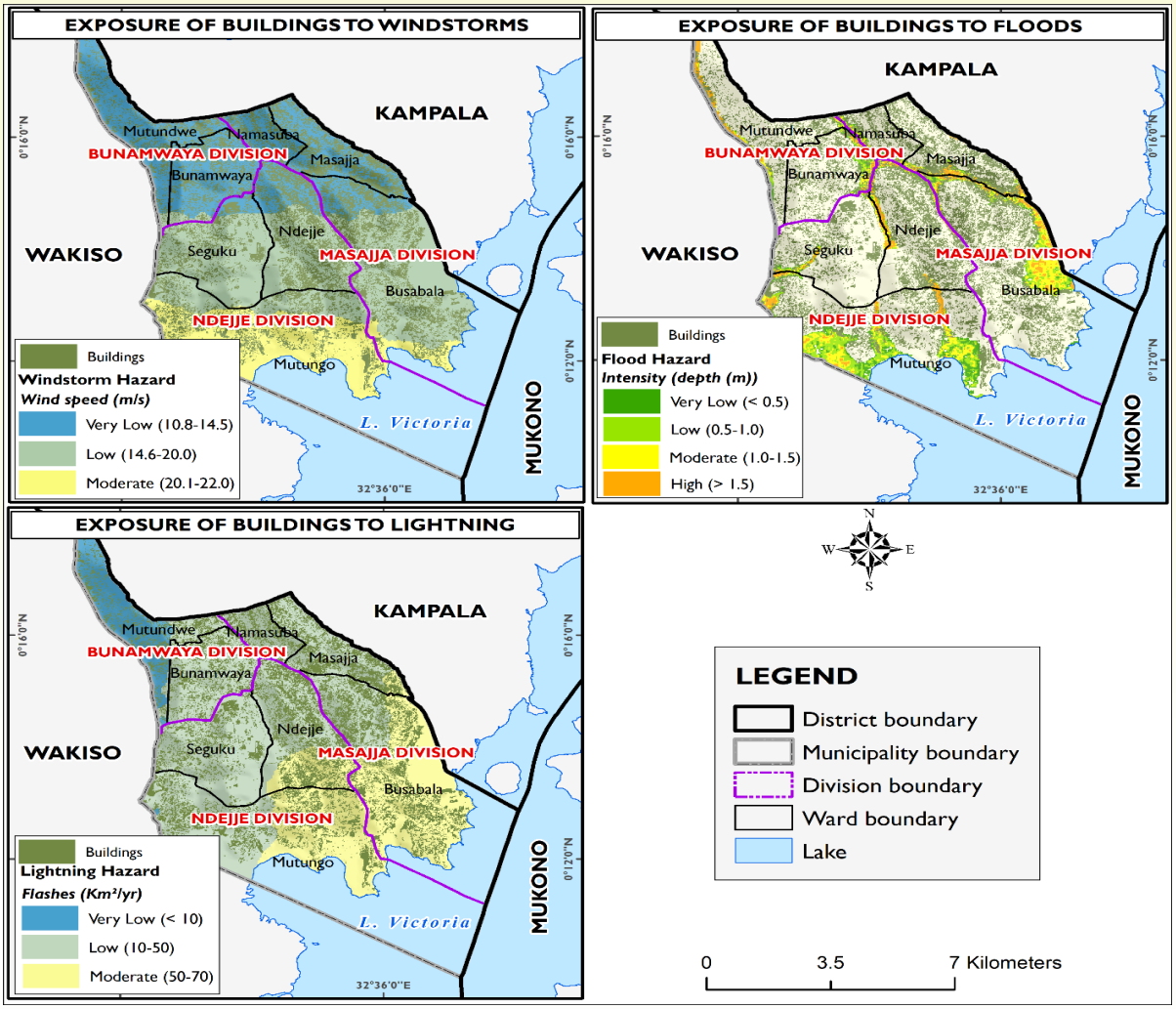


Figure 23: Exposure of buildings to lightning, flood, and windstorm hazards in Makindye Ssabagabo Municipality

2.4.6 Exposure of the social development sector to climate hazards

The social development sector element exposed to climate hazards in Makindye Ssabagabo Municipality is human population. The assessment showed that 117,650 people (41.5%) are exposed to drought hazard in Makindye Ssabagabo Municipality whereas 165,622 people (58.5%) are not exposed to drought hazard (**Table 14**). The biggest proportion of human population exposed to drought hazard is located in Masajja (65.3%) and Bunamwaya (41.7%) divisions.

A total of 31,027 people (11%) is exposed to floods whereas 252,245 people (89%) are not exposed to floods (**Table 14**). Masajja (13.5%) and Bunamwaya (11.1%) divisions have the biggest percentage of human population exposed to floods.

In Makindye Ssabagabo Municipality, 67,662 people (23.9%) are exposed to lightning hazard whereas 215,610 people (76.1%) are not exposed to lightning (**Table 14**). The people exposed to lightning hazard are located in Masajja (30.5%) and Ndejje (34.1%) divisions.

The analysis showed that 28,316 people (10%) in Makindye Ssabagabo Municipality are exposed to windstorm hazard whilst 254,956 people (90%) are not exposed to windstorms (**Table 14**). The human population exposed to windstorms is found in Ndejje (20.5%) and Masajja (7.8%) divisions.

Table 14: Human population exposed to drought, floods, lightning, and windstorms in Makindye Ssabagabo Municipality

Climate Hazards	Drought				Floods				Lightning				Windstorms			
Divisions	Exposed		Not Exposed		Exposed		Not Exposed		Exposed		Not Exposed		Exposed		Not Exposed	
	Popln	%	Popln	%	Popln	%	Popln	%	Popln	%	Popln	%	Popln	%	Popln	%
Bunamwaya	30,231	41.7	42,302	58.3	8,052	11.1	64,481	88.9	0	0.0	72,533	100	0	0.0	72,533	100
Masajja	76,509	65.3	40,739	34.7	15,812	13.5	101,436	86.5	35,743	30.5	81,505	69.5	9,148	7.8	108,100	92.2
Ndejje	10,910	11.7	82,581	88.3	7,164	7.7	86,327	92.3	31,919	34.1	61,572	65.9	19,168	20.5	74,323	79.5
Total	**117,650**	**41.5**	**165,622**	**58.5**	**31,027**	**11.0**	**252,245**	**89.0**	**67,662**	**23.9**	**215,610**	**76.1**	**28,316**	**10.0**	**254,956**	**90.0**

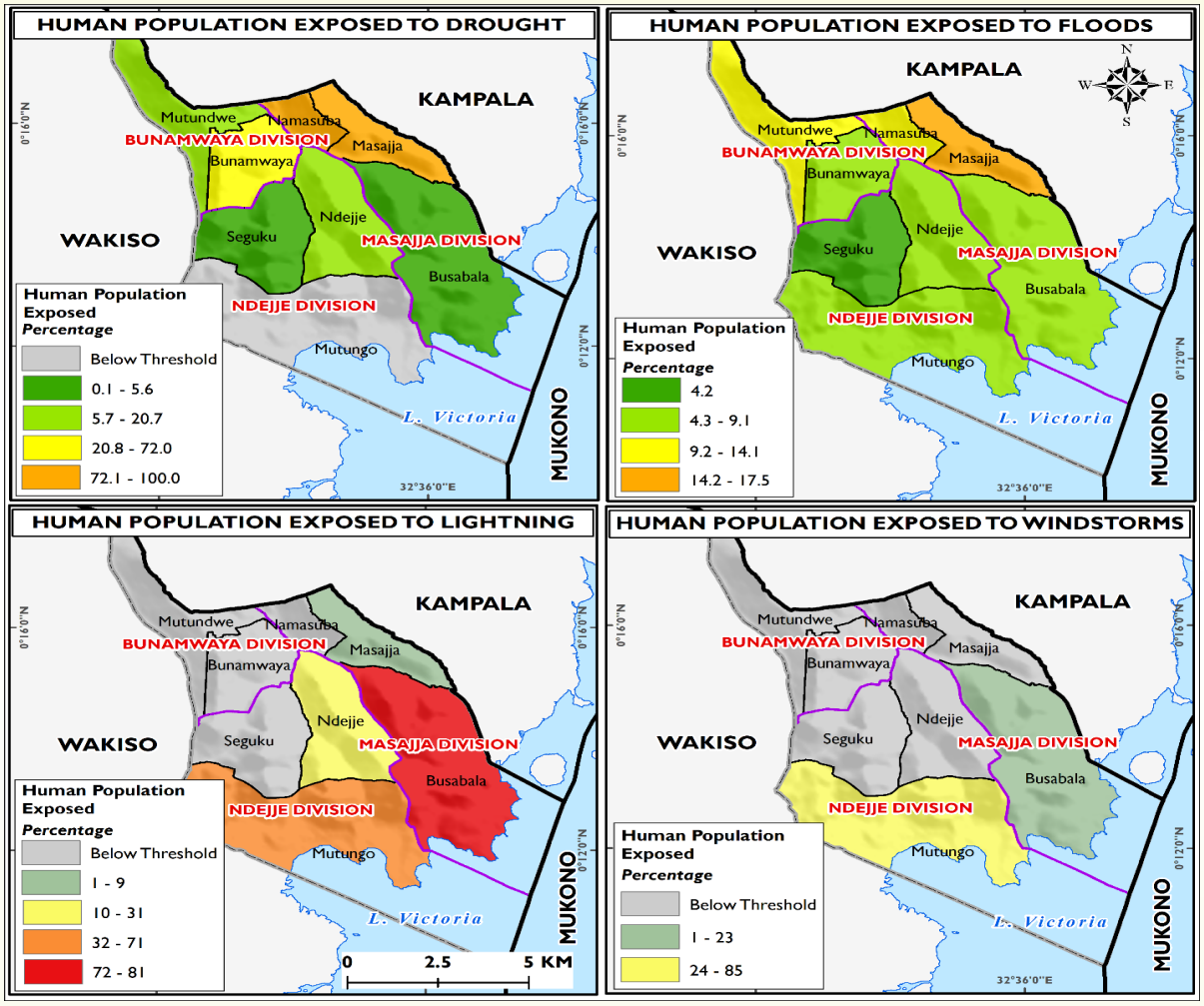


Figure 24: Human population exposure to drought, floods, lightning, and windstorms in Makindye Ssabagabo Municipality

All the people (100%) in Makindye Ssabagabo Municipality are exposed to hailstorm hazard. The wards with the biggest population exposed to hailstorms include Ndejje, Masajja, Busabala, Mutundwe, and Bunamwaya (**Figure 25**).

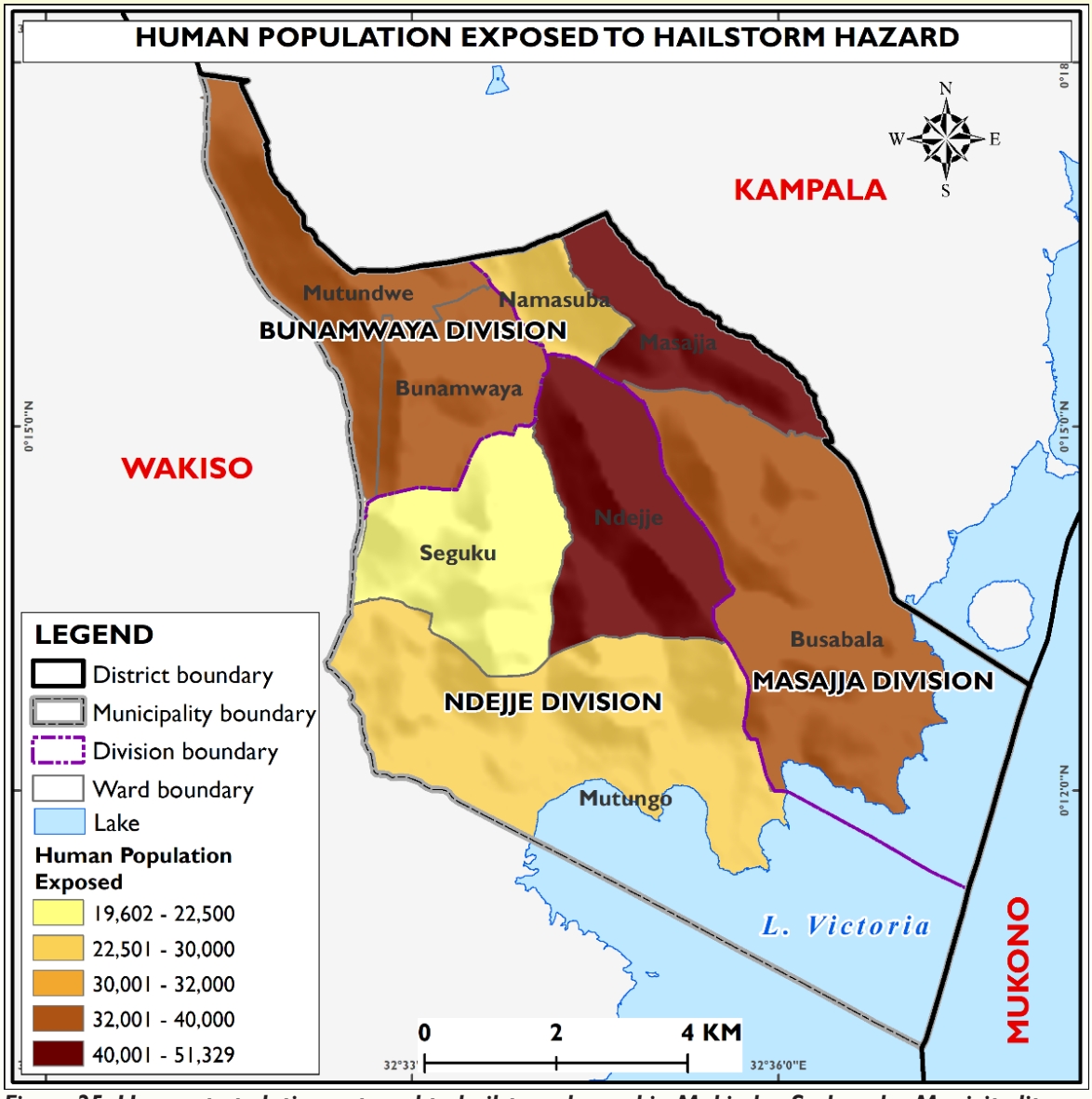


Figure 25: Human population exposed to hailstorm hazard in Makindye Ssabagabo Municipality

2.4.7 Exposure of the transportation sector to floods

The transportation sector element exposed to floods in Makindye Ssabagabo Municipality is roads.

The assessment showed that 7.6 kilometers (5.5%) of roads in Makindye Ssabagabo Municipality are exposed to floods whereas 129.9 kilometers (94.5%) of roads are not exposed to this hazard

Ndejje (6.8%) and Bunamwaya (5.1%) divisions have the biggest proportion of roads exposed to floods in Makindye Ssabagabo Municipality.

Some of the road sections that are highly exposed to floods include among others Ssebagala road in Kanaala road; Kibuye-Zana road in Lufuka cell; Nsalo-Lweza road in Lweza A and Zone 5; Natete- Nakawuka-Kisubi road in Mirembe cell; Zana-Ndejje road in Ndejje and Kirimanya cells; Florida-Ziwa road in Kirimanya cell; Kapande Ka Basiramu-Ndejje road in Lufuka and Kirimanya cells; BataBata-Misitwa-Para Suites road in Nakinga cell; and Najanankumbi-Busabala road in Ggangu C and Kibira A cells among others.

Table 15: Exposure of roads to floods in Makindye Ssabagabo Municipality

Divisions	Exposed		Not Exposed		Total	
	Length (Km)	%	Length (Km)	%	Length (Km)	%
Bunamwaya	2.2	5.1	41.8	94.9	44.0	32.0
Masajja	1.6	4.2	36.5	95.8	38.1	27.7
Ndejje	3.7	6.8	51.6	93.2	55.4	40.3
Total	**7.6**	**5.5**	**129.9**	**94.5**	**137.5**	**100**

The exposure of roads to floods in Makindye Ssabagabo Municipality is shown in **Figure 26**.

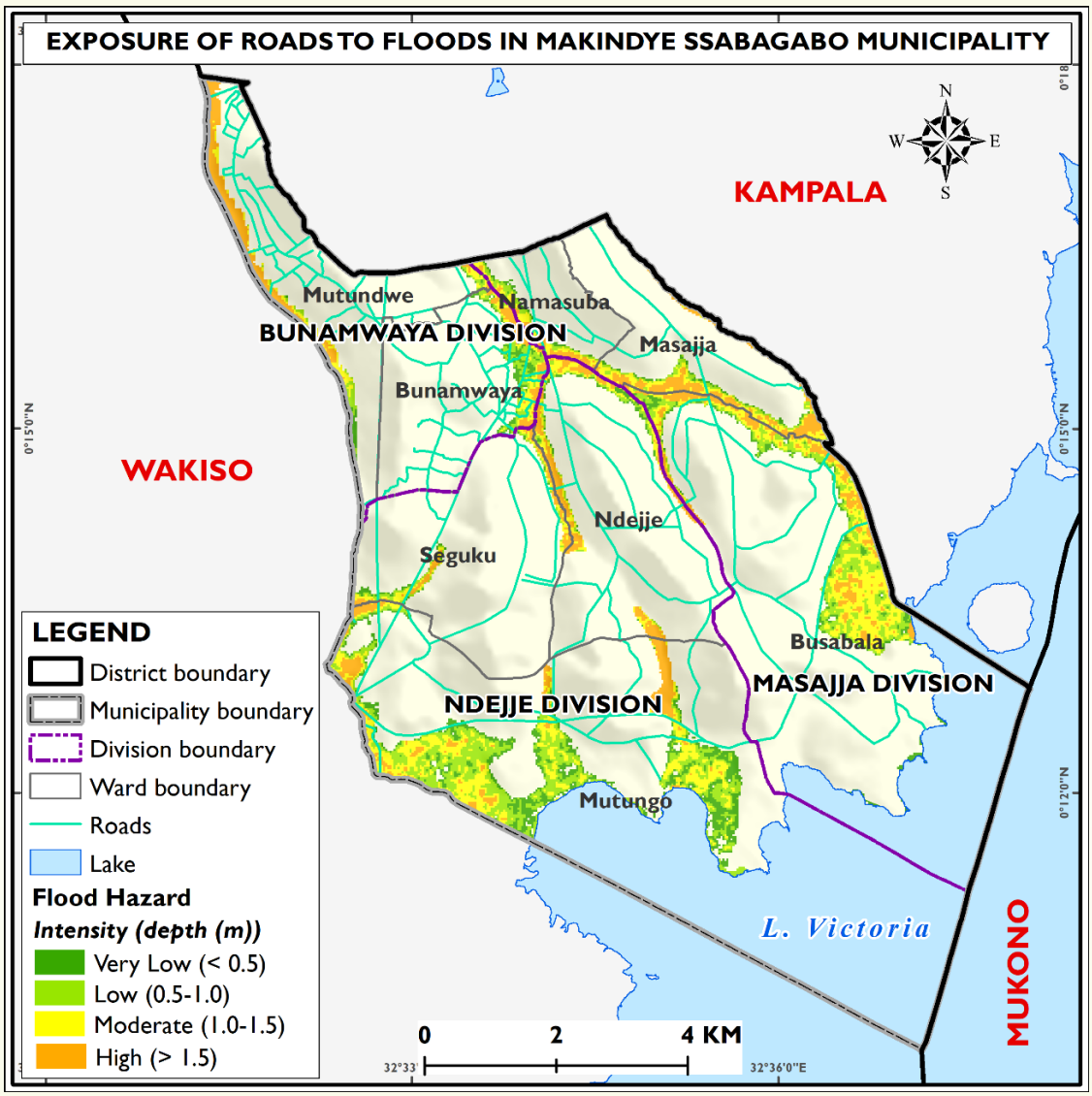


Figure 26: Exposure of roads to flood hazard in Makindye Ssabagabo Municipality

2.4.8 Exposure of the water and environment sector to climate hazards

The water and environment sector elements exposed to climate hazards in Makindye Ssabagabo Municipality include wetlands, and water sources.

Exposure of wetlands to drought and flood hazards

The analysis showed that 2.6 sq.km (16.6%) of wetlands are exposed to drought whereas 13.3 sq.km (83.4%) of wetlands are not exposed to drought hazard (**Table 16**). The biggest percentage of wetlands exposed to drought hazard in Makindye Ssabagabo Municipality is located in Bunamwaya (36.5%) and Masajja (32.1%) divisions.

A total of 5.5 sq.km (34.4%) of wetlands are exposed to floods while 10.5 sq.km (65.6%) of wetlands are not exposed to flood hazard (**Table 16**). The biggest proportion of wetlands exposed to floods is located in Bunamwaya (61.0%) and Masajja (52.3%) divisions.

Table 16: Exposure of wetlands to drought and floods in Makindye Ssabagabo Municipality

Climate Hazards	**Drought**				**Floods**			
Division	Exposed		Not Exposed		Exposed		Not Exposed	
	Area (Km²)	%	Area (Km²)	%	Area (Km²)	%	Area (Km²)	%
Bunamwaya	0.6	36.5	1.0	63.5	1.0	61.0	0.6	39.0
Masajja	1.6	32.1	3.4	67.9	2.7	52.3	2.4	47.7
Ndejje	0.5	4.8	8.9	95.2	1.9	20.1	7.4	79.9
Total	**2.6**	**16.6**	**13.3**	**83.4**	**5.5**	**34.4**	**10.5**	**65.6**

The exposure of wetlands to drought and flood hazards in Makindye Ssabagabo Municipality is shown in **Figure 27**.

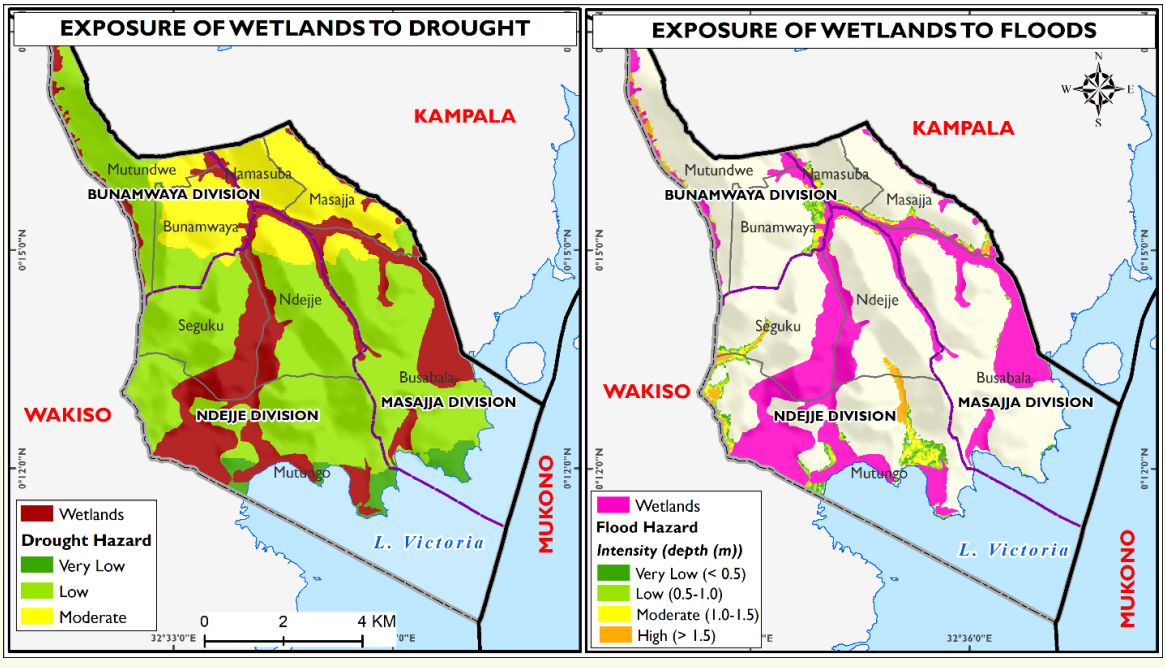


Figure 27: Exposure of wetlands to drought and flood hazards in Makindye Ssabagabo Municipality

Exposure of water sources to drought and flood hazards

The assessment showed that 113 water sources (34.6%) are exposed to drought hazard in Makindye Ssabagabo Municipality (**Table 17**). Masajja (61.2%) and Bunamwaya (31.0%) divisions have the biggest proportion of water sources exposed to drought hazard.

A total of 54 water sources (16.5%) are exposed to floods whereas 273 water sources (83.5%) are not exposed to flood hazard (**Table 17**). The biggest percentage of water sources exposed to floods in Makindye Ssabagabo Municipality is located in Ndejje (17.6%) and Masajja (16.5%) divisions.

Table 17: Exposure of water sources to drought and floods in Makindye Ssabagabo Municipality

Climate Hazards	**Drought**				**Floods**			
Division	Exposed		Not Exposed		Exposed		Not Exposed	
	Number	%	Number	%	Number	%	Number	%
Bunamwaya	31	31.0	69	69.0	15	15.0	85	85.0
Masajja	52	61.2	33	38.8	14	16.5	71	83.5
Ndejje	30	21.1	112	78.9	25	17.6	117	82.4
Total	**113**	**34.6**	**214**	**65.4**	**54**	**16.5**	**273**	**83.5**

In terms of exposure by water source type, kiosks (100%), yard taps for public use (35.2%), shallow wells (30.0%), public stand posts (33.3%), and protected springs (38.8%) have the biggest percentages exposed to drought hazard in Makindye Ssabagabo Municipality (**Table 18**). On the other hand, kiosks (50%), protected springs (28.1%), and shallow wells (20%) have the biggest proportional exposure to floods in Makindye Ssabagabo Municipality (**Table 18**).

Table 18: Exposure of the different types of water sources to drought and floods in Makindye Ssabagabo Municipality

Climate Hazards	**Drought**				**Floods**			
Types of water sources	Exposed		Not Exposed		Exposed		Not Exposed	
	Number	%	Number	%	Number	%	Number	%
Deep borehole	2	20.0	8	80.0	1	10.0	9	90.0
Kiosk	2	100	0	0.0	1	50.0	1	50.0
Protected spring	62	38.8	98	61.3	45	28.1	115	71.9
Public Stand Post	1	33.3	2	66.7	0	0.0	3	100
Rainwater Harvest Tank	2	8.3	22	91.7	0	0.0	24	100
Shallow well	6	30.0	14	70.0	4	20.0	16	80.0
Yard tap for public use	38	35.2	70	64.8	3	2.8	105	97.2
Total	**113**	**34.6**	**214**	**65.4**	**54**	**16.5**	**273**	**83.5**

The exposure of water sources to drought and floods is presented in **Figure 28**.

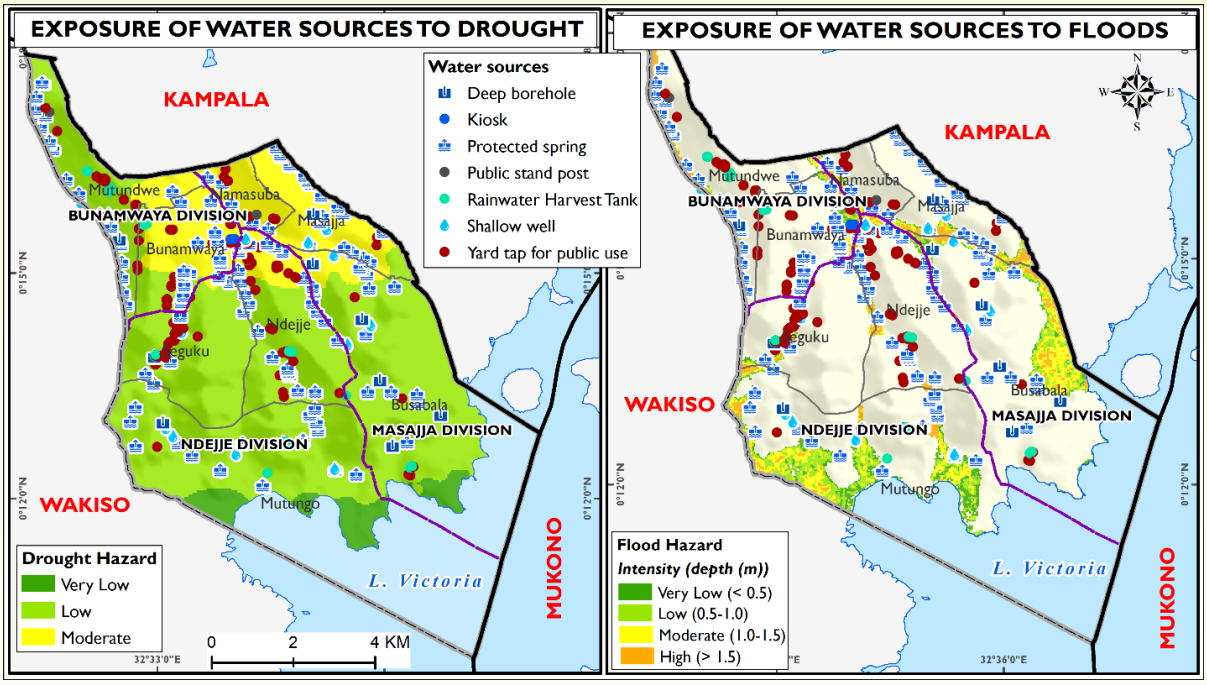


Figure 28: Exposure of water sources to drought and floods in Makindye Ssabagabo Municipality

2.6 Vulnerability Assessment

Vulnerability to climate change refers to the degree to which a system, community, or individual is susceptible to, and unable to cope with, the adverse impacts of climate change, including climate variability and extremes (IPCC, 2014). Vulnerability is shaped by physical, social, economic, and political factors that affect how prepared and resilient individuals or communities are to climate-related hazards. Vulnerability is a function of exposure, the sensitivity of a household or livelihood to the stress associated with that exposure, and the adaptive capacity to recover from the impacts of that exposure.

*Vulnerability = (Exposure * Sensitivity) – Adaptive Capacity*

2.5.1 Exposure of communities to multi-hazards

Exposure refers to the extent to which a system is exposed to climate change-related hazards. Makindye Ssabagabo Municipality is mostly exposed to drought, floods, hailstorms, lightning, and windstorms. These hazards were ranked, weighed and normalized to produce the exposure index. The assessment showed that the wards that have a high level of exposure (more than 80 percent) to drought, floods, hailstorms, lightning, and windstorms in Makindye Ssabagabo Municipality include Mutungo, Busabala, and parts of Masajja, Namasuba, Ndejje, Seguku, and Bunamwaya (**Figure 29**).

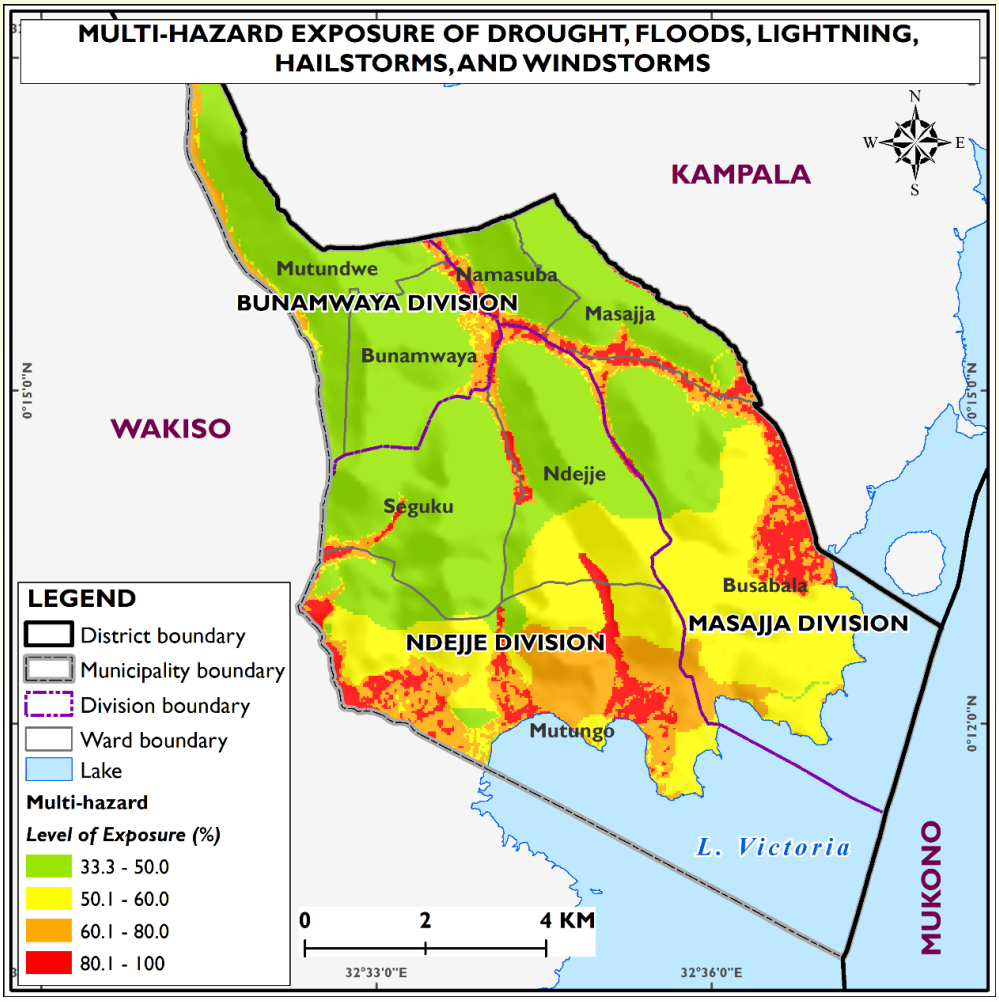


Figure 29: Multi-hazard exposure of communities to drought, floods, hailstorms, lightning, and windstorms in Makindye Ssabagabo Municipality

2.5.2 Sensitivity of communities to multi-hazards

Sensitivity describes the socio-economic or demographic characteristics that can make people susceptible to the negative effects of exposure (Declet-Barreto et al., 2020). The sensitivity indicators that were assessed at ward level are presented in **Table 19**. These indicators were ranked, weighed and normalized to produce the sensitivity index.

Table 19: Sensitivity indicators to multi-hazards in Makindye Ssabagabo Municipality

No	Sensitivity indicators	Description	Data source	Hotspot wards
1.	Access to health services	Access to health services reflects a community's ability to prevent, respond to, and recover from climate-related health impacts such as heat stress, disease outbreaks, and injuries from extreme weather.	Key Informant Interviews, 2025	Busabala
2.	Community awareness and preparedness	Community awareness and preparedness reflects how well individuals and groups understand climate risks and their capacity to take timely action.	Key Informant Interviews, 2025	Moderate in the entire municipality

No	Sensitivity indicators	Description	Data source	Hotspot wards
3.	Degradation of rivers	Degradation of rivers signifies the reduced ability of freshwater ecosystems to absorb and adapt to climate impacts, thereby increasing the vulnerability of communities that rely on these rivers for water, food, and livelihoods.	Key Informant Interviews, 2025	Busabala, Namasuba, and Masajja
4.	Dependency on climate-sensitive sectors e.g., agriculture	Dependency on climate-sensitive sectors highlights how heavily communities rely on industries like agriculture, fisheries, and tourism, which are directly affected by climate variability, making them more vulnerable to disruptions caused by extreme weather events, changing temperatures, and shifting ecosystems.	Key Informant Interviews, 2025	Namasuba and Mutundwe
5.	Dependence on surface water	Dependency on surface water reveals how reliant communities are on rivers, wetlands, and reservoirs that are highly vulnerable to changing climate, increasing their risk of water scarcity and reduced water quality under climate stress.	Key Informant Interviews, 2025	Low in the entire municipality
6.	Drainage systems	The design and capacity of drainage systems determine how well a community can manage increased rainfall, flooding, and storm surges, with inadequate or poorly maintained systems heightening vulnerability to climate-induced water-related hazards.	Key Informant Interviews, 2025	Poor in the entire municipality
7.	Housing quality	Housing quality reflects how well homes can protect occupants from climate-related hazards like floods, with poor-quality housing increasing the risk of damage, displacement, and health impacts during climate shocks.	Key Informant Interviews, 2025	Masajja
8.	Income levels	Income levels influence a community's ability to prepare for, respond to, and recover from climate impacts, with lower-income populations often lacking the resources needed for adaptation, making them more vulnerable to climate-related risks.	UBOS, 2021	Ndejje, Bunamwaya, Busabala, Namasuba, and Masajja
9.	Population density	High-density areas can amplify the impacts of climate hazards such as heatwaves, and disease spread while straining infrastructure and emergency response systems, increasing overall community vulnerability.	UBOS, 2024	Namasuba, Masajja, Ndejje
10.	Unemployment	Unemployment reflects limited economic resilience and reduced access to resources, making individuals and communities less able to adapt to or recover from climate-related events such as floods	Key Informant Interviews, 2025	Moderate in the entire municipality

No	Sensitivity indicators	Description	Data source	Hotspot wards
11.	Wetland degradation	Wetland degradation weakens the natural buffers that wetlands provide against climate impacts like flooding, storm surges, and droughts, reducing ecosystem resilience and increasing the vulnerability of both human and ecological communities that depend on them.	WMD, 2021	Bunamwaya, Masajja, Namasuba, Mutundwe, and Ndejje

The wards that have a high sensitivity to multi-hazards (i.e., drought, floods, hailstorms, lightning, and windstorms) include Namasuba, Masajja, and Mutundwe (**Figure 30**).

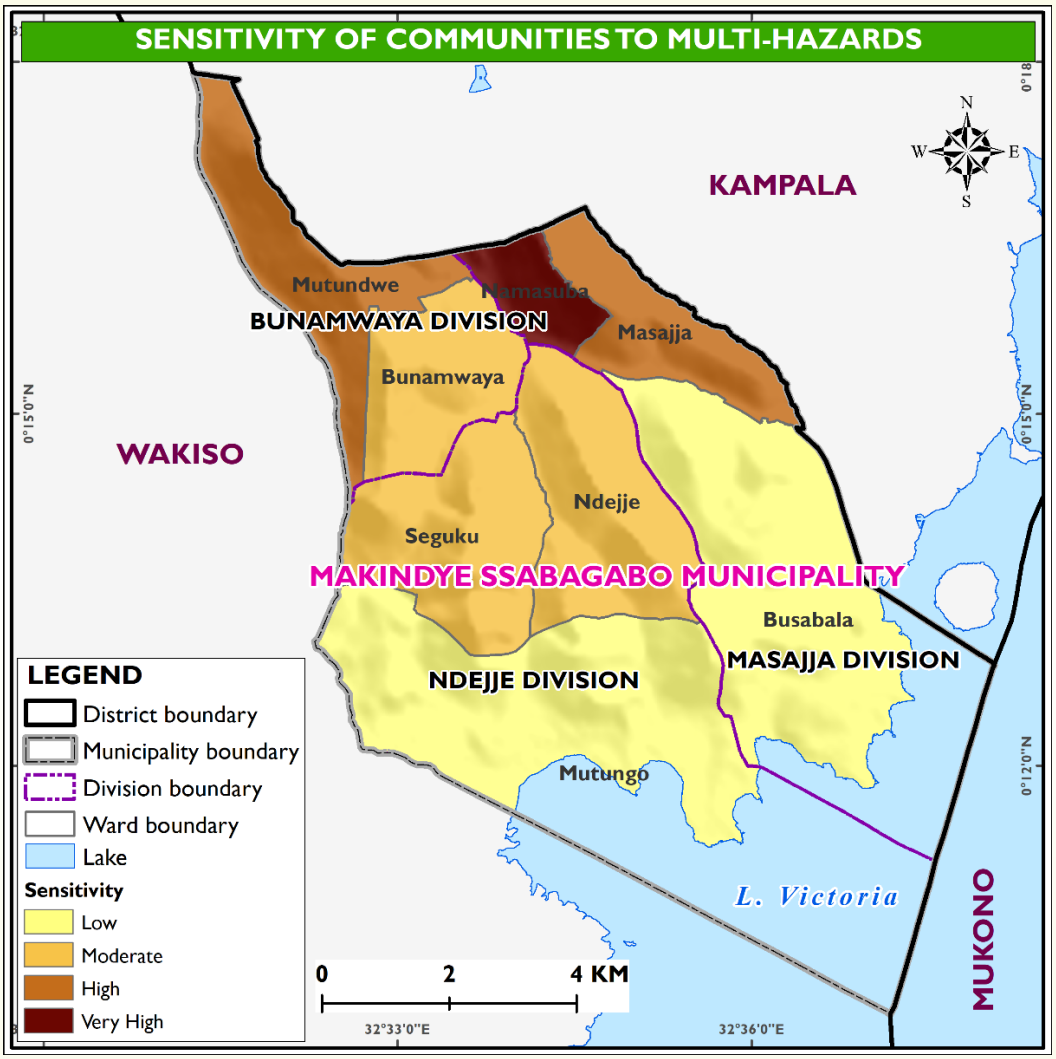


Figure 30: Sensitivity of communities to multi-hazards in Makindye Ssabagabo Municipality

2.5.3 Adaptive capacity of communities to multi-hazards

Adaptive capacity is defined as the ability of systems, institutions, humans, and other organisms to adjust to potential damage, to take advantage of opportunities, or to respond to consequences of climate change (IPCC, 2023). The adaptive capacity indicators that were assessed at the ward level are shown in **Table 20**. These adaptive capacity indicators were ranked, weighed and normalized to produce the adaptive capacity index.

Table 20: Adaptive capacity indicators to multi-hazards in Makindye Ssabagabo Municipality

No	Adaptive capacity indicators	Description	Data source	Hotspot wards
1.	Access to clean water and sanitation	Access to clean water and sanitation ensures that communities can maintain good health and hygiene, reducing the risk of waterborne diseases during extreme weather events, and enabling better resilience to climate-induced health challenges.	Key Informant Interviews, 2025	High in the entire municipality
2.	Access to credit	Access to credit allows individuals and communities to invest in resilience-building measures, such as climate-smart agriculture, infrastructure improvements, or emergency preparedness, providing the financial resources needed to recover and adapt to climate-related disruptions.	Key Informant Interviews, 2025	Moderate in the entire municipality
3.	Access to early warning information	Access to early warning information enables communities to prepare for and respond effectively to climate-related hazards, such as storms, floods, or heatwaves, by providing timely alerts and guidance that help reduce risks, protect lives, and minimize damage.	Key Informant Interviews, 2025	Low in the entire municipality
4.	Access to renewable energy e.g., solar	Access to renewable energy provides communities with a sustainable and reliable energy source, reducing dependence on fossil fuels, minimizing greenhouse gas emissions, and enhancing resilience by ensuring energy availability during climate-related disruptions like storms or power outages.	Key Informant Interviews, 2025	Ndejje, Busabala, Namasuba, and Masajja
5.	Awareness of climate risks and adaptation practices	Awareness of climate risks and adaptation practices reflects how well individuals and communities understand climate-related challenges and the actions they can take to mitigate risks, enabling them to make informed decisions and implement strategies that enhance resilience to changing climate conditions.	Key Informant Interviews, 2025	Moderate in the entire municipality
6.	Diverse livelihood options	Diverse livelihood options provide communities with multiple sources of income and resources, allowing them to better withstand climate shocks, and adapt by shifting to more resilient economic activities when needed.	Key Informant Interviews, 2025	Moderate in the entire municipality
7.	Education levels	Higher levels of education enable individuals and communities to better understand climate risks,	Key Informant Interviews, 2025	High in the entire municipality

No	Adaptive capacity indicators	Description	Data source	Hotspot wards
		access information, and adopt effective adaptation strategies, thereby enhancing their ability to respond to and recover from climate-related challenges.		
8.	Extension services	Extension services provide farmers, communities, and local businesses with essential knowledge, skills, and resources to adopt climate-resilient practices, improve productivity, and enhance the ability to cope with changing climate conditions.	Key Informant Interviews, 2025	High in the entire municipality
9.	Good environmental management practices	Good environmental management practices promote sustainable resource use, reduce environmental degradation, and enhance ecosystem resilience, enabling communities to better withstand and adapt to climate impacts.	Key Informant Interviews, 2025	Moderate in the entire municipality
10.	Healthcare infrastructure	Robust and well-maintained healthcare systems improve a community's ability to respond to climate-related health challenges, such as heatwaves, disease outbreaks, and extreme weather events, by providing timely medical care, disease prevention, and emergency services.	Key Informant Interviews, 2025	Busabala, Namasuba, and Masajja
11.	Proper waste disposal	Proper waste disposal helps prevent environmental degradation, reduces the risk of disease outbreaks, and ensures cleaner, healthier living conditions, enabling communities to better withstand the health and environmental impacts of climate change, such as flooding	Key Informant Interviews, 2025	Bunamwaya, Mutundwe, Mutungo, Ndejje, Busabala, Namasuba, and Masajja
12.	Rainwater harvesting	Rainwater harvesting provides an alternative and supplemental water source, reducing dependence on traditional sources such as rivers especially during the dry season.	Key Informant Interviews, 2025	Bunamwaya, Ndejje, Busabala, Namasuba, and Masajja
13.	Social networks	Strong, connected communities are better able to share resources, information, and support during climate events, helping individuals and groups recover more quickly and effectively from climate impacts	Key Informant Interviews, 2025	Moderate in the entire municipality
14.	Urban greening initiatives	Urban greening initiatives enhance cities' resilience by increasing green spaces, improving air quality, reducing heat island effects, and providing natural buffers against extreme weather, all of which help communities better cope with climate-related challenges.	Key Informant Interviews, 2025	Moderate in the entire municipality

No	Adaptive capacity indicators	Description	Data source	Hotspot wards
15.	Wetland and ecosystem restoration	Wetland and ecosystem restoration enhances the natural ability of wetlands to absorb climate impacts such as flooding, while supporting biodiversity, improving water quality, and strengthening the resilience of communities that rely on these ecosystems.	Key Informant Interviews, 2025	Low in the entire municipality

Seguku ward has a very high adaptive capacity to multi-hazards (i.e., drought, floods, hailstorms, lightning, and windstorms) while Namasuba, Masajja, and Busabala have a low adaptive capacity to multi-hazards (**Figure 24**).

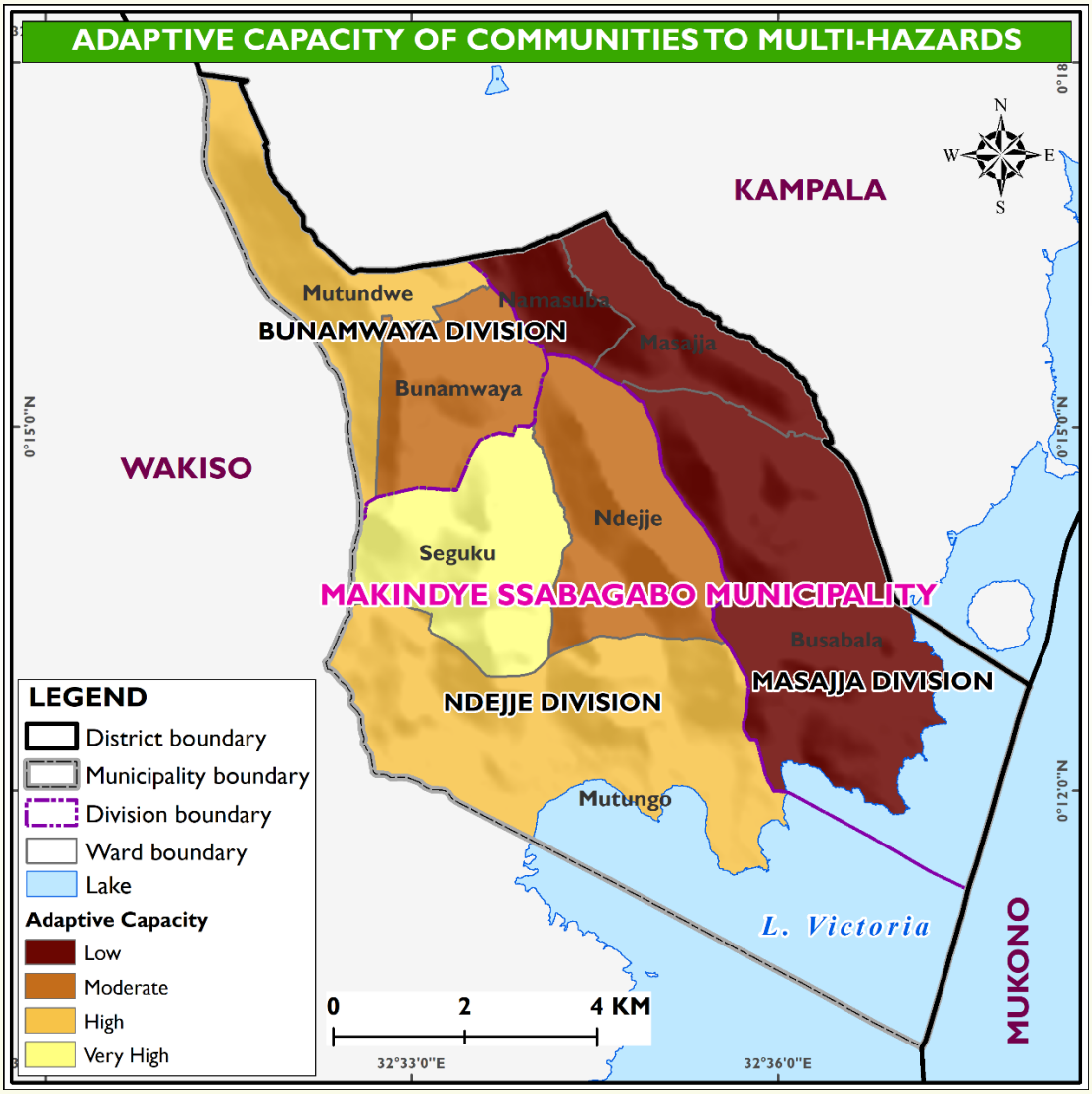


Figure 31: Adaptive capacity of communities to multi-hazards in Makindye Ssabagabo Municipality

2.5.4 Vulnerability of communities to multi-hazards

The communities that are highly vulnerable to multi-hazards (i.e., drought, floods, hailstorms, lightning, and windstorms) in Makindye Ssabagabo Municipality are found in Busabala, Masajja, Namasuba, Mutungo, Ndejje, and Bunamwaya (**Figure 32**).

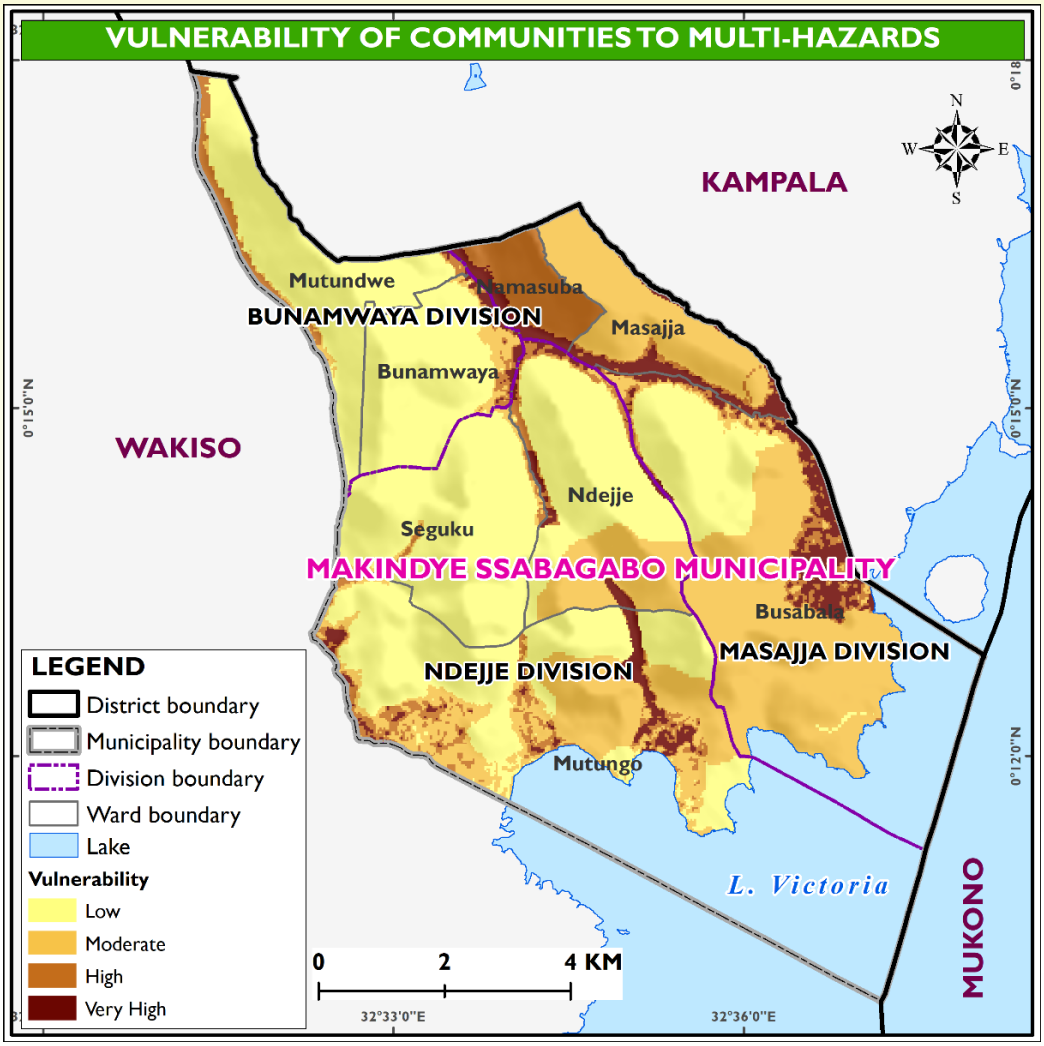


Figure 32: Vulnerability of communities to multi-hazards in Makindye Ssabagabo Municipality

2.5.5 Risk of communities to multi-hazards

Disaster risk is expressed as the likelihood of loss of life, injury or destruction and damage from a disaster in a given period (UNISDR, 2017). Disaster risk is widely recognized as the consequence of the interaction between a hazard and the characteristics that make people and places vulnerable and exposed (**Figure 33**).



Figure 33:The risk matrix
Source: de Brito et al., 2017

The multi-hazards experienced in Makindye Ssabagabo Municipality were combined, weighed, and normalized to produce the hazard index. The population of the communities in the different wards was considered as the exposed element. The population was normalized to produce the exposure index. The hazard, exposure, and vulnerability indices were analyzed and normalized to produce the risk index, which was mapped at the ward level.

Risk assessment showed that the wards that are at a high risk of multi-hazards (drought, floods, hailstorms, lightning, and windstorms) in Makindye Ssabagabo Municipality include Masajja, Ndejje, and Busabala (**Figure 34**).

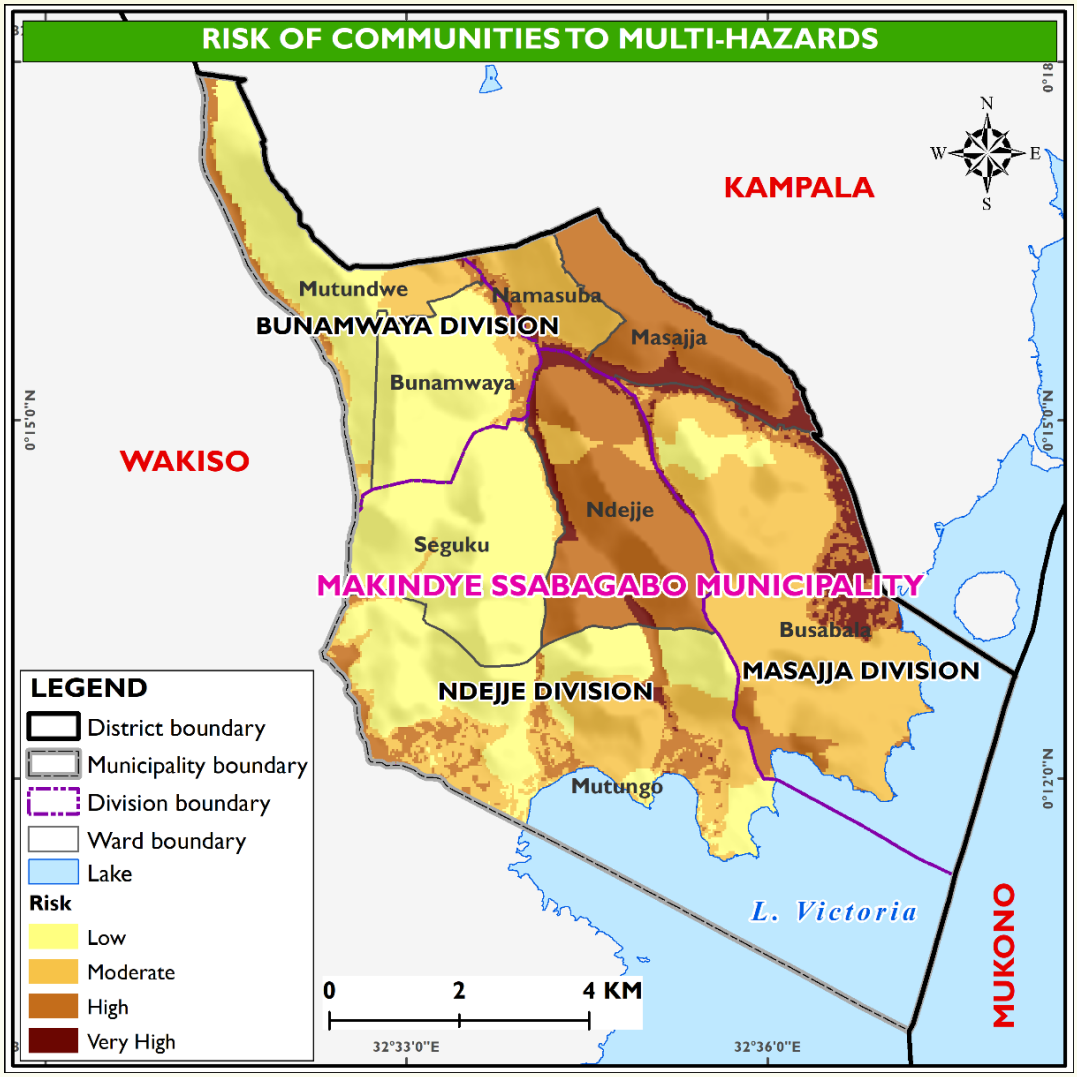


Figure 34: Risk of communities to multi-hazards in Makindye Ssabagabo Municipality

2.7 Impacts of climate change on sectors, livelihoods, and ecosystems

2.6.1 Impacts of climate change on the different sectors

Climate change has significantly impacted the agriculture, education, energy, health, land and housing, manufacturing, social development, trade, transportation, water and environment, and waste management sectors of Makindye Ssabagabo Municipality (**Table 21**).

Table 21: Impacts of climate change by sector in Makindye Ssabagabo Municipality

Sectors	Impacts of climate change by sector
Agriculture	• Reduced crop yield • Reduced livestock productivity • Destruction of croplands • Increased prevalence of pests, parasites, and diseases • Reduced quality of forage available to grazing livestock
Education	• Damage to school buildings, rendering them unsafe or unusable • Displacement disrupts children's education and causes psychosocial stress • Disruption of academic calendars, leading to increased absenteeism and dropout rates
Energy	• Power shortages due to reduced hydropower generation in hydroelectric plants • Damage to energy infrastructure • Higher energy consumption, especially for cooling
Health	• Increased prevalence of malaria • Reduced water quality and sanitation leads to outbreak of waterborne diseases such as typhoid • Food insecurity contributes to higher rates of malnutrition, particularly among children, pregnant women, and individuals with compromised immune systems
Land and housing	• Damage to homesteads • Displacement of communities • Loss of lives • Land degradation through increased soil erosion
Manufacturing	• Reduced agricultural outputs affect the availability of raw materials for manufacturing processes • Damage to the manufacturing facilities and related infrastructure such as factories, warehouses, and transportation networks • Power shortages affect manufacturing operations and increases reliance on alternative, often more expensive energy sources
Social development	• Displacement of communities • Destruction of school and health facilities infrastructure • Loss of lives • Changing climate patterns contribute to the spread of diseases, such as malaria and waterborne illnesses, due to increased temperatures and flooding
Trade	• Reduced export of key commodities such as coffee, tea, and cocoa • Damage to infrastructure such as roads and bridges disrupts supply chains, increases transportation costs, and hampers the timely delivery of goods to both domestic and international markets • Exporters face challenges meeting stringent environmental and sustainability standards imposed by trading partners, potentially limiting market access for non-compliant products
Transportation	• Erosion and degradation of roads, bridges, and culverts • Flooded or damaged roads hinder the movement of goods and people, affecting trade and access to essential services • Transport disruptions increase the cost of goods and services due to delays and the need for alternative, often longer, routes

Sectors	Impacts of climate change by sector
	• Damaged infrastructure and congestion lead to longer travel times and increased fuel consumption, contributing to higher greenhouse gas emissions
Water and environment	• Degradation of ecosystems such as wetlands, and forests • Habitat loss and a decline in biodiversity • Reduced availability of water resources • Deterioration of water quality • Soil degradation through erosion and nutrient depletion
Waste management	• Inadequate waste disposal practices result in solid waste clogging drainage systems, reducing their capacity to channel stormwater effectively • Organic waste decomposition in landfills emits methane, a potent greenhouse gas contributing to global warming • Poor waste management practices, such as open dumping and inadequate landfill operations, increase the emission of greenhouse gases • Increased use of plastics in agriculture, driven by changing climate, leads to microplastics infiltrating soils, posing risks to ecosystems and human health.

2.6.2 Perceived impacts of climate change on local livelihoods

Climate change has also impacted the local livelihoods in Makindye Ssabagabo Municipality. The major perceived impacts of climate change reported by communities in the various divisions of the municipality include increased food prices (16.8%), damage and/or loss of property during floods (15.7%), water shortages due to prolonged dry spells (13.1%), increased cost of water during the dry spells (11%), loss of income and unemployment due to floods (10.9%), and reduced crop yields for urban farmers (9.3%) (**Table 22**).

Table 22: Perceived impacts of climate change on local livelihoods in the different divisions of Makindye Ssabagabo Municipality

Impacts of climate change on local livelihoods	Bunamwaya		Masajja		Ndejje		MSMC	
	N	%	N	%	N	%	N	%
Increased food prices	63	31.0	131	12.3	63	24.0	257	16.8
Damage and/or loss of property during floods	58	28.6	138	13.0	45	17.1	241	15.7
Water shortages due to prolonged dry spells	55	27.1	105	9.9	40	15.2	200	13.1
Increased cost of water during the dry spells	9	4.4	124	11.6	36	13.7	169	11.0
Floods lead to loss of income and unemployment	8	3.9	128	12.0	31	11.8	167	10.9
Reduced crop yields for urban farmers	9	4.4	121	11.4	13	4.9	143	9.3
Increased occurrence of crop pests and diseases	1	0.5	111	10.4	31	11.8	143	9.3
Reduced livestock productivity	0	0.0	115	10.8	3	1.1	118	7.7
Increased occurrence of livestock pests, parasites and diseases	0	0.0	92	8.6	1	0.4	93	6.1
Total	**203**	**100**	**1,065**	**100**	**263**	**100**	**1,531**	**100**



Plate 1: Homes affected by floods in Namasuba ward, Makindye Ssabagabo Municipality

2.6.3 Perceived impacts of climate change on ecosystems

Climate change has not only impacted livelihoods but also ecosystems in Makindye Ssabagabo municipality. The major perceived impacts of climate change on ecosystems in Makindye Ssabagabo municipality include decline/loss of wetland benefits (21.6%), water pollution and decreased water quality (20.2%), reduced wetland biodiversity (14.8%), increased vulnerability of ecosystems to heat stress and drought (13.6%), and increased sedimentation in rivers and wetlands (12.6%) (**Table 23**).

Table 23: Perceived impacts of climate change on ecosystems

Impact of climate change on ecosystems	Bunamwaya		Masajja		Ndejje		MSMC	
	N	%	N	%	N	%	N	%
Decline/loss of wetland benefits	59	42.4	133	16.8	42	27.3	234	21.6
Water pollution and decreased water quality	37	26.6	136	17.2	46	29.9	219	20.2
Reduced wetland biodiversity	12	8.6	124	15.7	25	16.2	161	14.8
Increased vulnerability of ecosystems to heat stress and drought	20	14.4	117	14.8	11	7.1	148	13.6
Increased sedimentation in rivers and wetlands	0	0.0	129	16.3	8	5.2	137	12.6
Emergence of invasive species	1	0.7	103	13.0	9	5.8	113	10.4
Increased habitat fragmentation	10	7.2	50	6.3	13	8.4	73	6.7
Total	**139**	**100**	**792**	**100**	**154**	**100**	**1,085**	**100**

3. CLIMATE ACTION OPTIONS

3.1 Introduction

The climate action options proposed in this plan are meant to reduce greenhouse gas emissions (mitigation) and facilitate climate adaptation in Makindye-Ssabagabo Municipality. These actions are categorised into 12 result areas: Waste management, Energy, Transportation, Trade, Manufacturing, Agriculture, Environment, ICT, Land and housing, social development (Communities), Education and Sport and Health.

3.1.1 Waste Management

Unregulated waste dumping in wetlands like Nankinga and Lufuka Valley in the municipality exacerbates methane emissions and contaminates water sources. Flood-prone areas such as Masajja Division face worsened sanitation crises during rains, as plastic waste clogs drainage systems, amplifying flood risks. The municipality's reliance on undesignated open dump sits in areas like Kabuuma and Kisugula not only emits greenhouse gases (GHGs) but also threatens biodiversity in adjacent wetlands. Climate-driven heavy rainfall intensifies leachate pollution, directly impacting low-income communities in Kanala and Mirembe. This result area thus focuses on transforming waste handling systems to be sustainable, inclusive, and emissions-reducing. It seeks to promote waste segregation, recycling, composting, and biogas generation, while reducing open dumping and methane emissions from landfills. It also emphasizes behavioural change and policy enforcement at household, institutional, and municipal levels.

Outcome

A clean and sustainable waste management system that reduces emissions and enhances community health is promoted

Outputs

The outputs are;

1. Community awareness and participation in waste management enhanced
2. Organic and inorganic waste processing improved
3. Waste disposal infrastructure improved

Climate Actions (Mitigation and Adaptation)

Short-term actions: sensitizing communities on zero-waste policies, promoting waste segregation at the source, promoting return schemes for bottles/containers, setting up anaerobic digesters for biogas, building recycling facilities, promoting biodegradable packaging, promoting reuse of grey water, strengthening waste collection systems, and implementing waste audit systems.

Medium-term: Educating on compostable packaging use, establishing visible waste collection points, and establishing decentralized composting sites.

3.1.2 Energy

Apart from contributing to greenhouse gas emissions, the energy infrastructure in Makindye Ssabagabo Municipality is vulnerable to multiple climate hazards. Notably, 16% of distribution transformers are exposed to lightning, and 10.4% to windstorms. Substations are moderately exposed to flooding, particularly in flood-prone division like Masajja and Ndejje. The disruptions

caused by these increase the risk of blackouts, equipment damage, and safety hazards. Frequent droughts strain hydroelectric power, pushing households towards costly, polluting generators. Further, rising temperatures increase cooling demands, further stressing energy grids in commercial hubs like Ssabagabo Central. To enhance energy reliability and safety amid increasing climate risks, this result area aims to promote a transition from vulnerable, carbon-intensive infrastructure to clean and resilient energy systems. It prioritizes the adoption of solar, biogas, and energy-saving technologies, and encourages climate-resilient energy infrastructure across both public and private sectors.

Outcome

A low-carbon energy system that enhances efficiency and promotes clean energy is promoted

Outputs

Outputs for the energy sector interventions are

1. Energy efficiency improved
2. Access to renewable energy enhanced
3. Emissions from infrastructure reduced

Climate Actions (Mitigation and Adaptation)

Short-term actions: educating communities on energy conservation, advocating for the use of LED bulbs for lighting, training solar/biogas technicians, promoting energy-efficient clean cook stoves and use of briquettes.

Medium-term: Conducting energy audits, enforcing energy codes and standards in buildings, promoting solar and biogas, promoting energy cooperatives, installing solar lighting in public spaces, promoting energy-efficient technology in schools and medical facilities, advocating for energy-efficient technology in schools and health centres, and promoting solar water heating.

Long-term: Introducing green building codes, advocating for solar-powered irrigation, and retrofitting buildings for efficiency.

3.1.3 Transportation

The bustling boda boda (motorcycle taxi) culture and aging minibuses in Kisugula and Gangu emit high levels of CO_2, worsening air quality in densely populated areas. Flood-damaged roads in Masajja disrupt mobility, raising transport costs for traders. Plans for cycling lanes along Ndejje Road and electric vehicle charging points in Lubowa can cut emissions, but poor drainage in floodplains like Lufuka undermines infrastructure resilience. This result area thus aims to promote low-emission mobility solutions such as public transport, cycling, and walking. It also supports transitioning to electric and hybrid vehicles, improving transport infrastructure, and reducing emissions from road congestion through smart traffic management.

Outcome

Low-emission and efficient urban transport system is established

Output

The outputs are

1. Awareness and low-emission usage increased
2. Infrastructure for clean transport promoted
3. Integrated sustainable planning enhanced

Climate Actions (Mitigation and Adaptation)

Short-term actions: educate on non-motorised transport (like cycling, walking) and use of public transport; promoting shared mobility; restricting high-emission vehicles (set municipal level emission standards); and conducting low-carbon campaigns.

Medium-term: Introducing electric boda bodas in partnership with start-ups, developing pedestrian and cycling lanes, creating green corridors, implementing intelligent traffic lights reduce idling, transitioning to electric taxis and buses, integrating green transport in urban plans, deploying smart parking systems, and tracking emissions from transport.

Long-term: Providing e-vehicle charging points, promoting mixed-use neighbourhoods, and supporting transit-oriented development

3.1.4 Trade

Markets in Makindye-Ssabagabo Municipality, including Namasuba. Massajja, Mutundwe, and Bunamwaya Central markets generate massive non-recyclable waste, while goods transported via fossil-fuel trucks from these markets increase the trade sector's carbon footprint. Floods in Namasuba and Busabala disrupt supply chains. To reduce climate-related disruptions in trade and economic productivity, this result area focuses on circular economy practices (recycle, reuse, and reduce waste) and emissions reduction in the trade sector.

Outcome

A sustainable and climate-smart trade system that supports low-emission growth and local economic resilience is established.

Outputs

Outputs are

1. Improved efficiency and reduce emissions in trade logistics
2. Policy and incentives for sustainable trade strengthened

Climate Actions (Mitigation and Adaptation)

Short-term actions: adopting eco-friendly packaging (biodegradable/recyclable), engaging communities in sustainable trade, and tracking and reporting emissions in trade activities.

Medium-term: Providing platforms for green entrepreneurs, using hybrid vehicles for goods transport, transitioning trade hubs to renewable energy, and encouraging digital systems (e-invoicing).

Long-term: Designing energy-efficient warehouses, penalizing polluting trade activities, investing in carbon offset programs for trade (e.g., reforestation), and upgrading to energy-efficient machinery.

3.1.5 Manufacturing

Industries in Namasuba, Ngejje, and Bunamwaya (Agro-processing, food processing, construction, automotive repairs and parts) discharge untreated wastewater into wetlands like Lufuuka, degrading ecosystems that buffer floods. Further, energy-intensive brick kilns in Ndejje, Namasuba, Kisugula and Kabuuma emit black carbon, contributing to atmospheric warming. This result area seeks to green industrial operations by improving energy efficiency, adopting circular economy principles, treating industrial waste, and reducing emissions. It agitates for the use of

clean technologies, renewable energy, and sustainable sourcing of materials within manufacturing value chains in the municipality.

Outcome

A low-emission, resource-efficient industrial sector that promotes green innovation and sustainable growth is promoted

Outputs

Outputs are;

1. Industrial emissions and pollution reduced
2. Circular and energy-efficient production promoted
3. Sustainable supply chains promoted

Climate Actions (Mitigation and Adaptation)

Short-term actions: Training manufacturers on green technologies and sourcing local raw materials to reduce transport emissions.

Medium-term: Using solar thermal/biomass for heating, promoting water and material reuse, transitioning trade hubs to renewable energy, encouraging certification in green production, introducing climate criteria in procurement policies, digitizing inventory and shipping systems, conducting lifecycle assessments of products, and fostering public-private collaboration in green innovation.

Long-term: Implementing closed-loop recycling systems.

3.1.6 Agriculture

Wetland encroachment for farming in Lufuka, Masajja, Bunamwaya, and Mirembe destroys natural carbon sinks, while droughts in slash yields of drought-sensitive crops and post-harvest losses worsen food insecurity during extreme weather. This result area seeks to green industrial operations by improving energy efficiency, adopting circular economy principles, treating industrial waste, and reducing emissions. It promotes the use of clean technologies, renewable energy, and sustainable sourcing of materials within manufacturing value chains.

Outcome

Climate-resilient urban and peri-urban agriculture systems are established

Outputs

Outputs are

1. Climate-smart agriculture practices promoted
2. Water use and soil conservation improved
3. Value chains and access to green markets enhanced

Climate Actions (Mitigation and Adaptation)

Short-term actions: training farmers in CSA practices, promoting agroforestry and intercropping, encouraging water harvesting in farms, educating on soil conservation (terracing, mulching), promoting the use of compost and green manure, and training on post-harvest handling.

Medium-term: Promoting small-scale irrigation, constructing water storage ponds, monitoring soil health regularly, establishing climate-resilient storage facilities, and linking farmers to climate-smart markets

Long-term: Supporting climate certification for farmers.

3.1.7 Water and Environment

Encroachment on Kyetinda, Lufuuka, and Kaliddubi wetlands reduces flood regulation, exposing the areas like Masajja and Kabuuma to recurrent flooding. Deforestation in Mutungo Hills disrupts rainfall patterns, worsening water scarcity in Masajja A & B and Mutundwe. This result area centers on ecosystem restoration and conservation, including wetlands, forests, and urban greenery. It aims to strengthen environmental governance, promote community awareness, enforce laws, and enhance nature-based solutions to reduce risks like floods and heatwaves.

Outcome

The urban environment is restored and protected to support sustainable ecosystems and community wellbeing

Outputs

Outputs are

1. Environmental awareness and stewardship are promoted
2. Natural ecosystems restored and protected
3. Sustainable planning and enforcement strengthened

Climate Actions (Mitigation and Adaptation)

Short-term actions: sensitizing communities on environmental management, raising awareness on ecosystem conservation for DRR, educating on sustainable land use practices, engaging religious and cultural leaders in conservation, controlling illegal dumping in natural areas, enforcing environmental protection laws, and training enforcement teams on environmental compliance.

Medium-term: Restoring and protecting wetlands and forests, promoting large-scale urban tree planting, integrating environmental protection in urban plans, monitoring emissions and pollution hotspots, developing climate risk zoning maps, and demarcating and protecting wetland boundaries.

Long-term: Establishing green belts, parks, and gardens.

3.1.8 Information, Communication and Technology (ICT)

Access to early flood warnings in Nankinga and Lufuka prevents climate risk reductions, yet mobile apps can be used to not only deliver early warning information but also report illegal dumping. The result areas thus will ensure that ICT is leveraged to support early warning systems, data-driven planning, awareness campaigns, and digital access to climate-smart solutions. This area aims to enhance connectivity, digital literacy, and the integration of modern technology into climate resilience strategies across sectors.

Outcome

A digitally connected and informed municipality using ICT for climate adaptation, early warning, and resilience is leveraged

Outputs

Outputs are

1. Climate and disaster early warning systems improved
2. Community digital literacy enhanced
3. ICT integrated into climate planning and governance

Climate Actions (Mitigation and Adaptation)

Short-term actions: developing local early warning systems, using SMS alerts and community radios for dissemination, translating warnings into local languages/formats, training leaders and communities on ICT for climate, and using digital platforms to share adaptation practices.

Medium-term: Setting up ICT-based control rooms, promoting digital apps for farming/weather information, equipping schools and CBOs with ICT tools, supporting women and youth ICT inclusion, using GIS for risk mapping and planning, digitalizing permit and reporting systems, using AI to optimize transport and waste systems, and archiving local climate data for research.

Long-term: Developing a municipal climate portal.

3.1.9 Land and Housing

Housing structures particularly in informal settlements and illegal settlements in areas like Kalidubi's floodplains and Kabuuma wetlands face high displacement risks during storms. Weak zoning laws allow construction in high-risk areas like Mirembe. Many areas also face wetland encroachment and lack enforcement of building codes. The focus is on climate-resilient urban planning, flood-proof housing, sustainable land use, and enforcement of building codes. It also promotes equitable access to secure and climate-safe housing, especially in informal or vulnerable settlements, while protecting fragile ecosystems like wetlands from encroachment.

Outcome

A resilient and inclusive urban settlement system with climate-proof land use and housing infrastructure is developed

Outputs

Outputs are

1. Climate-resilient and inclusive housing promoted
2. Climate risk integrated into urban planning
3. Enforcement and monitoring strengthened

Climate Actions (Mitigation and Adaptation)

Short-term actions: Encouraging participatory planning for housing, using building materials with a low carbon footprint, conducting risk-based land assessments, enforcing planning and construction standards, monitoring informal development in risk zones, and establishing a land and housing climate task force.

Medium-term: Promoting resilient building design, elevating structures in flood-prone areas, supporting low-income housing retrofits, developing zoning that avoids high-risk zones, restricting development in wetlands and floodplains, including green spaces in urban planning, improving land tenure systems for resilience, and building local capacity for urban climate planning.

Long-term: Promoting mixed-use development to reduce emissions.

3.1.10 Social Development (Communities)

Marginalized groups in Nankinga and Lufuka bear the brunt of climate shocks, losing livelihoods to floods. Women in Ndejje lead recycling cooperatives, while youth in Masajja advocate for green jobs. However, limited funding for vocational training slows progress. The social development result area is cross-cutting and aims to empower communities, especially vulnerable groups—women, youth, and persons with disabilities—to participate in and benefit from climate action. It promotes inclusive planning, community-based adaptation, education, green jobs, and enhanced social safety nets to reduce vulnerability.

Outcome

Community resilience and inclusion in climate action is enhanced

Outputs

Outputs are

1. Community awareness and participation in climate adaptation and DRR increased
2. Community livelihoods and support systems strengthened
3. Public health, social infrastructure, and well-being improved

Climate Actions (Mitigation and Adaptation)

Short-term actions: promoting resilient building design, elevating structures in flood-prone areas, supporting low-income housing retrofits, developing zoning that avoids high-risk zones, restricting development in wetlands and floodplains, including green spaces in urban planning, improving land tenure systems for resilience, and building local capacity for urban climate planning.
Medium-term: Strengthening community safety nets (cash-for-work, food-for-work, micro-insurance), improving access to sanitation and clean water in vulnerable parishes, promoting rainwater harvesting and greywater recycling at the household level, promoting green jobs and entrepreneurship for marginalized groups, providing vocational training for youth and women in climate-smart sectors, establishing local funds for community-led adaptation and recovery projects, and advocating for integrating climate change education into school curricula.

Long-term: Establishing green social spaces (parks, gardens) for recreation and carbon sinks.

3.1.11 Education and Sports

Schools in Masajja (A&B) and Mutundwe can integrate climate literacy into their curricula, empowering students to lead tree-planting drives. "Green Games" in Ndejje can raise awareness, but outdated infrastructure in Ndejje schools limits hands-on learning. The actions in this result area aim to build adaptive capacities, raising awareness and instilling environmentally responsible behaviour among children, youth in institutions, and the entire community. It also provides for integration of climate change education in schools, using sports as a tool for mobilization and engagement to enhance knowledge, attitudes, and practices that support climate resilience. It will ensure inclusiveness, promoting participation by girls, persons with disabilities, and marginalized groups in climate action.

Outcome

Climate change awareness, environmental education, and adaptive capacity among learners and youth are improved.

Outputs

Outputs are

1. Climate education and environmental awareness integrated into school curricula and co-curricular activities
2. Capacity building in schools strengthened
3. Inclusive school and community-based climate action initiatives implemented

Climate Actions (Mitigation and Adaptation)

Short-term actions: conducting school-based climate change sensitization and awareness sessions, developing and distributing materials on climate tailored for learners and youth, organizing inter-school sports and climate awareness events (e.g., "Green Games"), and training teachers, sports coaches, and school administrators on integrating climate action into education and extracurricular activities.

Medium-term: Initiating talent development programs in sports and climate leadership targeting youth and establishing environment and climate clubs in all public and private schools.
Long-term: Promoting lifelong learning and community engagement through school-community partnerships.

3.1.12 Health

Flood-related cholera outbreaks in Lufuka and Kabuuma strain health facilities, while heatwaves in exacerbate respiratory illnesses. Medical waste from clinics in Ndejje, Busabala and Masajja pollutes wetlands. The health result area therefore focuses on building community resilience to climate change by addressing climate-sensitive diseases, strengthening health systems, and promoting climate-smart health practices. It provides for actions to protect vulnerable populations and ensures continued access to essential services through climate-resilient infrastructure, awareness campaigns, disease surveillance, and emergency preparedness.

Outcome

Resilience of the health system and communities to climate-related health risks and emergencies is improved.

Outputs

Outputs are

1. Health system capacity strengthened
2. Access to climate-resilient health infrastructure and services improved
3. Community awareness and adoption of climate-smart health practices increased

Climate Actions (Mitigation and Adaptation)

Short-term actions: training health workers on identification and management of climate-sensitive diseases, conducting community sensitization on disease prevention, sanitation, and hygiene in relation to climate change, providing clean water and improved sanitation in health centers, carrying out public health campaigns on nutrition, hygiene, and climate-related disease prevention, and promoting household-level climate-sensitive health practices (e.g., safe water storage, mosquito control, and clean cooking).
Medium-term: developing and implementing community-based early warning systems for climate-induced health outbreaks (e.g., cholera, malaria), improving disease surveillance and reporting systems in health facilities, upgrading health facilities with rainwater harvesting systems

and flood-proof infrastructure, and supporting community health workers with climate-health education toolkits and outreach materials.

Long-term: integrating climate change and public health response planning into the municipal health strategy, establishing permanent municipal platforms for public dialogue and knowledge sharing on climate and health, constructing new climate-resilient health centers in underserved and vulnerable areas, and establishing rapid response units and equipping health centers for health emergencies.

3.2 Implementation Plan

Table 24: Climate action implementation plan

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
RESULT AREA: 1. Waste Management	**Output 1.1** Community awareness and participation in waste management enhanced	1. Sensitize communities on zero-waste policies	Lufuka, Gangu	S			Municipal Env. Office, CBOs
Outcome 1: **A clean and sustainable waste management system that reduces emissions and enhances community health is promoted**		2. Promote waste segregation at source	Masajja A, Mirembe	S			CDO, NGOs
		3. Promote return schemes for bottles/containers	Kisugula, Kanala	S			Private sector, Env. Office
		4. Educate on compostable packaging use	Bunamwaya, Ndejje		M		NGOs, CBOs
		5. Establish visible waste collection points	Entire Municipality		M		Municipal Engineering Dept
			Gangu, Mirembe		M		Env. Office, CBOs
	Output 1.2 Organic and inorganic waste processing improved	1. Establish decentralized composting sites	Lweza B, Kisugula			L	Env. Office, MoWE
		2. Set up anaerobic digesters for biogas	Municipality-wide			L	Private sector, MoLG
		3. Build recycling facilities	Bunamwaya		M		Private Sector
		4. Promote biodegradable packaging	Ndejje			L	MoWE, Municipal Health
		5. Promote reuse of greywater	Gangu	S			Env. Office

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
	Output 1.3 Waste disposal infrastructure improved	1. Seal closed landfill sites	Mirembe Cell	S			Engineering Dept
		2. Manage landfill runoff	All Divisions		M		Municipal Health, Private Firms
		3. Strengthen waste collection systems	Kisugula		M		MoLG, Env. Committee
		4. Implement waste audit systems	Municipality-wide	S			Env. Office, NGOs
		5. Train waste workers	Lufuka, Gangu	S			Municipal Env. Office, CBOs
RESULT AREA: 2. Energy	**Output 2.1** Energy efficiency improved	1. Educate on energy conservation	Bunamwaya, Mutungo	S			Municipal Environment Office, MoES
Outcome 2: **A low-carbon energy system that enhances efficiency and promotes clean energy is promoted**		2. Advocate for use of LED bulbs for lighting	Masajja A, Ssabagabo Central	S			Private sector (lighting vendors), MEMD
		3. Promote clean cookstoves	Katereke, Ndejje	S			CBOs, MEMD, NGOs
		4. Conduct energy audits	Lweza A & B, Kigo		M		MEMD, Private Energy Auditors
		5. Retrofit buildings for efficiency	Lubowa, Kisugula			L	Engineering Dept, Physical Planning Unit
		6. Enforce energy codes and standards in buildings	Municipal-wide		M		Physical Planning, Building Inspectorate
	Output 2.2 Access to renewable energy enhanced	1. Promote solar & biogas	Mutundwe, Lufuka	S	M		MEMD, NGOs, Renewable Energy Companies
		2. Advocate for solar-powered irrigation	Kisugula, Masajja B		M		MoWE, Agriculture Extension Office
		3. Promote energy cooperatives	Gangu, Ndejje A		M		Cooperatives Office, MEMD

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		4. Train solar/biogas technicians	Ndejje Vocational Training Center	S			Vocational Institutes, NGOs, MEMD
	Output 2.3. Emissions from infrastructure reduced	1. Install Solar lighting in public spaces	Kirembe, Masajja	S			Engineering Dept, MEMD
		2. Energy-efficient tech in schools/medical facilities	Bunamwaya, Kisugula, Katereke		M		MoES, MoH, MEMD
		3. Advocate for energy-efficient tech in schools and health centers	Mutungo, Masajja B		M		MoES, Municipal Education Dept
		4. Promote solar water heating	Kigo		M		Private Sector, MEMD
		5. Establish off-grid systems in remote areas	Ssabagabo border zones			L	MEMD, NGOs
		6. Introduce green building codes	Lubowa, Lweza			L	Municipal Planning Dept, MoLHUD
RESULT AREA: 3. Transportation	**Output 3.1** Awareness and low-emission usage increased	1. Educate on cycling, walking & public transport	Masajja, Ndejje, Lufuka	S			Municipal Transport Dept, NGOs, KCCA
Outcome 3: **Low-emission and efficient urban transport system is established**		2. Promote shared mobility	Kabuuma, Mutungo	S			Municipal Engineering Department and Physical Planning Unit
		3. Restrict high-emission vehicles	Entire Municipality (focus on Masajja Central)	S			KCCA Traffic Police, NEMA, Municipal Law Enforcement
		4. Conduct low-carbon campaigns	Kisugula, Kanala	S			Municipal Environment Office, CBOs, Local Media
		5. Introduce electric boda bodas in partnership with startups	Ndejje, Ssabagabo Road				KCCA, Startups Association, MEMD

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
	Output 3.2 Infrastructure for clean transport promoted	1. Develop pedestrian & cycling lanes	Masajja, Ndejje		M		Municipal Engineering Dept, Physical Planning
		2. Create green corridors	Lufuka Valley, Kalidubi Stream		M		Municipal Environment Office, NEMA, CBOs
		3. Implement intelligent traffic lights to reduce idling	Masajja Central		M		KCCA, Municipal Engineering Department
		4. Transition to electric taxis/buses	Entire Municipality (pilot in Ndejje)		M		KCCA, Private Transport Companies, MEMD
		5. Provide e-vehicle charging points	Lubowa, Kisugula			L	KCCA, Private Energy Firms
	Output 3.3 Integrated sustainable planning enhanced	1. Promote mixed-use neighbourhoods	Kabuuma, Gangu			L	Municipal Planning Department, KCCA
		2. Transit-oriented development	Ssabagabo Road			L	KCCA, Municipal Council, Private Developers
		3. Green transport in urban plans	All divisions		M		Physical Planning Unit, Municipal Environment Office
		4. Smart parking systems	Commercial hubs (Mutundwe, Namasuba Markets)		M		KCCA, Municipal Engineering Department
		5. Track emissions from transport	Entire Municipality				Municipal Environment Office, NEMA, Research Institutions
RESULT AREA: 4. Trade	**Output 4.1** Climate-smart and low-carbon trade practices promoted	1. Promote trade of climate-friendly goods (e.g., solar tech, eco-products)	Masajja Market, Namasuba, Bunamwaya	S			Municipal Trade Office, Private Sector Associations, NGOs, NEMA

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
Outcome 4: **A sustainable and climate-smart trade system that supports low-emission growth and local economic resilience is established**		2. Encourage local production and consumption	Kabuuma, Lufuka, Masajja	S			Municipal Production Office, CBOs, MoTIC
		3. Adopt eco-friendly packaging (biodegradable/recyclable)	Entire Municipality (focus on markets: Namasuba)	S			Packaging companies, Municipal Environment Office, NEMA
		4. Engage communities in sustainable trade	Mutungo, Ndejje, Kisugula	S			CBOs, Municipal CDO, NGOs
		5. Provide platforms for green entrepreneurs	Lubowa Industrial Area, Bunamwaya		M		UIA, Private Sector Foundation Uganda, Municipal Trade Office
	Output 4.2 Improved efficiency and reduce emissions in trade logistics	1. Use hybrid vehicles for goods transport	Masajja, Namasuba		M		KCCA Transport Division, Private Logistics Companies, MEMD
		2. Transition trade hubs to renewable energy	Ndejje, Lubowa Industrial Area			L	UIA, MEMD, Renewable Energy Companies
		3. Track and report emissions in trade activities	Entire Municipality (focus on industrial zones: Bunamwaya, Lubowa)	S			NEMA, Municipal Environment Office, Private Sector
		4. Design energy-efficient warehouses	Lubowa Industrial Area, Bunamwaya			L	UMA, Private Sector, Municipal Engineering Dept
		5. Encourage digital systems (e-invoicing)	Entire Municipality (pilot in Bunamwaya Central Market)		M		MoICT, Private Tech Firms, Municipal Trade Office

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
	Output 4.3 Policy and incentives for sustainable trade strengthened	1. Develop incentives for green trade	Municipal-wide (focus on Lubowa, Bunamwaya industrial zones)		M		Municipal Council, MoT, NEMA
		2. Penalize polluting trade activities	Kabuuma, Mutungo, Ndejje			L	NEMA, Municipal Law Enforcement, Environment Police
		3. Support local climate-resilient markets	Masajja, Lufuka		M		CDO, NGOs, Municipal Agriculture Office
		4. Promote circular economy in business models	Lubowa Industrial Area, Bunamwaya		M		UMA, Private Sector, NEMA
		5. Invest in carbon offset programs for trade (e.g., reforestation)	Nankinga, Lufuka, Mirembe			L	NEMA, Municipal Environment Office, Forestry Department
			Lubowa Industrial Area, Bunamwaya				UMA, NEMA, Municipal Environment Office
RESULT AREA 5: Manufacturing	**Output 5.1** Industrial emissions and pollution reduced	1. Install filtration and ventilation systems	Lubowa Industrial Area		M		MEMD, UMA, Private Sector
Outcome 5: **A low-emission, resource-efficient industrial sector that promotes green innovation and sustainable growth is promoted**		2. Capture and reuse waste heat	Bunamwaya, Ndejje	S			NEMA, Municipal Health Department, MWE
		3. Prevent untreated wastewater discharge	Entire Municipality				UMA, Private Sector, Municipal Procurement Unit

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		4. Use low-carbon/recycled raw materials	Bunamwaya, Lubowa Industrial Area	S			UMA, MEMD, Private Sector
		5. Upgrade to energy-efficient machinery	Lubowa Industrial Area, Bunamwaya			L	UMA, NEMA, Private Sector
	Output 5.2 Circular and energy-efficient production promoted	1. Implement closed-loop recycling systems	Bunamwaya, Lubowa			L	MEMD, UMA, Renewable Energy Companies
		2. Use solar thermal/biomass for heating	Entire Municipality (focus on Ndejje, Masajja)		M		MWE, Municipal Environment Office
		3. Promote water and material reuse	Lubowa Industrial Area, Bunamwaya		M		UMA, Vocational Training Institutes, NGOs
		4. Train manufacturers on green technologies	Entire Municipality (pilot in Lubowa)	S			UNBS, UMA, NEMA
		5. Encourage certification in green production	Bunamwaya, Lubowa		M		Private Sector Foundation Uganda, Municipal Procurement Unit
	Output 5.3 Sustainable supply chains promoted	1. Source local raw materials to reduce transport emissions	Municipal-wide (focus on Lubowa, Bunamwaya)	S			Municipal Council, UMA, NEMA
		2. Introduce climate criteria in procurement policies	Lubowa Industrial Area, Ndejje Trading Center		M		MICT, Private Logistics Firms
		3. Digitize inventory and shipping systems	Bunamwaya, Lubowa		M		UMA, Research Institutions (e.g., Makerere University), NEMA
		4. Conduct lifecycle assessments of products	Entire Municipality (focus on Lubowa, Bunamwaya)		M		MST, UMA, Private Sector

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		5. Foster public-private collaboration in green innovation	Mutundwe Market, Ndejje Trading Center, Bunamwaya		M		Municipal Trade Office, Private Sector Associations, NGOs, NEMA
RESULT AREA 6: Agriculture	**Output 6.1** Climate-smart agriculture practices promoted	1. Train farmers in CSA practices	Kisugula, Lufuka, Masajja (high agricultural activity)	S			Municipal Agriculture Office, NGOs, Ministry of Agriculture
Outcome 6: **Climate-resilient urban and peri-urban agriculture systems are established**		2. Promote agroforestry and intercropping	Mutungo, Ndejje (areas with deforestation and soil erosion)	S			Forestry Department, CBOs, Municipal Environment Office
		3. Upscale drought-tolerant crop varieties	Kabuuma, Bunamwaya (drought-prone areas)	S			NARO, Municipal Agriculture Office
		4. Encourage crop rotation and conservation farming	Lufuka, (areas with degraded soils)	S			CBOs, Municipal Agriculture Office, MAAIF
		5. Integrate livestock with CSA practices	Ndejje, Kisugula (livestock-dependent communities)		M		MAAIF, Municipal Agriculture Office, NGOs
	Output 6.2 Water use and soil conservation improved	1. Water harvesting in farms	Masajja, Namasuba, Busabala (flood- and drought-prone parishes)	S			MoWE, CBOs
		2. Promote small-scale irrigation	Bunamwaya, Kabuuma (dryland farming zones)		M		MoWE, Municipal Engineering Department
		3. Educate farmers on soil conservation (terracing, mulching)	Lufuka, Mirembe (erosion hotspots)	S			Municipal Agriculture Office, NGOs, CBOs
		4. Construct water storage ponds	Nankinga, Kisugula (water-scarce areas)		M		MoWE, Municipal Engineering Department

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		5. Monitor soil health regularly	Entire Municipality (focus on Masajja, Bunamwaya)		M		NARO, Municipal Agriculture Office, Research Institutes
	Output 6.3 Value chains and access to green markets enhanced	1. Promote use of compost and green manure	Bunamwaya, Lubowa (peri-urban farming zones)	S			Municipal Environment Office, waste management firms
		2. Establish climate-resilient storage facilities	Masajja, Ndejje (post-harvest loss hotspots)		M		Ministry of Trade, Private Sector, CBOs
		3. Train farmers on post-harvest handling	Lufuka, Kabuuma (farmers' cooperatives)	S			NGOs, Municipal Agriculture Office
		4. Link farmers to climate-smart markets	Ndejje Trading Center, Bunamwaya		M		Uganda National Farmers Federation, Private Sector
		5. Support climate certification for farmers	Entire Municipality (pilot in Bunamwaya)			L	UNBS, NARO
RESULT AREA 7: Water and Environment	**Output 7.1** Environmental awareness and stewardship is promoted	1. Sensitize communities on environmental management	Kabuuma, Lufuka	S			Municipal Environment Office, CBOs, NEMA
Outcome 7: **The urban environment is restored and protected to support sustainable ecosystems and community wellbeing**		2. Raise awareness on ecosystem conservation for DRR	Busabala, Namasuba (flood-prone areas)	S			NEMA, Uganda Red Cross, Municipal Disaster Preparedness Office
		3. Educate on sustainable land use practices	Mutungo, Kisugula (encroachment hotspots)	S			Municipal Land Office, CBOs, NEMA

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		4. Conduct school and youth-led eco-campaigns	Entire Municipality (focus on schools in Bunamwaya, Gangu)	S			Municipal Education Office, Schools, NGOs
		5. Engage religious and cultural leaders in conservation	Ndejje, Bunamwaya	S			Religious Institutions, Cultural Leaders, NEMA
	Output 7.2 Natural ecosystems restored and protected	1. Restore and protect wetlands and forests	Kaliddubi, Lufuka, Mirembe (degraded wetlands)		M		NEMA, Municipal Environment Office, Forestry Department
		2. Promote large-scale urban tree planting	Lubowa, Masajja Central (low green cover areas)		M		KCCA, NGOs
		3. Establish green belts, parks, and gardens	Lubowa, Kisugula			L	Municipal Planning Department, KCCA
		4. Demarcate and protect wetland boundaries	Mirembe, Buggu		M		NEMA, Municipal Surveyors, Local Leaders
		5. Control illegal dumping in natural areas	Kabuuma, Ndejje	S			Municipal Health Department, NEMA, Environment Police
	Output 7.3 Sustainable planning and enforcement strengthened	1. Enforce environmental protection laws	Entire Municipality (focus on Lubowa, Bunamwaya)	S			NEMA, Municipal Law Enforcement, Judiciary
		2. Integrate environmental protection in urban plans	Municipal-wide (priority: Masajja)		M		Physical Planning Unit, KCCA
		3. Monitor emissions and pollution hotspots	Lubowa Industrial Area, Bunamwaya		M		NEMA, Municipal Environment Office
		4. Develop climate risk zoning maps	Flood-prone areas (Masajja, Ndejje)		M		GIS teams, Municipal Planning Department

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		5. Train enforcement teams on environmental compliance	Entire Municipality (focus on Municipal staff)	S			NEMA, Uganda Police, Judicial Training Institutes
RESULT AREA 8: ICT	**Output 8.1** Climate and disaster early warning systems improved	1. Develop local early warning systems	Masajja, Ndejje (flood-prone divisions)	S			ICT Department, UNMA
Outcome 8: **A digitally connected and informed municipality using ICT for climate adaptation, early warning, and resilience is leveraged**		2. Install automatic weather stations	Kisugula, Kanala (climate-vulnerable zones)		M		UNMA, Municipal Engineering Department
		3. Set up ICT-based control rooms	Municipal Headquarters		M		ICT Department, KCCA, NEMA
		4. Use SMS alerts and community radios for dissemination	Lufuka, Nankinga	S			Local Media (e.g., CBS, Simba), Telecom Companies (e.g., MTN, Airtel)
		5. Translate warnings into local languages/formats	Entire Municipality (focus on Luganda-speaking communities)	S			CBOs, Local Language Experts, Municipal ICT Office
	Output 8.2 Community digital literacy enhanced	1. Train leaders and communities on ICT for climate	Kisugula, Kanala (low-digital-literacy areas)	S			Municipal ICT Office, NGOs, Vocational Institutes
		2. Promote digital apps for farming/weather info	Bunamwaya, Masajja		M		Ministry of ICT, Agri-Tech Startups (e.g., Ensibuuko)
		3. Equip schools and CBOs with ICT tools	Schools in Bunamwaya, Mutundwe		M		MoES, NGOs, Private Sector (e.g., Huawei)

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		4. Use digital platforms to share adaptation practices	Entire Municipality (pilot in Lubowa)	S			ICT Department, CBOs, Research Institutes
		5. Support women and youth ICT inclusion	Ndejje, Lufuka		M		Gender Office, Youth Councils, NGOs
	Output 8.3 ICT integrated into climate planning and governance	1. Use GIS for risk mapping and planning	Masajja, Ndejje		M		Physical Planning Unit, Makerere University (GIS Labs)
		2. Digitalize permit and reporting systems	Municipal Headquarters		M		ICT Department, Municipal Council
		3. Develop a municipal climate portal	Entire Municipality			L	ICT Department, NEMA, Software Developers
		4. Use AI to optimize transport and waste systems	Lubowa		M		ICT Innovators, KCCA
		5. Archive local climate data for research	Entire Municipality		M		Makerere University, ICT Department, NEMA
RESULT AREA 9: Land and housing	**Output 9.1** Climate-resilient and inclusive housing promoted	1. Promote resilient building design	Namasuba, Busabala (flood-prone areas)		M		Municipal Engineering Department, Local Leaders, NGOs
Outcome 9: **A resilient and inclusive urban settlement system with climate-proof land use and housing infrastructure is developed**		2. Elevate structures in flood-prone areas	Lufuka, Kabuuma (high flood risk zones)		M		Municipal Engineering Department, Disaster Preparedness Office
		3. Encourage participatory planning for housing	Kisugula, Kanala (informal settlement hotspots)	S			Municipal Planning Department, Community Leaders

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		4. Use building materials with low carbon footprint	Entire Municipality (pilot in Lubowa)	S			Private Sector (construction firms), Municipal Procurement Unit
		5. Support low-income housing retrofits	Nankinga, Mirembe (vulnerable communities)		M		Municipal Housing Office, NGOs, CBOs
	Output 9.2 Climate risk integrated into urban planning	1. Develop zoning that avoids high-risk zones	Kabuuma, Mutungo (wetland/encroa chment areas)		M		Physical Planning Unit, NEMA
		2. Restrict development in wetlands and floodplains	Ndejje, Lufuka (degraded wetlands)		M		NEMA, Municipal Law Enforcement, Environment Police
		3. Promote mixed-use development to reduce emissions	Lubowa, Kisugula (urbanizing zones)			L	Municipal Planning Department, KCCA
		4. Include green spaces in urban planning	Gangu (low green cover areas)		M		KCCA, Municipal Environment Office
		5. Conduct risk-based land assessments	Entire Municipality (focus on Masajja, Ndejje)	S			Makerere University (GIS teams), Municipal Surveyors
	Output 9.3 Enforcement and monitoring strengthened	1. Enforce planning and construction standards	Entire Municipality (focus on Bunamwaya, Lubowa)	S			Municipal Law Enforcement, NEMA
		2. Improve land tenure systems for resilience	Kisugula, Kanala (informal settlements)		M		Ministry of Lands, Municipal Land Office
		3. Monitor informal development in risk zones	Masajja, Ndejje (floodplains)	S			Environment Police, Local Council Leaders
		4. Build local capacity for urban climate planning	Nankinga, Lufuka (low-capacity communities)		M		Town Clerk's Office, NGOs, CBOs

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		5. Establish a land and housing climate task force	Municipal-wide	S			Municipal Council, NEMA, Community Representatives
RESULT AREA 10. Social development (Communities)	**Output 10.1** Community awareness and participation in climate adaptation and DRR increased	1. Sensitize communities about climate change and disaster preparedness	Kabuuma, Lufuka (high-degradation parishes)	S			CBOs, Municipal Environment Office, Uganda Red Cross
Outcome 10: **Community resilience and inclusion in climate action is enhanced**		2. Train communities in sustainable practices like agroforestry	Mutungo, Ndejje (agroforestry/soil conservation zones)	S			Municipal Agriculture Office, NGOs
		3. Advocate for integrated climate change education into school curricula	Schools in Katwe, Gangu		M		Ministry of Education, Municipal Education Office
		4. Conduct regular community workshops on DRR and first aid	Masajja, Kisugula (flood-prone communities)	S			Uganda Red Cross, Municipal Health Department
		5. Promote inclusive planning by engaging youth, women, and persons with disabilities	Ndejje, Bunamwaya (areas with high youth/women populations)	S			Gender Office, Youth Councils, NGOs
	Output 10.2 Community livelihoods and support systems strengthened	1. Support vulnerable groups to diversify income sources (e.g., green enterprises)	Kisugula, Kanala (low-income communities)	S			Community Development Office (CDO), NGOs, Private Sector
		2. Support community-based enterprises in recycling, organic farming, eco-tourism	Nankinga, Mirembe (recycling/eco-tourism hubs)	S			Municipal Trade Office, CBOs
		3. Establish local funds for community-led adaptation & recovery projects	Masajja, Ndejje (flood-prone areas)		M		Municipal Council, Microfinance Institutions

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		4. Provide vocational training for youth and women in climate-smart sectors	Lufuka, Kabuuma (marginalized communities)		M		Vocational Institutes, NGOs, Gender Office
		5. Promote green jobs and entrepreneurship for marginalized groups	Entire Municipality (focus on Bunamwaya)		M		Private Sector, Ministry of Gender, Youth Councils
	Output 10.3 Public health, social infrastructure, and well-being improved	1. Promote rainwater harvesting and greywater recycling at household level	Nankinga, Lufuka (water-scarce areas)		M		MoWE, CBOs
		2. Establish green social spaces (parks, gardens) for recreation and carbon sinks	Ndejje, Gangu (urbanized low-green zones)			L	KCCA, Local Councils, CBOs
		3. Train communities in psychosocial support post-disaster	Masajja, Ndejje (disaster-prone areas)	S			Municipal Health Department, NGOs, Uganda Red Cross
		4. Improve access to sanitation and clean water in vulnerable parishes	Kabuuma, Kisugula (vulnerable parishes)		M		MoWE, Municipal Health Department
		5. Strengthen community safety nets (cash-for-work, food-for-work, micro-insurance)	Entire Municipality (focus on Lufuka, Mirembe)		M		Municipal Council, NGOs, Microfinance Institutions
RESULT AREA 10. Social development (Communities)	**Output 10.1** Community awareness and participation in climate adaptation and DRR increased	1. Sensitize communities about climate change and disaster preparedness	Kabuuma, Lufuka (high-degradation parishes)	S			CBOs, Municipal Environment Office, Uganda Red Cross
Outcome 10: Community resilience and inclusion in climate action is enhanced		2. Train communities in sustainable practices like agroforestry	Mutungo, Ndejje (agroforestry/soil conservation zones)	S			Municipal Agriculture Office, NGOs

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		3. Advocate for integrated climate change education into school curricula	Schools in, Gangu, Bunamwaya		M		Ministry of Education, Municipal Education Office
		4. Conduct regular community workshops on DRR and first aid	Masajja, Kisugula (flood-prone communities)	S			Uganda Red Cross, Municipal Health Department
		5. Promote inclusive planning by engaging youth, women, and persons with disabilities	Ndejje, Bunamwaya (areas with high youth/women populations)	S			Gender Office, Youth Councils, NGOs
	Output 10.2 Community livelihoods and support systems strengthened	1. Support vulnerable groups to diversify income sources (e.g., green enterprises)	Kisugula, Kanala (low-income communities)	S			Community Development Office (CDO), NGOs, Private Sector
		2. Support community-based enterprises in recycling, organic farming, eco-tourism	Nankinga, Mirembe (recycling/eco-tourism hubs)	S			Municipal Trade Office, CBOs
		3. Establish local funds for community-led adaptation & recovery projects	Masajja, Ndejje (flood-prone areas)		M		Municipal Council, Microfinance Institutions
		4. Provide vocational training for youth and women in climate-smart sectors	Lufuka, Kabuuma (marginalized communities)		M		Vocational Institutes, NGOs, Gender Office
		5. Promote green jobs and entrepreneurship for marginalized groups	Entire Municipality (focus on, Bunamwaya)		M		Private Sector, Ministry of Gender, Youth Councils
	Output 10.3 Public health, social infrastructure, and well-being improved	1. Promote rainwater harvesting and greywater recycling at household level	Nankinga, Lufuka (water-scarce areas)		M		Ministry of Water and Environment, CBOs

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		2. Establish green social spaces (parks, gardens) for recreation and carbon sinks	Gangu (urbanized low-green zones)			L	KCCA, Local Councils, CBOs
		3. Train communities in psychosocial support post-disaster	Masajja, Ndejje (disaster-prone areas)	S			Municipal Health Department, NGOs, Uganda Red Cross
		4. Improve access to sanitation and clean water in vulnerable parishes	Kabuuma, Kisugula (vulnerable parishes)		M		Ministry of Water and Environment, Municipal Health Department
		5. Strengthen community safety nets (cash-for-work, food-for-work, micro-insurance)	Entire Municipality (focus on Lufuka, Mirembe)		M		Municipal Council, NGOs, Microfinance Institutions
RESULT AREA 11. Education and Sports	**Output 11.1** Climate education and environmental awareness integrated into school curricula and co-curricular activities	1. Conduct school-based climate change sensitization sessions	Health centres in Kisugula, Kabuuma (high disease burden)	S			Municipal Education Office, Schools, NGOs (e.g., CBOs)
Outcome 11: Community resilience and inclusion in climate action is enhanced		2. Develop and distribute learner-friendly climate materials	Entire Municipality (pilot in Masajja, Ndejje)	S			Ministry of Education, Municipal Education Office, NGOs
		3. Organize inter-school sports and climate awareness events ("Green Games")	Flood-prone clinics (Lufuka, Nankinga)	S			Municipal Sports Office, Schools, NGOs
		4. Train teachers, sports coaches, and administrators on integrating climate action	Vulnerable parishes (Kabuuma, Kisugula)		M		Ministry of Education, Teacher Training Colleges, NGOs

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		5. Establish environment and climate clubs in schools	Masajja, Ndejje (high-risk zones)		M		Municipal Environment Office, Schools, NGOs
	Output 11.2 Capacity building in schools strengthened	1. Train school management teams on climate-resilient infrastructure	Clinics in Kabuuma, Kisugula		M		Municipal Engineering Department, School Management Boards
		2. Provide training for teachers on participatory climate learning	Flood-prone clinics (Lufuka, Nankinga)		M		Ministry of Education, NGOs, Vocational Institutes
		3. Integrate climate themes into sports events and clubs	Vulnerable areas (Masajja, Ndejje)		M		Municipal Sports Office, Schools, Youth Councils
		4. Facilitate climate leadership programs targeting youth	Entire Municipality (focus on Katwe, Gangu)		M		NGOs (e.g., UNICEF), Youth Councils, Schools
		5. Organize climate-focused debates and quizzes in schools	Health centers in Bunamwaya, Lubowa		M		Schools, Municipal Education Office
	Output 11.3 Inclusive school and community-based climate action initiatives implemented	1. Initiate talent development programs in sports and climate leadership	Kabuuma, Lufuka (high-risk communities)	S			Municipal Sports Office, NGOs, Community Leaders
		2. Promote school-community tree planting and clean-up drives	Entire Municipality (focus on, Ndejje)	S			Municipal Environment Office, Schools, CBOs
		3. Establish school-community partnerships for local climate projects	Nankinga, Kisugula (water-scarce areas)		M		Municipal Education Office, CBOs, Schools
		4. Engage youth groups in school-based climate monitoring	Masajja, Ndejje (flood-prone zones)		M		Youth Councils, Schools, Municipal Environment Office

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		5. Promote lifelong learning and community engagement events	Municipal-wide			L	Municipal Education Office, NGOs, Community Centers
RESULT AREA 12. Health	**Output 12.1** Health system capacity strengthened	1. Train health workers on climate-sensitive disease management	Masajja, Ndejje (flood-prone areas)	S			Municipal Health Department, Ministry of Health, WHO
Outcome 12: Resilience of the health system and communities to climate-related health risks and emergencies is improved		2. Improve disease surveillance and reporting systems	Lufuka, Kabuuma (high flood risk zones)		M		Uganda Ministry of Health, Municipal Health Office
		3. Equip health centers with climate-related emergency supplies	Kisugula, Kanala (informal settlement hotspots)		M		Red Cross, Municipal Health Department
		4. Support health workers with climate-health education toolkits	Entire Municipality (pilot in Lubowa)	S			NGOs (e.g., MSF), Municipal Health Office
		5. Develop rapid response units for climate-induced health emergencies	Nankinga, Mirembe (vulnerable communities)		M		Uganda Red Cross, Municipal Disaster Preparedness Office
	Output 12.2 Access to climate-resilient health infrastructure and services improved	1. Provide clean water and improved sanitation in health centers	Kabuuma, Mutungo (wetland/encroachment areas)	S			Ministry of Water and Environment, Municipal Health Office
		2. Upgrade health centers with rainwater harvesting and flood-proofing	Ndejje, Lufuka (degraded wetlands)		M		Municipal Engineering Department, KCCA
		3. Construct climate-resilient health centers in vulnerable areas	Lubowa, Kisugula (urbanizing zones)			L	Ministry of Health, Private Sector Contractors
		4. Equip health centers for climate-related emergencies	Gangu (low green cover areas)		M		Municipal Health Department, NGOs

Result Areas & Outcomes	Outputs/Objectives	Climate actions to reduce Greenhouse Gasses	Priority areas/Divisions/wards	Short-term	Mediu m-term	Long-term	Responsibility
		5. Improve waste management systems at health facilities	Entire Municipality (focus on Masajja, Ndejje)		M		NEMA, Municipal Health Office
	Output 12.3 Community awareness and adoption of climate-smart health practices increased	1. Conduct community sensitization on climate-related health risks	Entire Municipality (focus on Bunamwaya, Lubowa)	S			Municipal Health Office, CBOs
		2. Carry out public health campaigns on hygiene and disease prevention	Kisugula, Kanala (informal settlements)	S			Uganda Red Cross, Municipal Health Department
		3. Promote household climate-sensitive practices (safe water, mosquito nets, clean cooking)	Masajja, Ndejje (floodplains)	S			NGOs, Municipal Health Office
		4. Develop community-based early warning systems	Nankinga, Lufuka (low-capacity communities)		M		ICT Department, Community Leaders
		5. Establish municipal platforms for public dialogue on climate and health	Municipal-wide			L	Municipal Council, Health NGOs, Community Representatives

3.3 Implementation Arrangements

3.3.1 Overall Coordination Framework

Makindye Ssabagabo Municipal Council will serve as the lead entity responsible for implementing the Climate Action Plan. A Municipal Climate and Disaster Taskforce, chaired by the Department of Natural Resources, will be established. This taskforce will comprise technical representatives from relevant departments including Works, Health, Physical Planning, Community Development, and Education. The taskforce will be responsible for overseeing plan implementation, mobilizing resources, coordinating activities, conducting monitoring and evaluation, and reporting progress. It will report directly to the Town Clerk.

3.3.2 Division-Level Coordination

Implementation at the division level will be guided by Environment and Natural Resources Committees. Health and Sanitation Committees at the division level will also support implementation. These committees will mobilize local communities, enforce climate actions, and report to the municipal taskforce.

3.3.3 Roles of Stakeholders

The stakeholder in the implementation of the climate action plan includes government institutions, non-governmental organizations, community groups, academic institutions, and private sector actors. Each stakeholder will play a unique role ranging from leadership, policy formulation, enforcement, technical support, and funding, to community mobilization, education, innovation, and public awareness. This multi-stakeholder approach will ensure inclusive, participatory, and effective climate action by leveraging diverse expertise, resources, and community knowledge.

Table 25: Roles of stakeholders

Stakeholder	Category	Role in Climate Action Plan Implementation
National Environment Management Authority (NEMA)	Government Agency	Enforce environmental laws, monitor pollution, approve restoration projects, and regulate wetland/industrial activities.
Kampala Capital City Authority (KCCA)	Local Government	Oversee urban planning and infrastructure development (e.g., drainage, green corridors), and coordinate city-wide climate initiatives.
Makindye Ssabagabo Municipal Council	Local Government	Lead implementation, allocate budgets, coordinate stakeholders, enforce local bylaws, and report progress.
Municipal Environment Office	Local Government	Manage waste systems, conduct sensitization campaigns, and oversee wetland/forest restoration.
Community-Based Organizations (CBOs)	Community Organizations	Mobilize communities, lead grassroots initiatives (e.g., tree planting, waste segregation), and provide local feedback.
Non-Governmental Organizations (NGOs)	Civil Society	Provide technical/financial support, train communities, and implement projects (e.g., eco-tourism, clean energy).
Local Council Leaders (LC1s, LC3s)	Community Leadership	Enforce bylaws, mobilize communities, resolve land conflicts, and report encroachment/illegal activities.
Private Sector (e.g., manufacturers, logistics)	Private Sector	Adopt green technologies, fund CSR initiatives, and support renewable energy/eco-friendly trade.

Stakeholder	Category	Role in Climate Action Plan Implementation
Uganda Manufacturers Association (UMA)	Private Sector Association	Promote circular production, train industries on energy efficiency, and advocate for green policies.
Ministry of Water and Environment (MoWE)	National Government	Manage water resources, support wetland demarcation, and oversee climate-resilient infrastructure.
Ministry of Agriculture	National Government	Promote climate-smart agriculture (CSA), distribute drought-resistant seeds, and train farmers.
Uganda National Farmers Federation	Farmers' Association	Link farmers to markets, advocate for CSA policies, and organize cooperatives.
Religious/Cultural Leaders	Community Leadership	Advocate for conservation in communities, integrate climate messages into cultural/religious practices.
Academic Institutions (e.g., Makerere University)	Academia/Research	Conduct research, provide GIS mapping, and support data-driven planning (e.g., risk assessments).
Uganda Red Cross	International NGO	Lead disaster response, train communities on DRR, and distribute emergency supplies.
World Bank/Development Partners	International Organizations	Fund large-scale projects (e.g., wetland restoration, renewable energy) and provide technical expertise.
Youth/Women Groups	Marginalized Groups	Participate in green jobs, lead community projects (e.g., recycling), and advocate for inclusive planning.
Vocational Training Institutes	Education	Train youth/women in climate-smart skills (e.g., solar tech, agroforestry).
Media Outlets (e.g., CBS Radio)	Civil Society	Disseminate climate information, run awareness campaigns, and report environmental violations.
Private Nurseries/Agro-Dealers	Private Sector	Supply climate-resilient seeds, seedlings, and farming tools to communities.
Community Health Workers	Healthcare	Monitor climate-related diseases, conduct hygiene campaigns, and report outbreaks.

3.4 Costing of Climate Actions

The table below provides a detailed breakdown of costs to realize the climate action outputs to guide planning, resource mobilization, and effective implementation of climate mitigation and resilience-building activities.

Table 26: Costing of climate action outputs (in millions of UGX)

Result Areas	Outputs	2026	2027	2028	2029	2030	Total
		Uganda Shillings ('000)					
1. Waste Management	1.1 Community awareness programs	422.7	429.8	431.4	437.1	443.2	2164.2
	1.2 Organic/inorganic waste processing	352.8	361.3	368.6	373.5	387.3	1843.5
	1.3 Infrastructure improvement	2320.6	3332.4	3324.8	3357.3	3377.3	15712.4
2. Energy	2.1 Energy efficiency campaigns	356.4	366.4	376.4	386.4	396.4	1882
	2.2 Renewable energy access	649.8	650.7	653.3	663.4	674.2	3291.4
	2.3 Emission control tech	486.2	493.8	497.7	499.7	514.2	2491.6
3. Transportation	3.1 Awareness campaigns	381.6	393.3	395.5	398.4	399.6	1968.4
	3.2 Clean transport infrastructure	137.5	137.5	137.5	137.5	137.5	687.5
	3.3 Planning enhancement	246.8	246.8	246.8	246.8	246.8	1234
4. Trade	4.1 Climate-smart trade initiatives	342.3	342.3	342.3	342.3	342.3	1711.5
	4.2 Sustainable logistics	214.1	214.1	214.1	214.1	214.1	1070.5
	4.3 Trade policy support	136.7	136.7	136.7	136.7	136.7	683.5
5. Manufacturing	5.1 Emission reduction support	485.4	488.8	491.7	495.8	498.5	2460.2
	5.2 Circular production incentives	321.5	339.4	356.5	365.3	374.2	1756.9
	5.3 Sustainable supply chain programs	392.8	398.4	399.5	402.8	411.3	2004.8
6. Agriculture	6.1 Climate-smart agriculture	368.3	368.3	368.3	368.3	368.3	1841.5
	6.2 Water & soil conservation	426.9	437.6	448.3	462.5	473.6	2248.9
	6.3 Value chain development	324.1	380.3	394.2	414.8	426.8	1940.2
7. Environment	7.1 Environmental awareness	213.4	313.4	389.4	423.2	487.4	1826.8
	7.2 Ecosystem restoration	423.5	473.5	493.5	523.5	423.5	2337.5
	7.3 Urban planning enforcement	238.7	338.7	378.7	438.7	488.7	1883.5
8. ICT	8.1 Early warning systems	462.2	562.2	582.2	662.2	762.2	3031
	8.2 Digital literacy training	263.6	363.6	463.6	563.6	663.6	2318
	8.3 ICT integration into planning	242.2	362.2	372.2	482.2	492.2	1951

Result Areas	Outputs	2026	2027	2028	2029	2030	Total
				Uganda Shillings ('000)			
9. Land & Housing	9.1 Climate-resilient housing	683.2	693.2	718.2	744.8	883.2	3722.6
	9.2 Climate risk integration	209.5	247.5	269.5	289.5	299.5	1315.5
	9.3 Enforcement monitoring	260.4	289.4	366.4	369.4	410.4	1696
10. Social Development	10.1 Community participation	324.6	434.6	448.6	354.3	374.5	1936.6
	10.2 Livelihood support programs	653.2	693.2	733.2	759.2	773.2	3612
	10.3 Health & well-being	470.7	493.5	520.5	560.2	580.3	2625.2
11. Education and Sports	11.1 Climate education and environmental	312.3	342.3	365.3	385.3	392.3	1797.5
	11.2 Capacity building in schools	248.4	264.4	278.4	288.4	298.4	1378
	11.3 School-based climate action initiatives	354.2	364.2	374.2	384.2	394.2	1871
12. Health	12.1 Health system capacity strengthened	428.6	468.6	488.6	498.6	499.6	2384
	12.2 Climate-resilient health infrastructure	6245.3	6545.3	6845.3	6945.3	6989.3	33570.5
	12.3 Climate-smart health community awareness and adoption	328.1	428.1	368.1	388.1	398.1	1910.5

3.5 Resource Mobilization

The implementation of the climate action plan will rely on

1. Municipal budget allocations through annual budgets aligned to the Climate Action Plan
2. Central government transfers and sector conditional grants for environmental, health, and infrastructure projects
3. Development partner contributions through project-based and technical support programs
4. Private sector investments in renewable energy, housing, waste management, and urban agriculture
5. Community contributions in the form of labor, materials, or voluntary services for local initiatives
6. Innovative financing mechanisms such as climate bonds, carbon credit trading, and green funds

Table 27: Funding sources and use

Sources	Use
Local Revenue	Partial funding for implementation, e.g., emergency budgeting, quick response
Development Partners	Infrastructure, capacity building, early warning systems, innovation projects (e.g., Youth Climate Action Fund)
Government Grants & Sectoral Ministries (e.g., MEMD, MAAIF, MoLHUD)	Programmatic support for specific result areas like agriculture, housing, and energy
Private Sector Contributions	Investment in green businesses, public-private partnerships
Community Contributions	Volunteering, local labor, community-owned adaptation initiatives

3.6 Mechanism to Disseminate climate information

To ensure that timely, reliable, and actionable climate information reaches decision-makers, communities, and relevant stakeholders to enhance preparedness, resilience, and informed climate action in Makindye-Ssabagabo Municipality, the following climate information sharing channels will be utilized.

3.6.1 Early Warning Systems (EWS)

1. Automated SMS and Radio Alerts: Bulk SMS platforms and local FM radios will disseminate warnings on extreme weather events (storms, floods, and heatwaves).
2. Weather Stations: Installed stations will continuously monitor local conditions, providing real-time updates to municipal authorities and the public.
3. EWS Mobile Application: A municipal EWS app will deliver forecasts, alerts, and safety advice to residents, community leaders, and institutions.

3.6.2 Community-Based Information Channels

1. Community Meetings (Barazas): Local leaders and municipal staff will organize regular community sensitization meetings, especially before and during rainy seasons.
2. Religious and Social Institutions: Information shared through churches, mosques, schools, and social clubs using noticeboards, bulletins, and announcements.

3. Town Council Noticeboards: Climate updates, emergency alerts, and guidance notices posted at municipal and division offices.

3.6.3 ICT and Digital Platforms

Social Media (WhatsApp, Facebook, Twitter): Makindye-Ssabagabo Municipal Council official accounts will broadcast climate-related information, advisories, and risk alerts.
Website and Online Portals: The municipal website will host climate bulletins, advisories, seasonal forecasts, and emergency contact information.

3.6.4 Institutional Networks

Disaster Committees and Focal Persons: Sectoral and divisional DRR committees will cascade climate information to wards, schools, businesses, and local institutions.
Schools and Health Facilities: Climate alerts and preparedness information shared through school clubs and health centers.

3.6.5 Materials

Posters, brochures, and flyers are distributed in marketplaces, health centers, schools, and churches, carrying simplified climate forecasts, preparedness tips, and emergency contacts.
IEC (Information, Education, and Communication) Materials: Targeted messages in local languages for vulnerable groups like women, the elderly, and persons with disabilities.

3.6.6 Feedback and Two-Way Communication

The municipality will provide a feedback communication mechanism to allow communities to report local weather observations, hazards, and incidents; ask questions or seek clarifications through hotlines, social media, and EWS app features; and engage in participatory meetings to review the usefulness and clarity of information.

4. MONITORING,EVALUATION AND LEARNING

4.1 Introduction

Effective implementation of a climate action plan requires a strong and inclusive Monitoring, Evaluation, and Learning (MEL) system. The MEL framework will ensure that progress is systematically tracked, outcomes are evaluated, and valuable lessons are documented and integrated into future planning and decision-making processes. The specific objectives of the MEL include:

1. Monitor the progress of actions across all result areas and sectors.
2. Evaluate the effectiveness and efficiency of interventions.
3. Identify and share best practices and lessons learned.
4. Promote community feedback and inclusive learning.
5. Enable evidence-based decision-making for adaptive management

4.2 Institutional Responsibility

To achieve the objectives of the MEL, a coordinated structure has been established, assigning clear roles and responsibilities to key actors at different levels of the municipal system. These actors range from municipal leadership and departmental focal persons to grassroots community representatives, technical experts, and development partners. Each actor plays a distinct and complementary role, contributing to the overall effectiveness, accountability, and responsiveness of climate actions within the municipality.

Table 28: MEL structure and Roles

Actors	Category	Role in MEL Framework
Municipal Council (Climate Task Force)	Local Government	Oversee MEL implementation, allocate resources, and consolidate reports. Ensure alignment with municipal goals and national policies.
Municipal Environment Office	Technical Department	Monitor progress of environmental actions (e.g., wetland restoration, waste management). Submit quarterly progress reports.
Sector Heads (e.g., Health, Transport)	Municipal Departments	Track progress in their domains (e.g., health clinics, cycling lanes). Adjust sectoral plans based on MEL findings.
NEMA (National Environment Management Authority)	National Agency	Evaluate adherence to environmental laws and standards. Validate data accuracy and intervention impacts.
CBOs and NGOs	Civil Society	Gather grassroots insights on project effectiveness. Involve communities in data collection (e.g., tree survival rates).
Academic/Research Institutions	Academia	Conduct independent studies to assess outcomes (e.g., carbon sequestration, flood reduction). Train municipal staff on MEL tools.
Community Leaders (LC1s, Religious/Cultural Leaders)	Local Governance	Verify on-ground progress and challenges. Share MEL findings with communities in local languages.
Private Sector Partners	Business/Industry	Track green initiatives (e.g., energy-efficient factories, eco-trade). Suggest improvements based on MEL insights.
ICT Department	Technical Department	Maintain a municipal climate portal for real-time data tracking. Use AI/GIS tools to visualize trends (e.g., emission hotspots).
Uganda Bureau of Statistics (UBOS)	National Agency	Provide socio-economic and environmental indicators. Ensure MEL metrics align with national frameworks.
Youth/Women Groups	Marginalized Groups	Highlight gender-specific impacts and barriers. Lead community dialogues to share lessons.
Development Partners (e.g., World Bank)	International Agencies	Fund independent audits of high-impact projects. Introduce global best practices.
Media (Radio, Local Press)	Communication	Publicize MEL results and success stories. Report gaps or delays in implementation.

4.2.1 MEL Indicators

Indicator Type	Examples
Input Indicators	Number of climate projects funded; staff trained
Output Indicators	Number of trees planted; early warning systems installed
Outcome Indicators	Reduction in flood impacts, increased renewable energy adoption
Impact Indicators	Improved food security, reduced disaster-related losses

4.2.2 Tools and Approaches

1. Quarterly progress reports
2. Annual review workshops
3. Community scorecards
4. GIS and remote sensing
5. Mobile-based data collection (e.g., Kobo Toolbox)
6. Before-and-after assessments

4.2.3 Learning and Feedback Mechanisms

1. Community feedback sessions at division level
2. Knowledge sharing workshops with stakeholders
3. Integration of indigenous knowledge into adaptation strategies
4. Documentation of success stories and lessons learned

4.2.4 Reporting and Communication

1. Regular reports submitted to the Town Clerk and Municipal Council
2. Annual Climate Action Progress Reports shared with stakeholders
3. Dashboards or portals for real-time tracking (to be developed)
4. Public briefings via radio, community meetings, and social media

4.2.5 Adaptive Management

1. Use MEL findings to revise strategies and reallocate resources
2. Identify underperforming interventions for redesign or discontinuation
3. Integrate climate projections and emerging data to update plans

4.2.6 Risk Management

Potential implementation risks include limited funding, political interference, climate disasters, and low community participation. Risk mitigation strategies include:

1. Diversified financing and partnerships
2. Strengthening local ownership and participation
3. Institutionalizing climate action in municipal policies
4. Regular scenario planning and adaptive management approaches

Table 6: Climate Action Monitoring, Evaluation, and Learning Framework (2026-2030)

Outcome/Result Areas	Outputs	Indicators	Means of Verification	Baseline (2024)	Target 2026	Target 2027	Target 2028	Target 2029	Target 2030
1. Waste Management	1.1 Community awareness and participation enhanced	% of households participating in waste sorting and recycling	Household surveys, program records	20%	35%	50%	65%	75%	85%
	1.2 Organic and inorganic waste processing improved	Tons of waste processed annually	Waste processing plant reports	5,000	7,000	10,000	12,000	14,000	16,000
	1.3 Waste disposal infrastructure improved	No. of improved/rehabilitated waste disposal sites	Municipal reports	2	3	5	7	8	10
2. Energy	2.1 Energy efficiency improved	% reduction in municipal energy consumption	Energy audit reports	0%	5%	10%	15%	20%	25%
	2.2 Access to renewable energy enhanced	% of households using renewable energy	Household surveys, utility records	10%	15%	20%	30%	40%	50%
	2.3 Emissions from infrastructure reduced	% reduction in emissions from public infrastructure	Emissions monitoring reports	0%	5%	10%	20%	30%	40%
3. Transportation	3.1 Awareness and low-emission transport use increased	% of population using low-emission transport	Traffic surveys, commuter records	5%	10%	20%	30%	40%	50%
	3.2 Infrastructure for clean transport promoted	Km of new bicycle lanes and pedestrian paths	Infrastructure records	5	10	15	20	25	30
	3.3 Integrated sustainable planning enhanced	No. of integrated urban transport plans developed	Municipal planning documents	0	1	2	3	4	5

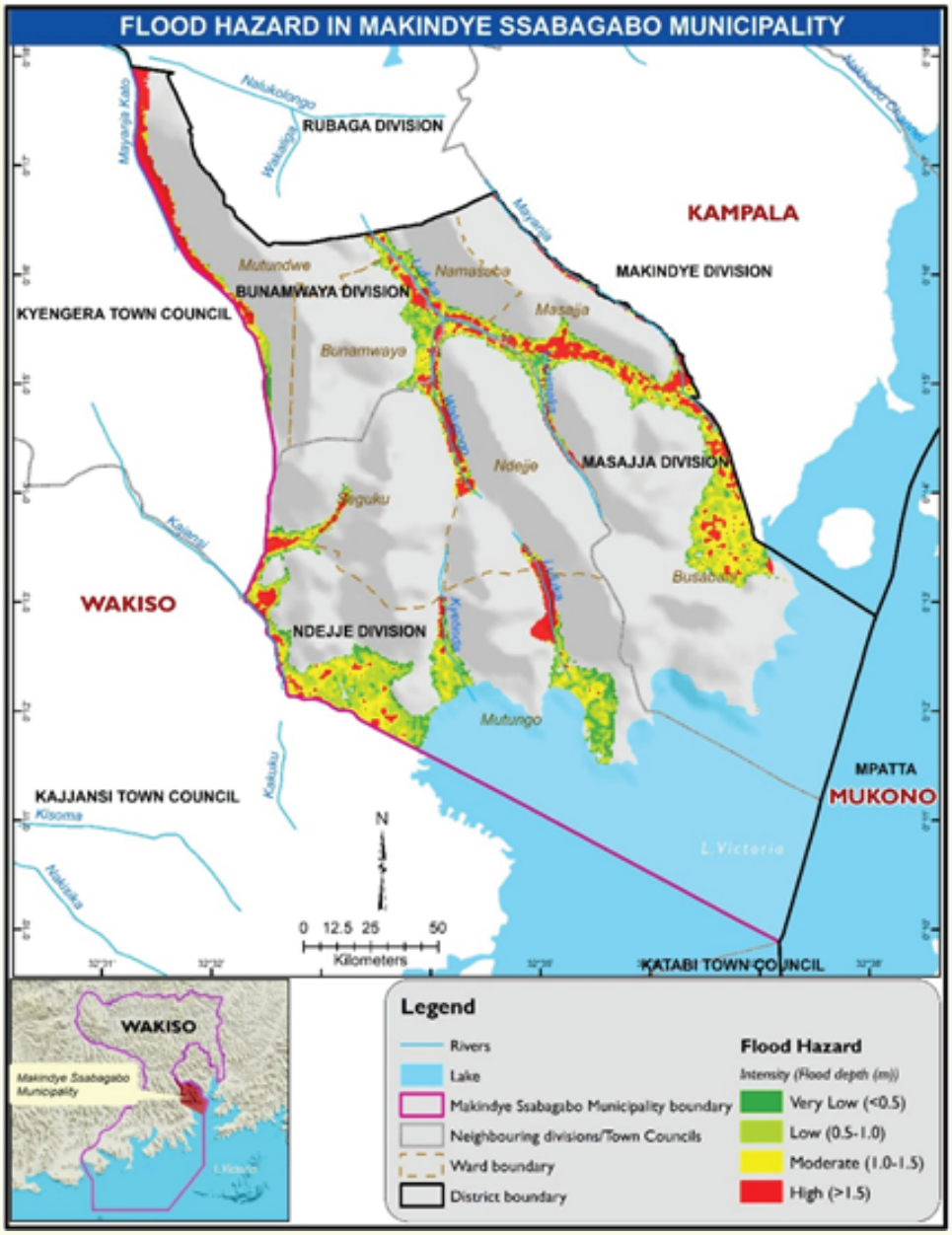
Outcome/Result Areas	Outputs	Indicators	Means of Verification	Baseline (2024)	Target 2026	Target 2027	Target 2028	Target 2029	Target 2030
4. Trade	4.1 Climate-smart and low-carbon trade practices promoted	% of markets adopting eco-friendly practices	Market surveys, inspection reports	5%	10%	20%	30%	40%	50%
	4.2 Improved efficiency and reduced emissions in logistics	% reduction in emissions from trade transport	Emissions audit reports	0%	5%	10%	20%	30%	40%
	4.3 Policies and incentives for sustainable trade strengthened	No. of policies developed and implemented	Policy review documents	0	1	2	3	4	5
5. Manufacturing	5.1 Industrial emissions and pollution reduced	% reduction in emissions from local industries	Industrial audit reports	0%	5%	10%	20%	30%	40%
	5.2 Circular and energy-efficient production promoted	% of industries applying circular production	Industry surveys	5%	10%	20%	30%	40%	50%
	5.3 Sustainable supply chains promoted	% of businesses adopting green supply chains	Trade association reports	5%	10%	20%	30%	40%	50%
6. Agriculture	6.1 Climate-smart agriculture practices promoted	% of farmers adopting climate-smart practices	Agricultural extension reports	10%	20%	30%	40%	50%	60%
	6.2 Water use and soil conservation improved	% of farms implementing soil/water conservation	Farm surveys	5%	15%	25%	35%	45%	55%
	6.3 Value chains and access to green markets enhanced	No. of farmers accessing green value chains	Market records	50	100	150	200	250	300

Outcome/Result Areas	Outputs	Indicators	Means of Verification	Baseline (2024)	Target 2026	Target 2027	Target 2028	Target 2029	Target 2030
7. Environment	7.1 Environmental awareness and stewardship promoted	% of residents aware of climate and environmental issues	Community surveys	30%	50%	60%	70%	80%	90%
	7.2 Natural ecosystems restored and protected	Area (ha) of green and natural spaces restored	Environmental agency reports	10	20	30	40	50	60
	7.3 Sustainable planning and enforcement strengthened	No. of enforcement actions on illegal environmental practices	Municipal records	10	20	30	40	50	60
8. ICT	8.1 Climate and disaster early warning systems improved	% of population receiving timely early warnings	Communication system reports	20%	40%	60%	70%	80%	90%
	8.2 Community digital literacy enhanced	% of citizens trained on digital tools for climate adaptation	Training reports	10%	20%	30%	40%	50%	60%
	8.3 ICT integrated into climate planning and governance	No. of plans integrating ICT-based climate tools	Planning reports	1	2	3	4	5	6
9. Land & Housing	9.1 Climate-resilient and inclusive housing promoted	% of new houses built to climate-resilient standards	Building permits, inspections	10%	20%	30%	40%	50%	60%
	9.2 Climate risk integrated into urban planning	% of plans incorporating climate risk assessments	Urban plans	10%	25%	40%	55%	70%	85%
	9.3 Enforcement and monitoring strengthened	No. of land use and housing violations acted upon	Municipal records	10	20	30	40	50	60
10. Social Development	10.1 Community awareness and participation in climate action increased	% of population participating in climate adaptation activities	Community activity records	15%	25%	35%	45%	55%	65%

Outcome/Result Areas	Outputs	Indicators	Means of Verification	Baseline (2024)	Target 2026	Target 2027	Target 2028	Target 2029	Target 2030
	10.2 Community livelihoods and support systems strengthened	No. of households benefiting from livelihood programs	Program reports	200	400	600	800	1000	1200
	10.3 Public health, social infrastructure and well-being improved	% of public spaces upgraded with climate-resilient features	Infrastructure upgrade reports	10%	20%	30%	40%	50%	60%
11. Education and Sports	11.1 Climate education and environmental awareness integrated into school curricula and co-curricular activities	No. of schools with integrated climate education	MoES curriculum records, inspection reports	5	20	35	50	65	80
	11.2 Capacity building in schools strengthened	No. of teachers and school leaders trained on climate-resilient education	Training reports, attendance sheets	0	150	300	450	600	750
	11.3 School-based climate action initiatives implemented	No. of functional school-community climate clubs/projects	School activity reports, MoES reports	3	10	25	40	55	70
12. Health	12.1 Health system capacity strengthened	No. of health workers trained in climate-health	MoH training reports, certification lists	20	80	120	150	180	200
	12.2 Climate-resilient health infrastructure improved	No. of health centres upgraded with climate-resilient infrastructure (e.g., water harvesting, solar, flood-proofing)	Infrastructure audit reports, MoH & NMC assessments	2	5	8	11	13	15
	12.3 Climate-smart health community awareness and adoption enhanced	% of households adopting safe water, sanitation, and clean energy practices	Household survey reports, MoH outreach logs	15%	25%	35%	45%	55%	65%

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